

Beyond the Status Quo: A Critical Assessment of Lifecycle Investment Advice

Aizhan Anarkulova*

Scott Cederburg[†]

Michael S. O'Doherty[‡]

August 27, 2026[§]

Abstract

We challenge two tenets of lifecycle investing: (i) diversify across stocks and bonds and (ii) reduce equity allocations with age. We estimate optimal age-based allocations in a lifecycle model that jointly optimizes saving, withdrawals, and portfolio choice. The model incorporates a nonparametric block bootstrap that preserves the time-series and cross-sectional dependence structure of asset returns. The optimal portfolio remains near one-third domestic stocks and two-thirds international stocks regardless of age, with no allocation to fixed income assets. This internationally diversified, all-equity strategy substantially outperforms conventional target-date and balanced strategies in building wealth, preserving capital, supporting retirement income, and providing bequests.

JEL classifications: C15, D14, G11, G17, G51

Key words: Lifecycle asset allocation, retirement saving, target-date funds, long-horizon returns

*Goizueta Business School, Emory University. Email: aizhan.anarkulova@emory.edu.

[†]Eller College of Management, University of Arizona. Email: cederburg@arizona.edu.

[‡]Trulaske College of Business, University of Missouri. Email: odohertym@missouri.edu.

[§]We thank Turan Bali, Sreedhar Bharath, David Brown, John Y. Campbell, David Cantor, Allison Cole, Filipe Correia, Tim de Silva, Victor Duarte, Benjamin Felix, Scott Findley, Carolina Fugazza, Andrei Gonçalves, Clifton Green, Michael Gropper, Narasimhan Jegadeesh, Veronika Krepely Pool, Zhongjin Gene Lu, Benjamin Moll, Terry Nixon, Kunal Sachdeva, Rik Sen, John Shim, Mitch Towner, Chuck Trzcinka, Harry Turtle, and seminar participants at the 2023 MSUFCU Conference on Financial Institutions and Investments, the 2023 Netspar Pension Day, the 2024 Eastern Finance Association, the 2024 SFS Cavalcade North America, the 2024 Southern Finance Association, the 2025 American Finance Association, the 2025 ASU Sonoran Winter Finance Conference, the 2025 Four Corners Index Investing Jamboree, the 2025 ICPM Research Awards Finale, the 5th LTI-Bank of Italy Workshop on Long-Term Investors' Trends: Theory and Practice, Bluemetric, the Brandes Center at the University of California, San Diego, Emory University, Erasmus University Rotterdam, the Georgia Institute of Technology, Lehigh University, Louisiana State University, the University of Arizona, the University of Arkansas, the University of Georgia, and Utah State University for helpful comments and suggestions. The authors acknowledge the financial support provided by a research grant from Goizueta Business School of Emory University.

1 Introduction

Every year, Americans contribute more than 5% of their total employee compensation to defined contribution (DC) pension plans, with contributions of \$768 billion in 2023 alone.¹ They then face a question that determines their financial fate: How should I invest my savings? Many consult financial advisors. These professionals impart two central tenets of lifecycle investing—people should diversify across stocks and bonds and the young should invest more heavily in stocks than the old—perhaps having learned them from investments textbooks [e.g., Bodie, Kane, and Marcus (2024)] or CFA study materials [e.g., Blanchett, Cordell, Finke, and Idzorek (2023)]. Self-directed savers who read a popular book by Dave Ramsey, Suze Orman, or Tony Robbins receive similar advice [Choi (2022)]. Academics may study the literature on lifecycle investing and reach the same conclusions [e.g., Viceira (2001); Campbell and Viceira (2002); and Cocco, Gomes, and Maenhout (2005)]. A great many others are uninterested or overwhelmed, so they invest in the default option of their employer-sponsored retirement plan. To safeguard these investors, the Pension Protection Act of 2006 (PPA) created safe harbors for employer DC plans. The most popular Qualified Default Investment Alternatives (QDIAs) are portfolios that provide “long-term appreciation and capital preservation through a mix of equity and fixed income exposures based on the participant’s age” [29 CFR § 2550.404c-5(e)(4)(i)].² As such, regulators rely on “generally accepted investment theories” [29 CFR § 2550.404c-5(e)(4)] that mirror the two principles to define QDIAs. In summary, these two tenets of investment advice—split investments across stocks and bonds and invest more in stocks while young than while old—are close to being uniformly given and universally followed.

In this paper, we challenge these principles of lifecycle investing. We consider a US couple

¹The total DC plan contribution is from the *Private Pension Plan Bulletin: Abstract of 2023 Form 5500 Annual Reports* from the Department of Labor. The contribution amount of \$767.6 billion represents 5.4% of 2023 compensation of employees (\$14,207.4 billion) from Table 2.1 of the National Income and Product Accounts (NIPA) published by the Bureau of Economic Analysis.

²See <https://www.law.cornell.edu/cfr/text/29/2550.404c-5>. Vanguard (2026) reports that target-date funds, which are QDIAs under 29 CFR § 2550.404c-5(e)(4)(i), make up 98% of the QDIAs in DC plans.

who jointly optimizes saving, withdrawals, and age-based portfolio allocations in the presence of labor income risk, Social Security income, and longevity risk. Our primary contribution lies in the modeling of investment opportunities. We depart from conventional modeling practice by adopting a nonparametric block bootstrap approach that avoids ex ante selection of which return features matter for portfolio choice and parametric assumptions about their functional forms. Our block bootstrap generates forward-looking holding-period returns by resampling long sequences of consecutive historical returns. These sequences are drawn from a comprehensive dataset of returns on domestic stocks, international stocks, government bonds, and government bills spanning over 2,600 years of country-month data from 39 developed countries.³ By construction, our approach preserves the historical time-series and cross-sectional properties of asset returns while capturing the latent dynamics of the investment opportunity set.

In the base case of our lifecycle model, the household chooses age-based allocations across domestic stocks, international stocks, bonds, and bills. We impose constraints—no leverage and non-negative portfolio weights—that reflect the opportunities available to most retirement savers. The optimal weights are remarkably stable throughout the lifecycle, with approximately one-third allocated to domestic stocks and two-thirds to international stocks at all ages, and no allocation to fixed income assets. The resulting strategy stands in sharp contrast to the status quo advice favoring age-based, stock-bond portfolios. Although explicit treatment of international stocks remains relatively uncommon in the lifecycle investing literature, the substantial allocation to this asset class underscores the importance of allowing for international diversification.⁴

³We consider multiple risky assets but no risk-free asset, similar to Campbell and Viceira (2002, 2005); Campbell, Chan, and Viceira (2003); Sangvinatsos and Wachter (2005); Hoevenaars, Molenaar, Schotman, and Steenkamp (2008); Koijen, Nijman, and Werker (2010); and Duarte, Fonseca, Goodman, and Parker (2024). A significant majority of studies include a risk-free asset, either with a single risky asset [e.g., Merton (1969); Samuelson (1969); Viceira (2001); Cocco, Gomes, and Maenhout (2005); Pástor and Stambaugh (2012); Dahlquist, Setty, and Vestman (2018); Gomes, Michaelides, and Zhang (2022); Gomes and Smirnova (2023); and Choukhmane and de Silva (2026)] or with multiple risky assets [e.g., Merton (1971); Brennan, Schwartz, and Lagnado (1997); Lynch (2001); Lynch and Tan (2010); and Catherine (2022)].

⁴Baxter and Jermann (1997) consider international diversification in the context of non-tradeable human capital, concluding that investors should short domestic markets to hedge labor income risk. Michaelides (2003); Hnatkovska (2010); Coeurdacier and Rey (2013); and Bretscher, Julliard, and Rosa (2016) study these hedging motives in lifecycle

We compare the optimal age-based strategy with two QDIA benchmarks: (i) a balanced strategy with 60% domestic stocks and 40% bonds and (ii) a representative target-date fund (TDF). The age-based structure of our optimal policy creates a natural comparison with TDFs, whose stock-bond allocations also vary with investor age. We make welfare comparisons using two complementary measures. The first is an equivalent annual fee, defined as the constant annual fee charged on assets under the optimal investment strategy that leaves the household indifferent between the optimal strategy and an alternative strategy. The second is the consumption certainty-equivalent loss (CEL), defined as the constant percentage reduction in lifetime consumption and bequest under the optimal strategy that leaves the household indifferent between the two strategies. Both measures indicate economically large welfare losses from following conventional lifecycle investment advice. Relative to the optimal strategy, the balanced strategy has an equivalent annual fee of 2.09% and a consumption CEL of 1.73%, while the corresponding values for the TDF are 1.90% and 1.56%, respectively.⁵

We further evaluate the optimal strategy by studying four retirement outcomes: wealth at retirement, the retirement income replacement rate, reliance on Social Security income, and bequest at death. The optimal age-based strategy outperforms the two QDIA benchmarks across all four dimensions even as households optimally choose lower saving rates. Although the superior long-term appreciation of an internationally diversified equity portfolio naturally produces greater retirement wealth, our simulations reveal a more striking result: the optimal strategy also provides stronger capital preservation throughout retirement despite maintaining a 100% equity allocation.

Households achieve higher retirement income replacement, are substantially less likely to rely

asset allocation models featuring foreign stocks, concentrating on the roles of frictions, incomplete markets, and the strength of the relation between human capital and domestic asset returns in rationalizing home bias. Several researchers examine international diversification in settings without lifecycle features [e.g., Solnik (1974), Jorion (1985), Eun and Resnick (1988, 1994), French and Poterba (1991), and Ang and Bekaert (2002, 2004)].

⁵The welfare comparisons are conducted in a partial equilibrium framework. Following convention in the lifecycle portfolio choice literature, we do not model the general equilibrium adjustments in asset prices or expected returns that could arise if many households simultaneously adopted the optimal strategy.

primarily on Social Security income, and leave larger bequests than under the benchmark strategies. These advantages hold not only on average but throughout the distribution of retirement outcomes. Overall, the optimal age-based portfolio beats the two QDIAs across the board in achieving the PPA goals of long-term appreciation and capital preservation.

Under the optimal strategy, households choose to diversify throughout the lifecycle using international stocks rather than bonds. This portfolio choice reflects, in part, the long-horizon properties of asset returns uncovered by our nonparametric block bootstrap. To illustrate, Table I reports the annualized mean and standard deviation of returns on bonds and international stocks (Panel A); variance ratios at horizons of one, ten, 20, and 30 years calculated as in Poterba and Summers (1988) (Panel B); and correlations of log returns and log inflation (Panel C). Bonds offer modest average real returns (0.95% annually) compared with international stocks (7.03%), requiring substantial diversification benefits to justify their inclusion in an optimal lifecycle portfolio. At the one-month horizon, bonds appear less risky with lower standard deviation (9.51% versus 23.26% annualized) and lower correlation with domestic stocks (0.21 versus 0.33). At the 30-year horizon, the picture changes. Bonds' per-year variance increases to 2.39 times the one-year variance, but international stocks' decreases to 0.75 times. The correlation of bonds with domestic stocks rises to 0.45, whereas international stocks maintain a steady correlation.⁶ International stocks also better preserve real buying power, exhibiting essentially no correlation with inflation (-0.02) compared with a correlation of -0.78 for bonds. In sum, bonds ultimately seem unattractive for long-horizon investors. They have low returns, high long-term variance, high long-term correlation with domestic stocks, and high exposure to inflationary periods.

Our analysis differs from the existing lifecycle literature primarily in its treatment of the

⁶Also see Campbell and Viceira (2002, 2005); Bali, Demirtas, Levy, and Wolf (2009); and Siegel (2014) for evidence on the risk of bonds over extended holding periods. The internet appendix confirms that our block bootstrap simulations preserve the variance ratio patterns in the actual data.

investment opportunity set. Building on foundational lifecycle portfolio choice models such as Cocco, Gomes, and Maenhout (2005), much of the recent literature has evaluated optimal asset market participation and asset allocation by incorporating additional sources of household risk and new margins of household choice, including labor income risk, housing, endogenous retirement, health expenditures, flexible labor supply, and portfolio participation frictions [see Gomes, Haliassos, and Ramadorai (2021) for a review]. These studies typically characterize investment opportunities using parameterized return distributions compatible with their solution methods. Any realistic characterization of investment opportunities, however, requires an infeasibly large number of parametric choices governing the levels and time variation in expected returns, volatility, higher-order moments, correlations, and higher-order cross-moments. These assumptions ultimately drive conclusions about optimal portfolio choice [e.g., Michaud (1989); Brandt (1999); Garlappi, Uppal, and Wang (2007); Wachter (2010); and Pástor and Stambaugh (2012)], such that omissions or inaccuracies can materially affect the resulting portfolio recommendations. Our results suggest that the economic implications of modeling long-horizon investment opportunities are at least as great as those of the household-side extensions emphasized in the recent lifecycle literature.

To explain why our findings differ from conventional lifecycle investment advice, we reconcile our model with the canonical framework of Cocco, Gomes, and Maenhout (2005). We first estimate a version of our model that closely mirrors their specification by restricting the investment opportunity set to domestic stocks and a hypothetical real risk-free asset, replacing the prevailing Social Security benefit formula with a stylized replacement-rate rule, and increasing the coefficient of relative risk aversion from our baseline value of $\gamma = 3.84$ to $\gamma = 10$. Under these assumptions, our model closely reproduces the traditional lifecycle pattern of declining equity exposure, with a glide path that exhibits, if anything, slightly lower equity allocations than that of Cocco, Gomes, and Maenhout (2005). We then relax these assumptions one at a time. Each change shifts the optimal allocation

substantially toward our baseline strategy. Expanding the investment opportunity set to include international stocks, replacing the hypothetical real risk-free asset with actual government bonds and bills, restoring the prevailing Social Security benefit formula, and lowering risk aversion all move the optimal portfolio markedly closer to an all-equity allocation. The assumptions driving these results largely reflect realistic features of the investment environment and retirement institutions: investors have access to international equity markets, fixed income assets are not risk free, and Social Security benefits are determined by established formulas.

Our conclusions prove robust across diverse model specifications. We consider the simpler optimization problem of choosing fixed weights throughout the lifetime. The optimal fixed-weight policy of 34% domestic stocks, 66% international stocks, 0% bonds, and 0% bills closely mirrors the optimal age-based policy and achieves virtually identical welfare, with an equivalent annual fee of 0.0008% and a consumption CEL of 0.0007%. Our results are also robust to the sample period (i.e., restricting to post-World War II), bootstrap design, risk aversion, bequest motive, retirement timing, household type (e.g., single versus couple), and household income profile. Our findings are not driven by data from the US, small countries, or small stock markets. We also introduce additional background risks, including correlation between labor income and stock returns as well as post-retirement health, mortality, and medical expenditure risk. Across specifications, the optimal allocation between domestic and international equities varies somewhat, but households remain entirely invested in stocks throughout the lifecycle.

We contribute to the recent normative literature on the optimal design of lifecycle investment strategies [e.g., Michaelides and Zhang (2017, 2022); Dahlquist, Setty, and Vestman (2018); Kraft, Munk, and Weiss (2019); Gomes, Michaelides, and Zhang (2022); and Duarte, Fonseca, Goodman, and Parker (2024)] through our approach to modeling the investment opportunity set. Several classic studies assume constant investment opportunities with returns that are normally or lognormally

distributed [e.g., Merton (1969); Viceira (2001); Cocco, Gomes, and Maenhout (2005); and Gomes and Michaelides (2005)]. Many other studies consider particular aspects of dynamic investment opportunities or non-normalities in returns, finding that optimal lifecycle allocations are affected by time-varying expected returns, time-varying return variance, and skewness in returns.⁷ Our block bootstrap instead preserves empirically relevant features of investment opportunities without requiring parametric model specification or ex ante assumptions about which return characteristics drive optimal portfolio choice.

Although studies with time-varying investment opportunities provide important insights into optimal intertemporal hedging demands, Cochrane’s (2014, 2022) alternative perspective for long-term investors focuses on asset payoffs rather than the modeling of state-dependent investment opportunities. Cochrane (2014) states, “the hedging demands emphasized by the portfolio approach are really means to an end—an optimal consumption stream—rather than the end itself.” Our lifecycle optimization approach embodies this perspective by focusing on the long-horizon asset payoffs that support retirement consumption and bequest. With our method, the expected utility for a given strategy reflects complex return dynamics that are implicit in holding-period returns without an explicit model of these complexities. Investors who approach the lifecycle problem with a focus on long-term payoffs optimally choose internationally diversified equity portfolios throughout their lifetimes.

⁷See, e.g., Campbell and Viceira (1999); Barberis (2000); Lynch (2001); Wachter (2002); Campbell, Chan, and Viceira (2003); Pástor and Stambaugh (2012); Michaelides and Zhang (2017, 2022); and Gomes, Michaelides, and Zhang (2022) for evidence on time-varying expected returns; Lynch and Balduzzi (2000) and Chacko and Viceira (2005) for time-varying variance; and Fagereng, Gottlieb, and Guiso (2017); Catherine (2022); Shen (2024); and Bonaparte, Korniotis, Kumar, Michaelides, and Zhang (2025) for skewness.

2 The status quo in lifecycle investing

In seminal studies, Merton (1969) and Samuelson (1969) provide a baseline for lifecycle asset allocation. They demonstrate that investors have constant optimal allocations to a risky asset and a risk-free asset under the conditions that human capital is tradeable and investment opportunities are constant. Subsequent studies addressing the investor’s lifecycle problem relax these assumptions.

Human capital, as a dominant asset for many working-age investors, is the focus of much of the literature on lifecycle portfolio choice. With complete markets, investors capitalize labor income and optimal asset allocation is unaffected [Merton (1971)]. With incomplete markets (e.g., non-tradeable, non-insurable labor income and borrowing constraints), in contrast, Cocco, Gomes, and Maenhout (2005) demonstrate that investors optimally choose age-based allocations.⁸ If labor income risk is idiosyncratic, human capital substitutes for the risk-free asset in the optimal portfolio and the young hold more in stocks than the old.⁹ Bodie, Merton, and Samuelson (1992) consider endogenous labor supply and retirement with utility over consumption and leisure, finding that labor supply flexibility increases optimal financial portfolio risk. Because the young have more labor flexibility than the old, the young should hold more in stocks.¹⁰ Reinforcing these human capital effects, mean reversion in stock returns further increases the attractiveness of stocks for young investors with long investment horizons [e.g., Barberis (2000), Wachter (2002), and Siegel (2014)].

Regulation stemming from the PPA builds upon these findings to favor an “investment fund product or model portfolio that applies *generally accepted investment theories* [emphasis added], is

⁸See also Heaton and Lucas (1997), Koo (1998), and Viceira (2001) for treatments of non-insurable labor income risk in infinite-horizon models.

⁹Human capital can substitute for the risky asset with a systematic component in labor income [e.g., Viceira (2001); Campbell and Viceira (2002); Benzoni, Collin-Dufresne, and Goldstein (2007); and Lynch and Tan (2011)], such that the optimal risky allocation decreases in the presence of labor income if the correlation between labor income shocks and stock returns is sufficiently high.

¹⁰Farhi and Panageas (2007); Chai, Horneff, Maurer, and Mitchell (2011); and Hubener, Maurer, and Mitchell (2016), among others, also study the implications for portfolio choice of flexible labor supply and endogenous retirement. Other studies consider the effects of nonemployment on portfolio decisions [e.g., Bremus and Kuzin (2014); Fagereng, Guiso, and Pistaferri (2018); and Bagliano, Fugazza, and Nicodano (2019)].

diversified so as to minimize the risk of large losses and that is designed to provide varying degrees of long-term appreciation and capital preservation through a mix of equity and fixed income exposures based on the participant's age, target retirement date (such as normal retirement age under the plan) or life expectancy" [29 CFR § 2550.404c-5(e)(4)(i)]. TDFs meet these design criteria by following age-based, stock-bond asset allocation policies. They adopt allocations with higher exposures to equities for younger investors and increase exposures to fixed income assets as the investors age. Figure 1 illustrates this design with the advertised unconditional glide path weights in domestic stocks, international stocks, bonds, and bills from a TDF offered by a major investment firm.

TDFs have exploded in popularity since the passage of the PPA, with total assets under management (AUM) reaching \$4.8 trillion at year-end 2025 [Parker, Schoar, and Sun (2023) and Roy (2026)]. The growth reflects widespread adoption of automatic enrollment by plan sponsors, the overwhelming tendency to select TDFs as default funds, and participants' propensity to retain their default elections [e.g., Madrian and Shea (2001) and Mitchell and Utkus (2022)].¹¹ For a typical employee, the default option is a TDF that matches employee age to a target retirement date without consideration of other employee characteristics [Balduzzi and Reuter (2019)]. Across all participants in Vanguard plans, 84% use TDFs and 61% hold their entire account balance in a single TDF [Vanguard (2026)]. In short, age-based TDF strategies now represent the status quo for retirement saving.

¹¹As a default option, TDFs benefit investors who are inattentive, have behavioral biases, or lack financial literacy by offering diversification benefits and automatic reallocations. A large literature shows suboptimal investment decisions related to international diversification [Bekaert, Hoyem, Hu, and Ravina (2017)], asset allocation [Benartzi and Thaler (2001)], contribution levels [Lusardi and Mitchell (2007, 2011) and Goda, Levy, Manchester, Sojourner, and Tasoff (2020)], stock market participation [van Rooij, Lusardi, and Alessie (2011)], and account concentration in employer stock [Poterba (2003)]. Campbell (2016); Beshears, Choi, Laibson, and Madrian (2018); and Gomes, Haliassos, and Ramadorai (2021) provide comprehensive reviews of this evidence.

3 Data

In this section, we describe the data on asset class returns used in our analyses. We take the perspective of a US couple saving for retirement. The investment horizons of retirement savers are long, extending 75 years or more for young households. The relatively short history of US financial markets poses a small sample problem given this setting, so we model forward-looking returns using the history of asset class returns from a broad cross section of developed economies. Investors in our model use the historical track record of asset class performance in these countries to form beliefs about the *forward-looking* joint distribution of returns. Accordingly, we abstract from the frictions that *historically* constrained investment in diversified asset class portfolios (e.g., the absence of low-cost diversification vehicles, high trading costs, capital controls, and restrictions on the repatriation of foreign investments). A modern US investor saving through an employer-sponsored DC plan would typically have access to low-cost index funds covering major asset classes. The small fee differentials across the index products on a typical DC plan menu would have a negligible effect on optimal asset allocation.

We follow Anarkulova, Cederburg, and O'Doherty (2022) in classifying countries as developed. The dataset includes monthly real returns for domestic stocks, international stocks, government bonds, and government bills for 39 developed countries. The data cover the period from 1890 to 2023, but the sample periods for individual countries vary based on data availability and the timing of economic development (i.e., a given country is included in the sample only for the period after it achieves developed status).

Our primary data source is the GFDatabase from Global Financial Data (GFD). For each sample country, the database contains time series of total return indexes, price indexes, dividend-price ratios, and total market capitalization for stocks; yields for ten-year government bonds and short-term bills; consumer price indexes; and exchange rates. The internet appendix provides detailed descriptions

of the appropriate GFD data series for each country, alternative sources used to fill gaps in the GFD database, calculations of asset class returns, adjustments to these calculations for several periods surrounding major market disruptions (e.g., the closure of the New York Stock Exchange in 1914 at the onset of World War I and the Greek government bond default in 2012), and dataset validation.

The dataset is a balanced panel in the sense that each country-month has non-missing returns for domestic stocks, international stocks, bonds, and bills. The nominal returns for domestic stocks, bonds, and bills for a given country are measured in the local currency; the nominal returns are then converted to real returns using the local inflation rate. The nominal international stock returns for a given country are market-capitalization-weighted averages of the nominal returns for all non-domestic stock markets, with appropriate adjustments for changes in exchange rates. Analogous to the calculations for the other asset classes, the nominal international stock returns are converted to real returns based on local inflation.¹² As such, all asset class returns for a given country-month reflect the real investment outcomes of local investors in that month.

The broad sample provides a more comprehensive characterization of potential investment outcomes than samples based on single countries (e.g., the US or the UK). Although single-country samples are commonly used to calibrate inputs for investment simulations, such samples contain few independent observations of long-horizon investment outcomes. Moreover, these samples suffer from both survivor bias [e.g., Brown, Goetzmann, and Ross (1995)] and easy data bias [e.g., Dimson, Marsh, and Staunton (2002)]. These biases can contribute to long historical periods over which realized performance exceeds ex ante expectations [e.g., Fama and French (2002) and Avdis and Wachter (2017)].¹³

¹²Anarkulova, Cederburg, and O'Doherty (2025) show that the value-weighted average of foreign stock market real returns and changes in real exchange rates each contribute about half of the overall volatility of the monthly international stock returns in our dataset.

¹³The dataset construction methods are designed to mitigate survivor bias and easy data bias. The sample inclusion dates for individual countries are based on ex ante measures of economic activity (e.g., the proportion of a country's labor force employed in the manufacturing and services sectors and the country's membership in global policy organizations like the Organisation for Economic Co-operation and Development), and we take significant steps to construct continuous

In Table II, we list each sample country and the corresponding data coverage. Five countries—Denmark, France, Germany, the UK, and the US—are included in the sample over the full 1890 to 2023 period. The sample periods for the other countries are shorter owing to data availability and development classification status. Our data cover 91% of the potential country-months in the developed country sample. Table II also presents the geometric average real return and the standard deviation of real return for each combination of country and asset class. For the pooled sample of all 31,801 country-month observations, the geometric average return is 0.37% for domestic stocks, 0.44% for international stocks, 0.04% for bonds, and -0.03% for bills (untabulated). Based on comparisons with the pooled sample, the average real returns in the US sample are higher for domestic stocks, bonds, and bills and lower for international stocks. But the US is not an extreme outlier relative to other countries for any of the four asset classes.¹⁴

4 Methods

In this section, we present the household’s lifecycle problem and our solution method. We first introduce the lifecycle structure and the household’s utility optimization problem (Section 4.1). We then specify the stochastic process for labor income and the calculation of Social Security benefits (Section 4.2). Finally, we describe the return environment and the Monte Carlo procedure used to optimize and evaluate policies (Section 4.3).

4.1 Household utility and portfolio choice problem

Households in our base specification are composed of a female (f) and a male (m) of equal age. The model periods are in months indexed by $t = 1, 2, \dots, T_{max}$, where T_{max} is the month of death of the monthly data series for each country.

¹⁴The standard deviation of bond returns for Germany is extreme relative to the estimates for other sample countries. This volatility is attributable to the well-known period of hyperinflation in Germany in the 1920s. Excluding all German data from our model has no material impact on the optimal asset allocation policies described in Section 5.

last remaining survivor from the couple. Each member of the household is eligible to work starting from the first month of age 25 ($t = 1$), and the household splits income between consumption and saving from that date until retirement. The retirement date is denoted T_{ret} , and our base case specifies an exogenous retirement age of 65, such that $T_{ret} = 480$. An individual may experience nonemployment during the 40-year potential working period. At time $T_{ret} + 1$, each individual leaves the workforce and begins to draw from retirement savings and Social Security.

The Social Security Administration (SSA) reports conditional death probabilities at each age for females and males.¹⁵ Our simulations incorporate gender-specific longevity risk, and the lifespan of each individual is randomly determined. Both the female and the male in each couple are alive at age 25, but one or both may die before retirement at age 65. The household's size H_t equals two while both members are alive and one after the first death. There is considerable uncertainty over longevity outcomes. The 5th percentile of age at death for the couple (i.e., the last survivor) is 70.8 years, and the 95th percentile is 100.0 years (see the internet appendix for further details on the distribution of age at death).

Household utility is determined by monthly consumption and a bequest:

$$U(C, B) = \sum_{t=1}^{T_{max}} \delta^t \frac{(C_t / \sqrt{H_t})^{1-\gamma}}{1-\gamma} + \delta^{T_{max}} \theta \frac{(B+k)^{1-\gamma}}{1-\gamma}, \quad (1)$$

where C_t is real consumption in month t , B is the real bequest, δ is the subjective discount factor, θ and k are bequest utility parameters, and γ is the coefficient of relative risk aversion. Following Duarte, Fonseca, Goodman, and Parker (2024), we scale household consumption by the square root of household size to reflect differences in consumption needs for couples versus singles. We set the annual subjective discount factor to 0.98, so the monthly factor is $\delta = 0.98^{1/12}$. Our bequest utility specification follows De Nardi, French, and Jones (2010). We use their estimate for risk aversion of

¹⁵See <https://www.ssa.gov/oact/HistEst/PerLifeTables/2022/PerLifeTables2022.html>.

$\gamma = 3.84$, and we assume that this risk aversion coefficient applies to both consumption and bequest. De Nardi, French, and Jones (2010) estimate a bequest intensity of $\theta = 2,360$ when studying bequest utility alongside utility from annual consumption, and we multiply this parameter estimate by 12^γ to reflect the mechanical difference in scaling between monthly and annual consumption levels. Finally, we inflation-adjust their bequest curvature parameter k , which determines the extent to which bequests are viewed as luxury goods, and use $k = \$490,000$ in 2022 USD.¹⁶

The household jointly chooses three policies: saving during the working period, withdrawals during the retirement period, and portfolio weights throughout the lifecycle. Each choice is governed by a policy rule based on the household's state and applied monthly. During the working period, the household saves a share of its savable income, which is household labor income in excess of a subsistence level of $Y_{min} = \$15,000$ per year (in 2022 USD):

$$\tilde{Y}_\tau = \mathbb{1} \left\{ \max\{Y_\tau^f, Y_\tau^m\} \geq Y_{min} \right\} \left(Y_\tau^f + Y_\tau^m - Y_{min} \right), \quad (2)$$

where Y_τ^i is individual i 's annual labor income in year τ . The indicator function reflects that households save only when at least one member earns Y_{min} or more, consistent with evidence of low retirement saving rates among low-income workers [e.g., Vanguard (2026)]. The saving rate varies with the household's age and its wealth relative to permanent labor income. During the retirement period, the household withdraws a share of its current wealth each month, and the withdrawal rate varies with age and wealth relative to the couple's Social Security benefit. The household allocates wealth across domestic stocks, international stocks, bonds, and bills with weights that depend on age. Portfolio weights are constant within the age windows 25–34, 35–39, ..., 85–89, and 90 and older. This design mitigates the effects of simulation error on the estimated optimal weights.

¹⁶We do not consider housing, which is an important asset for many households. Venti and Wise (1991) and Poterba, Venti, and Wise (2011) show that few households use reverse mortgages or otherwise decrease their home equity late in life.

The two wider windows correspond to the youngest ages (when accumulated retirement wealth is minimal) and the oldest ages (when relatively few simulated households remain alive), because the optimal portfolio weights are estimated least precisely in these age ranges.

We parameterize the saving rate and the withdrawal rate with logistic functions, so monthly saving and withdrawals are

$$S_t = \frac{1}{12} \bar{Y}_\tau \Lambda(b_0 + b_1 a_t^w + b_2 \widehat{W}_t) \quad \text{for } 12(\tau - 1) < t \leq 12\tau, \quad (3)$$

$$D_t = \frac{1}{12} \Lambda(c_0 + c_1 a_t^r + c_2 \widehat{W}_t^{SS}) W_{t-1}, \quad (4)$$

where $\Lambda(x) = 1/(1 + e^{-x})$ is the logistic function, W_t is the household's end-of-month invested wealth (such that W_{t-1} is wealth on hand at the beginning of month t), a_t^w is working household age (normalized to range from zero to one over ages 25 to 65), $\widehat{W}_t = W_{t-1}/\bar{Y}_\tau$ is beginning-of-month wealth relative to the household's annualized permanent labor income $\bar{Y}_\tau$ (Section 4.2), a_t^r is retirement household age (normalized to range from zero to one over ages 65 to 120), and $\widehat{W}_t^{SS} = W_{t-1}/(12 SS_t)$ is beginning-of-month wealth relative to the household's annualized Social Security benefit, where SS_t is the couple's combined monthly benefit. The logistic form bounds both rates to be positive and ensures saving never exceeds savable income and withdrawals never exceed wealth, and the coefficients are unrestricted in sign.

The household chooses the saving parameters $b = (b_0, b_1, b_2)$, the withdrawal parameters $c = (c_0, c_1, c_2)$, and the age-window portfolio weights $\{w_j\}_{j=1}^{13}$ for the four asset classes to maximize

expected lifetime utility from equation (1):

$$\max_{\{b, c, \{w_j\}_{j=1}^{13}\}} \mathbb{E}_0[U(C, B)] \quad (5)$$

$$\text{s.t. } R_t^p = w'_{j(t)} R_t, \quad (6)$$

$$\mathbf{1}' w_j = 1 \quad \text{for } j = 1, \dots, 13, \quad (7)$$

$$w_j \geq 0 \quad \text{for } j = 1, \dots, 13, \quad (8)$$

$$W_0 = 0, \quad (9)$$

$$W_t = (W_{t-1} - D_t) R_t^p + S_t, \quad (10)$$

$$C_t = \begin{cases} \max\{\frac{1}{12}(Y_\tau^f + Y_\tau^m) - S_t, SSI_t\} & \text{for } t \leq T_{ret}, \\ \max\{D_t + SS_t, SSI_t\} & \text{for } t > T_{ret}, \end{cases} \quad (11)$$

$$B = W_{T_{max}}, \quad (12)$$

where w_j is a 4×1 vector of portfolio weights for age window j , $j(t)$ denotes the window containing month t , R_t is a 4×1 vector of gross real monthly returns on the four asset classes (Section 4.3), R_t^p is the gross portfolio return, $\mathbf{1}$ is a 4×1 vector of ones, saving S_t follows equation (3) during the working period with $S_t = 0$ thereafter, withdrawals D_t follow equation (4) during the retirement period with $D_t = 0$ beforehand, SSI_t is a monthly consumption floor equal to the Supplemental Security Income (SSI) guarantee for the household's size, and τ denotes the year containing month t (i.e., $12(\tau - 1) < t \leq 12\tau$). Equations (7) and (8) represent the constraints faced by the couple on taking levered positions and short positions in risky assets. In the internet appendix, we relax the restriction on portfolio leverage.

4.2 Lifecycle income

We model labor income using the flexible framework of Guvenen, Karahan, Ozkan, and Song (2021), which allows for worker heterogeneity, persistent and transitory income shocks, and employment and nonemployment states. They estimate the model to fit a large number of cross-sectional moments and time-series properties of lifecycle earnings data on millions of US workers from the SSA.

The annual income level for individual i ($i \in \{f, m\}$) at age $\tau + 24$ (for $\tau = 1, 2, \dots, 40$) is given by

$$Y_\tau^i = (1 - v_\tau^i) e^{(g(\tau) + \alpha^i + \beta^i f(\tau) + z_\tau^i + \varepsilon_\tau^i)}, \quad (13)$$

where $g(\tau)$ is a quadratic polynomial that fits the well-known hump shape of lifecycle earnings, $f(\tau)$ is a linear function increasing in τ , α^i and β^i are individual-specific parameters that affect the expected level and slope of earnings, z_τ^i is a persistent earnings component following

$$z_\tau^i = \rho z_{\tau-1}^i + \eta_\tau^i, \quad (14)$$

and ε_τ^i is a transitory earnings shock. Both the persistent shocks (η_τ^i) and the transitory shocks (ε_τ^i) follow normal mixture distributions. Finally, $v_\tau^i = 0$ represents full-year employment, whereas $v_\tau^i = 1$ is full-year nonemployment. This nonemployment variable takes values as follows:

$$v_\tau^i = \begin{cases} 0 & \text{with prob. } 1 - p_v(\tau, z_\tau^i), \\ \min\{1, \phi_\tau\} & \text{with prob. } p_v(\tau, z_\tau^i), \end{cases} \quad (15)$$

where ϕ_τ follows an exponential distribution with mean $1/\lambda$, $p_v(\tau, z_\tau^i) = \frac{e^{\xi_\tau^i}}{1 + e^{\xi_\tau^i}}$ is the nonemployment probability, and $\xi_\tau^i = a_v + b_v f(\tau) + c_v z_\tau^i + d_v z_\tau^i f(\tau)$ with $b_v < 0$, $c_v < 0$, and $d_v < 0$. Thus, the

probability of nonemployment decreases with the persistent earnings component, generating persistence in the nonemployment state.

We assume annual income is divided evenly over months in the year. The permanent labor income that enters the saving rule in equation (3) is the systematic component of equation (13),

$$\bar{Y}_\tau^i = e^{(g(\tau) + \alpha^i + \beta^i f(\tau) + z_\tau^i)}, \quad (16)$$

which excludes the transitory shock and the nonemployment state. Household permanent income $\bar{Y}_\tau$ is the sum of $\bar{Y}_\tau^i$ across living members.

The heterogeneity in earnings processes across individuals is captured by two income parameters, α^i and β^i , and the initial state of the persistent income component, z_0^i . High (low) values for α^i and β^i designate types with high (low) levels and growth rates, respectively, for expected lifetime earnings, whereas high (low) z_0^i captures a tendency for high (low) early-career earnings. In our base case analyses, we set all three parameters for both members of the couple equal to their median values in the Guvenen, Karahan, Ozkan, and Song (2021) calibration, i.e., $(\alpha^i, \beta^i, z_0^i) = (0, 0, 0)$.

We simulate from the labor income model using the parameter estimates from the replication code of Guvenen, Karahan, Ozkan, and Song (2021) with the additional assumption that the income model applies equally to females and males. We scale the simulation output (which initially has no standard unit of measurement) to match the level of average log earnings in 2010 USD [Figure C.36 in Guvenen, Karahan, Ozkan, and Song (2021)] and then convert to 2022 USD by adjusting for the change in the consumer price index (CPI). Panel A of Figure 2 plots the distribution of household income from our base case simulation as a function of household age. The median exhibits the well-known hump shape of lifecycle earnings [e.g., Cocco, Gomes, and Maenhout (2005)], while the 25th and 75th percentiles indicate substantial earnings uncertainty.

Our simulations incorporate retirement income from Social Security benefits. Benefits are calculated based on earnings during the working years, and we use the formulas effective in 2022, incorporating spousal and survivor benefits in the retirement period. In the internet appendix, we provide full details of the Social Security benefit calculations. Finally, the consumption floor in equation (11) reflects Supplemental Security Income, whose maximum monthly benefit in 2022 is \$1,261 for couples and \$841 for singles.

4.3 Return environment and Monte Carlo optimization

We adopt a stationary block bootstrap approach in the spirit of Politis and Romano (1994) to draw time series of monthly real returns for the four asset classes. For each draw, we randomly select a starting country-month observation in the full sample and draw a block of consecutive returns from the same country to capture time-series dependencies in asset returns. The block length is drawn from a geometric distribution with an average length of 120 months in the base case, so the blocks reflect long-term time-series properties of returns. We jointly draw returns for all four asset classes from each selected country-month to preserve cross-sectional dependencies across assets, and we denote a monthly real return vector as

$$R_t = [R_t^{Domestic\ stocks} \quad R_t^{International\ stocks} \quad R_t^{Bonds} \quad R_t^{Bills}]'. \quad (17)$$

In the event that the asset return data from a country-period are insufficient to fill a return block, we draw a random country and continue to fill the return block with return data from the beginning of that country's sample. We repeatedly draw blocks of returns from random countries and periods until we produce a time series of asset class returns that spans the couple's lifetime. This nonparametric approach preserves the time-series and cross-sectional properties of historical returns, including return predictability, volatility dynamics, and tail risk, without imposing a

parametric return model.¹⁷

For a given policy $(b, c, \{w_j\}_{j=1}^{13})$, we simulate 1,000,000 lifecycle outcomes, with the following steps in each draw:

1. We determine the lifespan of the household. We generate random longevity using conditional mortality probabilities, and we assume that the probability of death is equal across the 12 months at a given age. In the base case, we denote the realized lifespans of the female and the male as T_f and T_m and the couple's lifespan as $T_{max} = \max\{T_f, T_m\}$.
2. We simulate labor income for each individual from the model in Section 4.2 and calculate the couple's Social Security benefits from their realized earnings histories.
3. We draw a time series of asset class returns spanning the couple's lifetime using the block bootstrap, and we compute the monthly portfolio return $R_t^p = w'_{j(t)} R_t$.
4. The couple begins with no wealth, $W_0 = 0$. During the working period, monthly saving follows the saving rule in equation (3), consumption follows equation (11), and wealth evolves according to equation (10).
5. During the retirement period, monthly withdrawals follow the withdrawal rule in equation (4), consumption follows equation (11), and wealth evolves according to equation (10). The couple's bequest is all remaining wealth at the death of the last surviving spouse.
6. The household's time series of monthly consumption and its bequest determine utility according to equation (1).

¹⁷The academic literature proposes dozens of stock market return predictors [e.g., Goyal, Welch, and Zafirov (2024)] that capture distinct expected return patterns at short and long horizons [e.g., Cederburg, Johnson, and O'Doherty (2023)], a host of methods for forecasting volatility [e.g., Engle (1982) and Bollerslev (1986)], and many approaches to estimating time-varying tail risk [e.g., Kelly and Jiang (2014)]. The bootstrap allows these return dynamics to influence optimal policies without requiring that they be explicitly modeled.

The average household utility across draws is the Monte Carlo estimate of the expected utility of the policy. We hold the simulated draws fixed across candidate policies, so that differences in estimated expected utility across policies are paired comparisons that difference out common simulation noise. For each policy, we compile statistics for variables of interest, such as wealth at retirement and retirement-period drawdown.¹⁸

Because the bootstrap imposes no parametric conditional return model, the household's problem admits no analytic conditional expectations from which to build a Bellman recursion, so we instead jointly optimize the policy parameters directly by simulation. Our procedure has three stages. First, we implement a coarse global search jointly over the saving parameters, withdrawal parameters, and portfolio weights with the Nelder–Mead simplex method from a large set of starting points using 250,000 lifecycle draws per evaluation. Second, we refine the best solutions with a pattern search jointly over all policy parameters, using 1,000,000 lifecycle draws matched to the final evaluation set. Third, we verify the solution on a one-percentage-point portfolio-weight grid. Starting from the refined solution, we evaluate all neighboring weight vectors on the 1,000,000 lifecycle draws, moving to the best neighbor that improves expected utility and re-optimizing the saving and withdrawal parameters after each move, until no neighboring weight vector offers an improvement.

We make welfare comparisons across strategies using two measures based on expected lifetime utility from consumption and bequest. The first is an equivalent annual fee. Given the expected lifetime utility from an alternative asset allocation strategy (e.g., the TDF), we find the constant annual fee, charged as a percentage of assets under the optimal age-based strategy, that yields the same expected lifetime utility. Households re-optimize their saving and withdrawal policies under both the alternative strategy and the fee-paying optimal strategy, so the fee captures the welfare

¹⁸Maximum portfolio drawdowns are calculated as the largest real negative cumulative return relative to the previous peak. The working-period drawdown occurs entirely within the working years. The retirement-period drawdown begins with a peak that could occur during either the working years or the retirement years. That is, falling asset prices in the late working years can contribute to our measured retirement drawdown.

cost of the asset allocation strategy itself rather than a suboptimal saving or withdrawal response to it. The second measure is the consumption CEL, which is the constant percentage reduction in consumption and bequest under the optimal age-based strategy that yields the same expected lifetime utility as the alternative strategy.

5 Results

In this section, we characterize the optimal lifecycle investment policies and evaluate their economic implications. We begin by presenting the optimal age-based and fixed-weight strategies and comparing their welfare with that of four benchmark strategies (Section 5.1). We then examine lifecycle investment outcomes under the optimal and alternative investment strategies (Section 5.2). Next, we reconcile our lifecycle investment policies with those implied by traditional lifecycle portfolio choice models (Section 5.3). We conclude by examining how optimal portfolio choice is affected by additional background risks, conditioning on the market state, investor beliefs, and a broad range of alternative model specifications (Sections 5.4–5.7).

5.1 Optimal lifecycle investment strategies

Panel A of Table III summarizes the portfolio weights of the optimal age-based strategy, and Figure 3 illustrates the sequence of these weights from age 25 to 95. Our key finding is that the optimal age-based policy eschews fixed income investments in favor of an all-equity portfolio at all ages, with an average split of 34% domestic stocks and 66% international stocks. This result may appear counterintuitive, given the vaunted diversification benefits and safety offered by bonds. As discussed in the context of Table I, however, bonds become riskier and more correlated with domestic stocks as the investment horizon lengthens. The optimal allocation reflects the superior diversification benefits and growth potential offered by international stocks.

We compare the optimal age-based strategy against four benchmark strategies with portfolio weights reported in Panel A of Table III: a 100% allocation to bills, a 100% allocation to domestic stocks, a balanced strategy with a 60% allocation to domestic stocks and a 40% allocation to bonds, and the TDF shown in Figure 1. The bills strategy was a common default in the pre-PPA era, and the balanced and TDF strategies are QDIAs under the PPA. Panel A also reports the optimal strategy among those that maintain fixed weights throughout the lifecycle. This strategy allocates 34% to domestic stocks and 66% to international stocks at all ages, matching the average split of the optimal age-based strategy. We include this strategy to measure the economic value of varying the portfolio weights with age.

Panels B and C of Table III show the estimated saving rate and withdrawal rate policy coefficients for each strategy. Under the optimal age-based strategy's saving rule, the saving rate increases with age and decreases with wealth, all else equal. Panel B of Figure 2 shows the resulting lifecycle profile of the saving rate. The median saving rate is 3.2% of income at age 25, rises to a peak of 10.5% around age 40, and falls back toward zero in the final working years as declining labor income (Panel A) combines with accumulated wealth to reduce the desire for further saving. The estimated withdrawal rule for the optimal strategy implies that the withdrawal rate increases with age and is largely insensitive to wealth. Panel C of Figure 2 shows the corresponding lifecycle profile: the median withdrawal rate rises from 5.4% at age 65 to 19.5% at age 95, so couples consume their wealth at an increasing rate as the expected remaining lifetime shortens. Panel D of Figure 2 plots the lifecycle profile of household wealth under the optimal strategy. Median wealth among surviving households rises from \$0.09 million at age 40 to \$0.69 million at retirement and declines to \$0.14 million by age 95.

Although the optimal age-based strategy provides higher expected utility than the benchmark strategies by construction, Panel D of Table III reveals that the welfare gains are economically large.

We report the equivalent annual fee and the consumption CEL for each strategy. The benchmark strategies perform poorly. The equivalent annual fee is 5.79% for bills, 1.22% for domestic stocks, and 2.09% for the balanced strategy. The status quo TDF adopts an age-based strategy in an attempt to optimize over the lifecycle, but it substantially underperforms the optimal strategy: an annual fee of 1.90% of assets under the optimal age-based strategy would reduce its expected lifetime utility to that of the TDF. The consumption CELs convey the same message. A household adopting the TDF fares as well as one following the optimal age-based strategy with consumption and bequest reduced by 1.56%, and the consumption CELs are 5.02% for bills, 0.97% for domestic stocks, and 1.73% for the balanced strategy. Overall, these findings imply that a household currently heeding conventional investment advice can achieve a large increase in utility by reallocating to an internationally diversified equity portfolio.¹⁹

The optimal fixed-weight and age-based strategies deliver effectively identical economic outcomes. Beyond holding the same average portfolio, the fixed-weight household adopts saving rate and withdrawal rate policies that are nearly indistinguishable from those under the age-based strategy. The welfare cost of fixing the weights is negligible, with an equivalent annual fee of 0.0008% and a consumption CEL of 0.0007%. The gains from optimal investing therefore arise from holding an all-equity, internationally diversified portfolio rather than from varying portfolio weights with age.²⁰

Given the similarity in economic outcomes across the optimal age-based and fixed-weight strategies, the fixed-weight design provides a convenient setting for quantifying utility losses from

¹⁹As is traditional in the literature [e.g., Campbell, Cocco, Gomes, and Maenhout (2001)], we take a partial equilibrium view of lifecycle investing and do not consider equilibrium effects of investor shifts across investment strategies. As Heaton and Lucas (2000) state, “Although one could look at portfolio choice in a general equilibrium framework, unless the model were capable of generating a realistic returns process the results on portfolio choice would also be suspect.”

²⁰In the internet appendix, we also consider an asset allocation policy that conditions on household wealth in addition to age. We find that the optimal policy remains fully invested in equities at every age and wealth level, the optimal domestic-international equity split varies only modestly across wealth levels, and the welfare gain from conditioning on wealth is small.

deviating from the optimal allocation. As shown in the internet appendix, expected lifetime utility as a function of the allocation to domestic stocks is relatively flat around the optimum of 34% domestic stocks and 66% international stocks. All allocations ranging from 15% domestic and 85% international to 53% domestic and 47% international have equivalent annual fees below 0.10% (relative to the optimal fixed-weight strategy). This finding gives real-world investors considerable latitude in choosing equity strategies. For example, a US investor may feel comfortable investing over half of their wealth in the domestic market given the US's large global weight. As long as investors remain invested in stocks and avoid overly large domestic equity allocations, the utility costs of deviating from the optimum are small. The utility losses incurred by deviating from the optimal portfolio by adding fixed income, in contrast, are substantially greater.

5.2 Lifecycle investment strategy performance

In this section, we examine simulation results for couples who adopt the optimal age-based investment strategy or one of the four benchmark strategies from Table III. For each strategy, we study six retirement saving outcomes: (i) the distribution of the saving rate during the working years, (ii) the distribution of wealth at retirement, (iii) the distribution of the withdrawal rate during retirement, (iv) the distribution of the income replacement rate, (v) the probability that Social Security income accounts for more than 75% of retirement income, and (vi) the distribution of wealth at death. We also consider two intermediate outcomes: (i) the distribution of the maximum drawdown during the working years and (ii) the distribution of the maximum drawdown during the retirement years. In the internet appendix, we demonstrate that the optimal age-based and fixed-weight strategies provide nearly identical performance according to the metrics covered in this section.

5.2.1 Pre-retirement period

Panel A of Figure 4 illustrates the distribution of the total dollar-weighted saving rate for the optimal age-based strategy and the four benchmark strategies across 1,000,000 bootstrap simulations. A household's saving rate is total dollars saved during the working years divided by total gross labor income over the same period. In each box-and-whiskers plot in Figure 4, the middle line is the median, the box designates the interquartile range, and the whiskers extend to the 10th and 90th percentiles. Couples following the optimal age-based strategy save the least: the average saving rate is 8.1% under the optimal strategy versus 13.4% for bills, 8.8% for domestic stocks, 10.4% for the balanced strategy, and 10.1% for the TDF. Saving rates under the optimal strategy are lower than those under the four benchmark strategies at every percentile of the distribution, with the exception of the extreme left tail for domestic stocks (i.e., the 1st and 2nd percentiles).

Panel A of Table IV summarizes the distributions of retirement wealth. For each strategy, the table reports the mean, standard deviation, and percentiles of the wealth distribution at the beginning of the retirement period across the bootstrap simulations; Panel B of Figure 4 displays the corresponding distributions graphically. The exhibits report wealth at retirement in millions of 2022 USD. About 2.3% of households have both members die prior to retirement age. For these couples, bequests occur prior to retirement and wealth at retirement is \$0.

The benchmark strategies differ substantially in the retirement wealth they generate. The bills strategy performs poorly in wealth accumulation, with an average retirement wealth balance of only \$0.38 million despite the highest saving rates. The poor performance demonstrates that money market and stable value funds are ineffective retirement saving tools when used in isolation and provides support for the changes to default DC plan options brought about by the PPA. Investing in domestic stocks produces \$0.99 million in wealth on average, which is higher than the averages for the two QDIAs of \$0.80 million for the balanced portfolio and \$0.86 million for the TDF. The

optimal age-based strategy also generates \$0.99 million in average retirement wealth, even though households following this strategy save less than those using any of the benchmark strategies. The differences in retirement wealth therefore understate the differences in investment performance because households following the benchmark strategies partially compensate for lower portfolio growth by saving more.

More important, the optimal strategy provides substantially better downside protection. The 5th percentile of retirement wealth is \$0.13 million under the optimal strategy, compared with \$0.04 million for bills, \$0.08 million for domestic stocks, \$0.08 million for the balanced strategy, and \$0.10 million for the TDF. Thus, international diversification improves the left tail of the retirement wealth distribution without sacrificing average retirement wealth.

5.2.2 Retirement period and bequest

Panel C of Figure 4 shows the distribution of the withdrawal rate during retirement. For each household, the withdrawal rate is the average annualized share of wealth withdrawn across the retirement years. The distributions of withdrawal rates are similar across strategies, with average rates between 9.1% and 9.3% for every strategy except bills. Couples investing in bills withdraw at a higher average rate of 10.9%, supporting retirement consumption by drawing down a larger share of a smaller balance.

Panel B of Table IV and Panel D of Figure 4 summarize the distribution of the income replacement rate during retirement for each strategy. The replacement rate is the ratio of average monthly consumption during retirement to average monthly household income during the working ages of 25 to 64. Couples who are deceased before retirement have a replacement rate of zero. The two QDIA strategies (along with Social Security benefits) allow couples to exceed full income replacement, on average. The mean replacement rate is 1.14 for the TDF, for example, with a

median of 0.99. The optimal strategy generates the highest average replacement rate of 1.59, and 72% of couples achieve full income replacement or better. Replacement rates under the optimal strategy exceed those under the two QDIA strategies at every percentile of the distribution, and they exceed those for all four benchmark strategies except in the far right tail, where domestic stocks generate slightly higher replacement rates. Because withdrawal rates are similar across strategies, the higher replacement rates for the optimal strategy reflect larger retirement balances rather than more aggressive withdrawals.

Preserving wealth in retirement is crucial, and retirees who deplete their savings must rely primarily on their Social Security and SSI benefits. Panel E of Figure 4 plots the probability that Social Security income accounts for more than 75% of retirement income. Couples investing in bills face this outcome with a 22.9% probability. The probabilities for domestic stocks and the two QDIA strategies are far lower, clustering between 4.0% and 5.4%. The fixed income allocations of the QDIA strategies therefore provide no material reduction in this risk relative to a portfolio invested entirely in domestic stocks. The optimal age-based strategy reduces the probability to just 0.7%. Equity offers strong potential for additional investment gains during retirement, and international diversification is crucial for capital preservation.

Panel C of Table IV and Panel F of Figure 4 display distributional statistics for real wealth upon the death of the last survivor in the household. The optimal strategy also generates substantial bequests. The mean wealth at death is \$0.76 million for the optimal strategy versus \$0.07 million for bills, \$0.85 million for domestic stocks, \$0.51 million for the balanced strategy, and \$0.42 million for the TDF. The means are heavily affected by right-tail outcomes. The median couple generates a bequest of \$0.26 million with the optimal strategy, which is larger than for bills (\$0.02 million), domestic stocks (\$0.18 million), the balanced strategy (\$0.13 million), or the TDF (\$0.11 million). The favorable bequests under the optimal strategy reflect both the superior wealth accumulation of

equities and the ability of international diversification to preserve wealth during retirement.²¹

5.2.3 Portfolio drawdowns

The focus on the six retirement saving outcomes summarized in Figure 4 glosses over informative intermediate portfolio performance measures. Investors and regulators may also view short-term losses as an important measure of the suitability of retirement strategies. To study this issue, Table V and Figure 5 report the largest portfolio drawdowns during the working years and retirement years for each strategy. The reported drawdowns are the largest peak-to-trough declines in real asset values for a given strategy, and they are expressed in decimal form in the table.

During the working years, each strategy produces large real drawdowns on average. Panel A of Table V shows average maximum drawdowns of 42% for bills, 67% for domestic stocks, 54% for the balanced strategy, 52% for the TDF, and 55% for the optimal strategy. An average drawdown of 55% under the optimal strategy would cause discomfort for even the most stalwart investors, but each strategy that attempts to provide long-term appreciation is subject to similarly large average intermediate losses. Investors, advisors, and regulators are likely most concerned about the largest potential drawdowns, and the optimal strategy outperforms the alternatives in the right tail of the drawdown distribution. The 95th percentile drawdown of 77% for the optimal strategy is favorable relative to drawdowns of 96% (bills), 96% (domestic stocks), 92% (balanced), and 81% (TDF).

Retirement-period drawdowns are likely an even larger concern for households. Panel B of Table V reports average maximum drawdowns during retirement of 46% for bills, 62% for domestic stocks, 50% for the balanced portfolio, 40% for the TDF, and 48% for the optimal strategy. According to this metric, the TDF performs better than the optimal portfolio. Based on avoiding the largest

²¹In the internet appendix, we further study household retirement outcomes. We show that the favorable performance of the optimal age-based strategy relative to the benchmarks in achieving retirement outcomes (i) is partially attributable to continued wealth generation in retirement to combat longevity risk, (ii) persists even if the domestic stock market has a poor realization during retirement, (iii) is strongest if realized inflation in retirement is high, and (iv) occurs even if the realized correlation between domestic and international stock markets is high.

drawdowns, however, the optimal strategy is the best. The 95th percentile for the optimal portfolio is a 74% drawdown versus an 89% drawdown for the TDF.

The superior retirement outcomes (generating wealth at retirement, producing retirement income, avoiding dependence on Social Security, and providing a bequest) of the optimal age-based strategy therefore come despite the potential for relatively large intermediate drawdowns. Experiencing large drawdowns is painful for investors, and many will have a natural inclination to abandon their strategies when these losses occur. A relevant policy issue is how to manage this behavior, as the optimal strategy relies on maintaining high equity allocations through good times and bad. Another important consideration is the relative emphasis placed on average drawdowns versus the largest potential drawdowns. The TDF provides lower average and median retirement drawdowns than the optimal age-based approach, but it also exposes households to substantially larger worst-case drawdowns when high inflation erodes the real value of its bond holdings.

5.3 Modeling assumptions and the all-equity optimum

Our finding that the optimal age-based strategy remains fully invested in equities throughout the lifecycle may be surprising given conventional wisdom. To reconcile this finding with the literature and identify the assumptions that produce it, we compare our approach with the canonical study of Cocco, Gomes, and Maenhout (2005). Several of their modeling assumptions differ from ours, including (i) an investment opportunity set containing only domestic stocks and a hypothetical real risk-free asset paying 2%; (ii) a coefficient of relative risk aversion of 10, set to “the upper bound for risk aversion considered reasonable by Mehra and Prescott (1985)” (p. 500); and (iii) retirement income from Social Security and other sources that equals 68.2% of ending permanent labor income, calibrated to individuals with a high school education.

We begin by incorporating all three modeling choices into our approach. Panel A of Figure 6

shows the optimal age-based strategy from our model (black line) and the average equity share by age reported in Figure 3(C) of Cocco, Gomes, and Maenhout (2005) (purple dots). Under these three assumptions, our model closely reproduces their age profile of equity allocations. In our model, the household is fully invested in stocks until age 40. The stock allocation falls below half by retirement, then rises slowly thereafter. This alignment holds despite remaining differences in the income process, domestic stock return dynamics, and solution method, suggesting that these differences do not drive the divergence between our all-equity result and their interior solution.

The remaining panels of Figure 6 depart from the Panel A specification by changing one key modeling choice at a time. The Panel A solution (black line) is repeated for reference. Panel B examines the effects of risk aversion and Social Security. Lowering the coefficient of relative risk aversion to our base value of 3.84 delays and reduces allocations to the risk-free asset, but preserves the lifecycle pattern. Replacing the stylized retirement income replacement-ratio specification with the actual Social Security formulas also has a large effect, and the allocation to stocks is 68% or higher at all ages. Benefits under the actual Social Security formulas depend on average lifetime earnings and are progressive, so retirement income is far less sensitive to permanent income shocks than under the replacement-ratio specification. Social Security therefore functions as a large, nearly risk-free endowment in our model, as it does in practice.

Panel C examines sensitivity to the assumed real risk-free rate. The assumed 2% real rate is high relative to the average real returns in our historical sample, which are about 1% for bonds and 0% for bills. Reducing the assumed real rate materially increases the optimal allocation to equity, but the household still prefers to invest up to 31% in the safe asset when the real rate of return is 0%.

Panel D adjusts the investment opportunity set relative to the Cocco, Gomes, and Maenhout (2005) benchmark specification consisting of domestic stocks and a hypothetical risk-free asset. Adding international stocks nearly eliminates the allocation to the risk-free asset, with the optimal

equity allocation of 86% or higher at all ages. Alternatively, we replace the risk-free asset with actual bonds and bills. Because these assets are risky in real terms, unlike the hypothetical risk-free asset, the household shifts toward domestic stocks while maintaining an equity allocation of at least 87% throughout the lifecycle.

In untabulated results, our model produces an all-equity allocation throughout the lifecycle under many combinations of assumptions. Two cases illustrate. First, holding risk aversion at $\gamma = 10$ and retaining Cocco, Gomes, and Maenhout's (2005) retirement income specification, the optimum is all equity once domestic stocks, international stocks, bonds, and bills are all available. Second, with $\gamma = 10$ and the actual Social Security formulas, a household with access to domestic and international stocks holds 100% equity regardless of the fixed income alternative offered (actual bonds and bills or a hypothetical risk-free asset with a 0%, 1%, or 2% real rate).

5.4 Background risks

Our baseline model incorporates important sources of background risk in the forms of stochastic labor income and uncertain longevity. In this section, we examine whether introducing additional background risks affects the optimal lifecycle investment strategy.

The first extension introduces correlation between innovations to the persistent earnings component in the labor income model and domestic stock returns. Because these innovations occur at the annual frequency, we relate them to compounded annual domestic stock returns generated by the bootstrap simulation. We use a Gaussian copula to induce correlation between annual domestic stock returns and the earnings innovations. We consider correlations ranging from 0.1 to 0.5.²² The

²²The literature diverges on the magnitude of the correlation between income and stock returns. Fama and Schwert (1977) and Cocco, Gomes, and Maenhout (2005), among others, find little relation. Davis and Willen (2000) and Campbell, Cocco, Gomes, and Maenhout (2001) estimate heterogeneous correlations with respect to educational attainment, with highly educated individuals having correlations as high as 0.3 to 0.5. Benzoni, Collin-Dufresne, and Goldstein (2007) model cointegration between dividends and labor income, and their model implies a correlation around 0.5 between stock returns and returns to human capital throughout most of the working years.

internet appendix shows that, when the correlation is 0.1, the Gaussian copula combined with heavy-tailed shock distributions produces earnings growth for employed workers that varies with market returns in close accordance with Catherine’s (2022) countercyclical labor income risk model, along with rates of new nonemployment that are higher in low-return states.

Table VI summarizes the optimal asset allocation policies under the alternative background risk specifications. Panel A reproduces the baseline results, and Panel B reports the optimal policies with correlation between labor income and domestic stock returns. As the correlation increases, the optimal domestic stock allocation declines under both the fixed-weight and age-based strategies. For the age-based strategy, this effect is concentrated in the early working years, when human capital constitutes the largest share of total household wealth. Even a modest correlation of 0.1 eliminates the domestic stock allocation at ages 25–34. By contrast, asset allocation after retirement is largely unaffected by the assumed correlation between labor income and domestic stock returns. Across all specifications in Panel B, households respond to higher labor income–stock return correlations by shifting from domestic to international stocks rather than to bonds or bills.

The second extension introduces health status, health-dependent mortality, and medical expenditures. Beginning at age 70, each living spouse follows the health, mortality, and medical expenditure processes estimated by De Nardi, French, and Jones (2010) for retired singles. Following Duarte, Fonseca, Goodman, and Parker (2024), we apply these processes separately to each spouse and aggregate medical expenditures at the household level. Before age 70, mortality follows the SSA life tables used in the baseline model and medical expenditures are zero.

Health status follows a two-state Markov process with healthy and sick states, where the transition probabilities depend on sex, age, and permanent income. Survival from age 70 depends on sex, age, health status, and permanent income, and individuals die by age 102. Medical expenditures follow a lognormal distribution with persistent and transitory shocks, and the

mean and volatility depend on sex, age, health status, and permanent income. Annual medical expenditures are spread evenly throughout the year and paid from retirement wealth up to the household's available retirement account balance. The internet appendix provides the full specification of the health, mortality, and medical expenditure processes.

Panel C reports the optimal asset allocation policies under this specification. Incorporating health, mortality, and medical expenditure risks affects the household's optimal saving and withdrawal decisions, but these risks have little effect on optimal asset allocation. The optimal fixed-weight strategy is identical to the baseline specification, allocating 34% to domestic stocks and 66% to international stocks. Likewise, the optimal age-based strategy remains fully invested in equities throughout the lifecycle, with domestic stock allocations nearly identical to those in the baseline specification.

5.5 Portfolio choice conditional on the market state

We study whether households benefit from conditioning portfolio allocations on the market state, using the domestic stock market price-dividend ratio as a state variable. We augment the dataset of asset class returns with the lagged domestic price-dividend ratio for each country-month before implementing the bootstrap. Thus, at the beginning of each month, the household observes the current valuation level of the domestic stock market when selecting its portfolio.

We divide market states into quintiles, ranging from low price-dividend ratios (0.00 to 18.76) to high price-dividend ratios (43.67 to infinity). The conditional allocation applies a quintile-specific tilt to the age-based weights,

$$w_{j,pd} = \frac{\max(w_j + \eta_{pd}, 0)}{\mathbf{1}' \max(w_j + \eta_{pd}, 0)}, \quad (18)$$

where w_j is the allocation vector for age window j and η_{pd} is the tilt vector for price-dividend quintile pd . This parameterization preserves the lifecycle profile of the age-based strategy while

allowing portfolio weights to vary systematically across market states, with assets potentially excluded in some states and held in others. We normalize the middle-quintile tilt to zero so that the weight vector equals w_j in this state. We also fix the first element of the remaining tilt vectors to zero for identification, leaving 12 free parameters that are optimized jointly with the age-based weights and the saving and withdrawal rules.

Table VII summarizes the optimal conditional age-based strategy. When domestic valuations are low, the household substantially overweights domestic stocks relative to the baseline age-based strategy. As valuations rise, the domestic stock allocation generally declines at all ages, reaching its lowest level in the highest valuation quintile. Despite these changes within the equity portfolio, the household remains fully invested in equities across the first four valuation quintiles and allocates only a modest 8% to bonds in the highest valuation quintile. Thus, conditioning on the market state primarily affects the composition of the equity portfolio rather than the overall allocation to equities.

5.6 Investor beliefs and American exceptionalism

As described in Section 3, we model forward-looking returns using a broad sample of developed countries. Some contend that the US stock market is “special,” such that the historical US sample alone provides the appropriate basis for modeling future investment opportunities for US investors. One view of American stock market exceptionalism is that high historical returns reflect investors’ past learning about US structural advantages, with these beliefs now reflected in high valuation ratios and lower future expected returns [e.g., van Binsbergen, Hua, Peeters, and Wachter (2025)]. Reliance on the historical US sample requires a different view—common among practitioners—that the US will continue to outperform foreign markets and generate unusually high future returns.²³

We examine how beliefs about American exceptionalism affect optimal portfolio choice. We

²³See, for example, <https://www.advisorperspectives.com/commentaries/2024/11/01/why-stay-overweight-u-s-equities> and <https://www.bloomberg.com/opinion/articles/2025-01-13/us-exceptionalism-doubting-america-can-still-cost-you-a-lot-of-money>.

combine the return distributions implied by the developed-country sample and the historical US sample using weights that reflect investors' beliefs that the US has more favorable long-run return characteristics. Specifically, we simulate $(100 - x)\% \times 1,000,000$ bootstrap return paths using the developed-country sample and $x\% \times 1,000,000$ bootstrap return paths using the historical US sample, where $x\%$ is the investor's subjective probability that the US return distribution better represents future investment opportunities.

Figure 7 shows that optimal age-based asset allocation policies are highly sensitive to these beliefs. Under the baseline specification, the household allocates approximately one-third of its portfolio to domestic stocks throughout the lifecycle. As the perceived likelihood of American exceptionalism increases, the optimal domestic stock allocation rises at all ages. Modest beliefs in American exceptionalism ($x = 25$) produce domestic stock allocations close to current global market weights, whereas complete conviction in American exceptionalism leads to a portfolio invested entirely in US stocks. These results suggest that the substantial home bias documented by French and Poterba (1991) is difficult to reconcile with optimal portfolio choice unless investors hold strong beliefs that the historical US experience is a better guide to future returns than the broader developed-country experience. At every belief level, the optimal portfolio includes no bonds or bills.

5.7 Robustness

Table VIII examines the robustness of the optimal fixed-weight and age-based asset allocation policies under a broad range of alternative design assumptions. Panel A reproduces the baseline results, and Panels B–I consider the following alternative specifications:

- **Alternative samples.** We restrict the return sample to the post-World War II period (October 1945 to December 2023), countries with relatively large populations, countries with large

market capitalization-to-GDP ratios, and countries other than the US (Panel B).²⁴

- **Calculation of asset class returns.** We examine two modifications to the calculation of asset class returns: an equal-weighted rather than value-weighted international stock index and US dollar-denominated real returns rather than local-currency real returns (Panel C).
- **Average block length.** We consider average block lengths of 60 months and 240 months in the stationary block bootstrap (Panel D).
- **Risk aversion.** We vary the coefficient of relative risk aversion from $\gamma = 0.5$ to $\gamma = 10$ (Panel E).²⁵
- **Household utility specification.** We modify the household utility specification by eliminating the bequest motive and scaling household consumption by household size rather than its square root (Panel F).
- **Retirement timing.** We consider fixed retirement ages of 62, 67, and 70, as well as endogenous retirement timing (Panel G).
- **Household type.** We consider single females, single males, female couples, and male couples (Panel H).
- **Investor type.** We examine four investor types differing along two dimensions: low or high human capital $[(\alpha^i, \beta^i)]$ and low or high initial income (z_0^i) , resulting in a broad range of lifetime earnings profiles and total lifetime earnings levels (Panel I).

²⁴The large-population sample includes a country's data only after its population first reaches 0.5% of the world population, and the developed-equity-market sample includes a country's data only after its market capitalization-to-GDP ratio first reaches 0.5. The internet appendix reports the corresponding sample periods for each country.

²⁵Calvet, Campbell, Gomes, and Sodini (2025) estimate risk aversion in the cross section of Swedish households. The risk aversion parameter of 7.5 is identical to their median estimate, and their standard deviation of 0.97 implies that the parameters of 5 and 10 are roughly two standard deviations from the median. Therefore, the large majority of households lie in this range according to Calvet, Campbell, Gomes, and Sodini's (2025) estimates.

Across all specifications, our central conclusion is unchanged. The optimal fixed-weight strategy remains fully invested in equities with no allocation to bonds or bills, and the optimal age-based strategy maintains a 100% equity allocation throughout the lifecycle. Thus, the preference for internationally diversified equity portfolios is robust to changes in the return sample, return calculations, bootstrap specification, household preferences, retirement assumptions, household composition, and lifetime earnings characteristics. None of the alternative specifications generates any demand for fixed income assets.²⁶ The internet appendix also examines optimal portfolio choice when households are permitted to borrow and invest with leverage [e.g., Asness (1996)]. Even under this extension, the optimal portfolio remains almost entirely invested in domestic and international stocks.

Several alternative specifications produce noticeable departures from the baseline domestic-international allocation while remaining fully invested in equities. The most prominent departure corresponds to the equal-weighted international stock index (Panel C). Under this specification, the optimal fixed-weight strategy allocates just 10% to domestic stocks, and the optimal age-based strategy places relatively little weight on domestic stocks throughout the lifecycle. The equal-weighted international stock index provides substantially greater diversification than the value-weighted index, which can be dominated by a small number of large equity markets. The additional diversification increases the optimal allocation to international stocks.

In a few specifications, the optimal age-based strategy departs materially from the baseline

²⁶We also consider untabulated robustness specifications in which we augment the baseline investment opportunity set with either Treasury Inflation-Protected Securities (TIPS) or international bonds. In contrast with the hypothetical real risk-free asset considered in Section 5.3, TIPS are risky, and their exposures (e.g., real interest rate risk and liquidity risk) link their returns to those of other asset classes. Because TIPS are unavailable for most countries and much of our sample period, we estimate a regression of monthly real returns for the Vanguard Inflation-Protected Securities Fund (VIPSX) on contemporaneous real returns for domestic stocks, international stocks, bonds, and bills in the US over the period July 2000 to December 2023 ($R^2 = 66\%$). We simulate real TIPS returns for the full sample as the fitted values from this regression plus a normally distributed shock with standard deviation equal to that of the regression residuals. International bond returns are computed by forming an equal-weighted portfolio of foreign government bonds with exchange rate adjustments and explicit treatment of sovereign default episodes. Full details on the calculation of international bond returns are provided in the internet appendix. Under both robustness specifications, the additional asset receives zero weight in both the optimal fixed-weight and age-based strategies.

domestic-international allocation only during the earliest working years. The domestic stock allocation is higher for the ex-US sample (Panel B) and US dollar-denominated returns (Panel C), but lower for the sample restricted to countries with developed equity markets (Panel B), very low risk aversion (Panel E), and the investor type with low initial income and high human capital (Panel I). These deviations are concentrated at ages 25 through 34, when accumulated retirement wealth is minimal and human capital constitutes nearly all of the household's lifetime resources. Consequently, the precise allocation between domestic and international stocks has little effect on expected lifetime utility during this stage of the lifecycle. Consistent with this interpretation, the corresponding optimal fixed-weight strategies typically remain close to the baseline allocation of 34% domestic stocks and 66% international stocks.

6 Conclusion

We study lifecycle portfolio choice using a framework that jointly optimizes saving, withdrawals, and age-based portfolio allocations. Our key contributions are to consider a broader investment opportunity set and to model long-horizon returns using a nonparametric block bootstrap that preserves their historical time-series and cross-sectional properties. The optimal strategy remains fully invested in equities throughout the lifecycle, with approximately one-third allocated to domestic stocks and two-thirds to international stocks.

The recent lifecycle portfolio choice literature has emphasized specifications that enrich the household's optimization problem by incorporating additional sources of household risk and additional margins of household choice. Our results highlight the complementary importance of modeling long-horizon investment opportunities. Empirically grounded modifications to the return environment alone can fundamentally alter optimal portfolio allocations, pointing to a much larger role for international stocks and a much smaller role for fixed income assets than implied by

standard lifecycle models.

References

- Anarkulova, Aizhan, Scott Cederburg, and Michael S. O'Doherty, 2022, Stocks for the long run? Evidence from a broad sample of developed markets, *Journal of Financial Economics* 143, 409–433.
- , 2025, Long-horizon losses in stocks, bonds, and bills, Working paper, Emory University.
- Ang, Andrew, and Geert Bekaert, 2002, International asset allocation with regime shifts, *Review of Financial Studies* 15, 1137–1187.
- , 2004, How regimes affect asset allocation, *Financial Analysts Journal* 60, 86–99.
- Asness, Clifford S., 1996, Why *not* 100% equities, *Journal of Portfolio Management* 22, 29–34.
- Avdīs, Efstathios, and Jessica A. Wachter, 2017, Maximum likelihood estimation of the equity premium, *Journal of Financial Economics* 125, 589–609.
- Bagliano, Fabio C., Carolina Fugazza, and Giovanna Nicodano, 2019, Life-cycle portfolios, unemployment and human capital loss, *Journal of Macroeconomics* 60, 325–340.
- Balduzzi, Pierluigi, and Jonathan Reuter, 2019, Heterogeneity in target date funds: Strategic risk-taking or risk matching?, *Review of Financial Studies* 32, 300–337.
- Bali, Turan G., K. Ozgur Demirtas, Haim Levy, and Avner Wolf, 2009, Bonds versus stocks: Investors' age and risk taking, *Journal of Monetary Economics* 56, 817–830.
- Barberis, Nicholas C., 2000, Investing for the long run when returns are predictable, *Journal of Finance* 55, 225–264.
- Baxter, Marianne, and Urban J. Jermann, 1997, The international diversification puzzle is worse than you think, *American Economic Review* 87, 170–180.
- Bekaert, Geert, Kenton Hoyem, Wei-Yin Hu, and Enrichetta Ravina, 2017, Who is internationally diversified? Evidence from the 401(k) plans of 296 firms, *Journal of Financial Economics* 124, 86–112.
- Benartzi, Shlomo, and Richard H. Thaler, 2001, Naive diversification strategies in defined contribution saving plans, *American Economic Review* 91, 79–98.
- Benzoni, Luca, Pierre Collin-Dufresne, and Robert S. Goldstein, 2007, Portfolio choice over the life-cycle when the stock and labor markets are cointegrated, *Journal of Finance* 62, 2123–2167.
- Beshears, John, James J. Choi, David Laibson, and Brigitte C. Madrian, 2018, Behavioral household finance, in B. Douglas Bernheim, Stefano DellaVigna, and David Laibson, eds.: *Handbook of Behavioral Economics: Foundations and Applications* 1 (Elsevier), Amsterdam, pp. 177–276.
- Blanchett, David M., David M. Cordell, Michael S. Finke, and Thomas M. Idzorek, 2023, Risk management for individuals, 2024 CFA Program Refresher Reading.
- Bodie, Zvi, Alex Kane, and Alan Marcus, 2024, *Investments*. McGraw Hill, New York, NY, 13th edn.
- Bodie, Zvi, Robert C. Merton, and William F. Samuelson, 1992, Labor supply flexibility and portfolio choice in a life cycle model, *Journal of Economic Dynamics and Control* 16, 427–449.

- Bollerslev, Tim, 1986, Generalized autoregressive conditional heteroskedasticity, *Journal of Econometrics* 31, 307–327.
- Bonaparte, Yosef, George M. Korniotis, Alok Kumar, Alexander Michaelides, and Yuxin Zhang, 2025, Stock market ownership transitions, *Management Science* 71, 4977–5000.
- Brandt, Michael W., 1999, Estimating portfolio and consumption choice: A conditional Euler equations approach, *Journal of Finance* 54, 1609–1645.
- Bremus, Franziska M., and Vladimir Kuzin, 2014, Unemployment and portfolio choice: Does persistence matter?, *Journal of Macroeconomics* 40, 99–113.
- Brennan, Michael J., Eduardo S. Schwartz, and Ronald Lagnado, 1997, Strategic asset allocation, *Journal of Economic Dynamics and Control* 21, 1377–1403.
- Bretscher, Lorenzo, Christian Julliard, and Carlo Rosa, 2016, Human capital and international portfolio diversification: A reappraisal, *Journal of International Economics* 99, 578–596.
- Brown, Stephen J., William N. Goetzmann, and Stephen A. Ross, 1995, Survival, *Journal of Finance* 50, 853–873.
- Calvet, Laurent E., John Y. Campbell, Francisco Gomes, and Paolo Sodini, 2025, The cross-section of household preferences, Forthcoming in *Journal of Finance*.
- Campbell, John Y., 2016, Restoring rational choice: The challenge of consumer financial regulation, *American Economic Review: Papers & Proceedings* 106, 1–30.
- Campbell, John Y., Yeung Lewis Chan, and Luis M. Viceira, 2003, A multivariate model of strategic asset allocation, *Journal of Financial Economics* 67, 41–80.
- Campbell, John Y., João F. Cocco, Francisco J. Gomes, and Pascal J. Maenhout, 2001, Investing retirement wealth: A life-cycle model, in John Y. Campbell, and Martin Feldstein, eds.: *Risk Aspects of Investment-Based Social Security Reform* (University of Chicago Press, Chicago), pp. 439–482.
- Campbell, John Y., and Luis M. Viceira, 1999, Consumption and portfolio decisions when expected returns are time varying, *Quarterly Journal of Economics* 114, 433–495.
- , 2002, *Strategic Asset Allocation: Portfolio Choice for Long-Term Investors*. Oxford University Press, Oxford, UK.
- , 2005, The term structure of the risk–return trade-off, *Financial Analysts Journal* 61, 34–44.
- Catherine, Sylvain, 2022, Countercyclical labor income risk and portfolio choices over the life cycle, *Review of Financial Studies* 35, 4016–4054.
- Cederburg, Scott, Travis L. Johnson, and Michael S. O’Doherty, 2023, On the economic significance of stock return predictability, *Review of Finance* 27, 619–657.
- Chacko, George, and Luis M. Viceira, 2005, Dynamic consumption and portfolio choice with stochastic volatility in incomplete markets, *Review of Financial Studies* 18, 1369–1402.
- Chai, Jingjing, Wolfram Horneff, Raimond Maurer, and Olivia S. Mitchell, 2011, Optimal portfolio choice over the life cycle with flexible work, endogenous retirement, and lifetime payouts, *Review of Finance* 15, 875–907.

- Cheng, Minder, and Ananth Madhavan, 2009, The dynamics of leveraged and inverse exchange-traded funds, *Journal of Investment Management* 7, 43–62.
- Choi, James J., 2022, Popular personal financial advice versus the professors, *Journal of Economic Perspectives* 36, 167–192.
- Choukhmane, Taha, and Tim de Silva, 2026, What drives investors’ portfolio choices? Separating risk preferences from frictions, *Journal of Finance* 81, 5–48.
- Cocco, João F., Francisco J. Gomes, and Pascal J. Maenhout, 2005, Consumption and portfolio choice over the life cycle, *Review of Financial Studies* 18, 491–533.
- Cochrane, John H., 2014, A mean-variance benchmark for intertemporal portfolio theory, *Journal of Finance* 69, 1–49.
- , 2022, Portfolios for long-term investors, *Review of Finance* 26, 1–42.
- Coeurdacier, Nicolas, and Hélène Rey, 2013, Home bias in open economy financial macroeconomics, *Journal of Economic Literature* 51, 63–115.
- Dahlquist, Magnus, Ofer Setty, and Roine Vestman, 2018, On the asset allocation of a default pension fund, *Journal of Finance* 73, 1893–1936.
- Davis, Steven J., Felix Kubler, and Paul Willen, 2006, Borrowing costs and the demand for equity over the life cycle, *Review of Economics and Statistics* 88, 348–362.
- Davis, Steven J., and Paul Willen, 2000, Using financial assets to hedge labor income risks: Estimating the benefits, Working paper, University of Chicago.
- De Nardi, Mariacristina, Eric French, and John B. Jones, 2010, Why do the elderly save? The role of medical expenses, *Journal of Political Economy* 118, 39–75.
- Dimson, Elroy, Paul R. Marsh, and Michael Staunton, 2002, *Triumph of the Optimists: 101 Years of Global Investment Returns*. Princeton University Press, Princeton, NJ.
- Duarte, Victor, Julia Fonseca, Aaron Goodman, and Jonathan A. Parker, 2024, Simple allocation rules and optimal portfolio choice over the lifecycle, Working paper, University of Illinois.
- Duggan, Julián Antonio, 1963, La colocación de los empréstitos externos: La Argentina y los mercados de capitales, Working paper, Universidad de Buenos Aires.
- Engle, Robert F., 1982, Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation, *Econometrica* 50, 987–1007.
- Eun, Cheol S., and Bruce G. Resnick, 1988, Exchange rate uncertainty, forward contracts, and international portfolio selection, *Journal of Finance* 43, 197–215.
- , 1994, International diversification of investment portfolios: U.S. and Japanese perspectives, *Management Science* 40, 140–161.
- Fagereng, Andreas, Charles Gottlieb, and Luigi Guiso, 2017, Asset market participation and portfolio choice over the life-cycle, *Journal of Finance* 72, 705–750.
- Fagereng, Andreas, Luigi Guiso, and Luigi Pistaferri, 2018, Portfolio choices, firm shocks, and uninsurable wage risk, *Review of Economic Studies* 85, 437–474.

- Fama, Eugene F., and Kenneth R. French, 2002, The equity premium, *Journal of Finance* 57, 637–659.
- Fama, Eugene F., and G. William Schwert, 1977, Human capital and capital market equilibrium, *Journal of Financial Economics* 4, 95–125.
- Farhi, Emmanuel, and Stavros Panageas, 2007, Saving and investing for early retirement: A theoretical analysis, *Journal of Financial Economics* 83, 87–121.
- Fischer, Gustav, 1923, Geld, Kredit, Währung, *Volkswirtschaftliche Chronik Jahresübersicht 1923*, 679–708.
- , 1924, Geld, Kredit, Währung, *Volkswirtschaftliche Chronik* December 1924.
- , 1925, Geld, Kredit, Währung, *Volkswirtschaftliche Chronik Jahresübersicht 1925*.
- Frazzini, Andrea, and Lasse Heje Pedersen, 2014, Betting against beta, *Journal of Financial Economics* 111, 1–25.
- French, Kenneth R., and James M. Poterba, 1991, Investor diversification and international equity markets, *American Economic Review: Papers & Proceedings* 81, 222–226.
- Garlappi, Lorenzo, Raman Uppal, and Tan Wang, 2007, Portfolio selection with parameter and model uncertainty: A multi-prior approach, *Review of Financial Studies* 20, 41–81.
- Goda, Gopi Shah, Matthew R. Levy, Colleen Flaherty Manchester, Aaron Sojourner, and Joshua Tasoff, 2020, Who is a passive saver under opt-in and auto-enrollment?, *Journal of Economic Behavior & Organization* 173, 301–321.
- Gomes, Francisco, Michael Haliassos, and Tarun Ramadorai, 2021, Household finance, *Journal of Economic Literature* 59, 919–1000.
- Gomes, Francisco, and Alexander Michaelides, 2005, Optimal life-cycle asset allocation: Understanding the empirical evidence, *Journal of Finance* 60, 869–904.
- Gomes, Francisco, Alexander Michaelides, and Yuxin Zhang, 2022, Tactical target date funds, *Management Science* 68, 3047–3070.
- Gomes, Francisco, and Oksana Smirnova, 2023, Stock market participation and portfolio shares over the life-cycle, Working paper, London Business School.
- Goyal, Amit, Ivo Welch, and Athanasse Zafirov, 2024, A comprehensive 2022 look at the empirical performance of equity premium prediction, *Review of Financial Studies* 37, 3490–3557.
- Graf von Luckner, Clemens M., Josefin Meyer, Carmen M. Reinhart, and Christoph Trebesch, 2025, Sovereign haircuts: 200 years of creditor losses, *IMF Economic Review* 73, 150–195.
- Guvenen, Fatih, Fatih Karahan, Serdar Ozkan, and Jae Song, 2021, What do data on millions of U.S. workers reveal about lifecycle earnings dynamics?, *Econometrica* 89, 2303–2339.
- Guvenen, Fatih, Serdar Ozkan, and Jae Song, 2014, The nature of countercyclical income risk, *Journal of Political Economy* 122, 621–660.
- Heaton, John, and Deborah J. Lucas, 1997, Market frictions, savings behavior, and portfolio choice, *Macroeconomic Dynamics* 1, 76–101.

- , 2000, Portfolio choice in the presence of background risk, *Economic Journal* 110, 1–26.
- Hnatkovska, Viktoria, 2010, Home bias and high turnover: Dynamic portfolio choice with incomplete markets, *Journal of International Economics* 80, 113–128.
- Hoevenaars, Roy P.M.M., Roderick D.J. Molenaar, Peter C. Schotman, and Tom B.M. Steenkamp, 2008, Strategic asset allocation with liabilities: Beyond stocks and bonds, *Journal of Economic Dynamics and Control* 32, 2939–2970.
- Hubener, Andreas, Raimond Maurer, and Olivia S. Mitchell, 2016, How family status and Social Security claiming options shape optimal life cycle portfolios, *Review of Financial Studies* 29, 937–978.
- Jeanneau, Serge, and Carlos Pérez Verdia, 2005, Reducing financial vulnerability: The development of the domestic government bond market in Mexico, *BIS Quarterly Review* December 2005, 95–107.
- Jordà, Òscar, Katharina Knoll, Dmitry Kuvshinov, Moritz Schularick, and Alan M. Taylor, 2019, The rate of return on everything, 1870–2015, *Quarterly Journal of Economics* 134, 1225–1298.
- Jorion, Philippe, 1985, International portfolio diversification with estimation risk, *Journal of Business* 58, 259–278.
- Kang, Kenneth, Geena Kim, and Changyong Rhee, 2005, Developing the government bond market in South Korea: History, challenges, and implications for Asian countries, *Asian Economic Papers* 4, 91–113.
- Kelly, Bryan, and Hao Jiang, 2014, Tail risk and asset prices, *Review of Financial Studies* 27, 2841–2871.
- Koijen, Ralph S.J., Theo E. Nijman, and Bas J.M. Werker, 2010, When can life cycle investors benefit from time-varying bond risk premia?, *Review of Financial Studies* 23, 741–780.
- Koo, Hyeng Keun, 1998, Consumption and portfolio selection with labor income: A continuous time approach, *Mathematical Finance* 8, 49–65.
- Kraft, Holger, Claus Munk, and Farina Weiss, 2019, Predictors and portfolios over the life cycle, *Journal of Banking and Finance* 100, 1–27.
- Kuznets, Simon, 1973, Modern economic growth: Findings and reflections, *American Economic Review* 63, 247–258.
- Longin, François, and Bruno Solnik, 2001, Extreme correlation of international equity markets, *Journal of Finance* 56, 649–676.
- Lusardi, Annamaria, and Olivia S. Mitchell, 2007, Baby Boomer retirement security: The roles of planning, financial literacy, and housing wealth, *Journal of Monetary Economics* 54, 205–224.
- , 2011, Financial literacy and retirement planning in the United States, *Journal of Pension Economics & Finance* 10, 509–525.
- Lynch, Anthony W., 2001, Portfolio choice and equity characteristics: Characterizing the hedging demands induced by return predictability, *Journal of Financial Economics* 62, 67–130.
- Lynch, Anthony W., and Pierluigi Balduzzi, 2000, Predictability and transaction costs: The impact on rebalancing rules and behavior, *Journal of Finance* 55, 2285–2310.

- Lynch, Anthony W., and Sinan Tan, 2010, Multiple risky assets, transaction costs, and return predictability: Allocation rules and implications for U.S. investors, *Journal of Financial and Quantitative Analysis* 45, 1015–1053.
- , 2011, Labor income dynamics at business-cycle frequencies: Implications for portfolio choice, *Journal of Financial Economics* 101, 333–359.
- Macaulay, Frederick R., 1938, *Some theoretical problems suggested by the movements of interest rates, bond yields and stock prices in the United States since 1856*. National Bureau of Economic Research, New York, NY.
- Madrian, Brigitte C., and Dennis F. Shea, 2001, The power of suggestion: Inertia in 401(k) participation and savings behavior, *Quarterly Journal of Economics* 116, 1149–1187.
- Mehra, Rajnish, and Edward C. Prescott, 1985, The equity premium: A puzzle, *Journal of Monetary Economics* 15, 145–161.
- Merton, Robert C., 1969, Lifetime portfolio selection under uncertainty: The continuous-time case, *Review of Economics and Statistics* 51, 247–257.
- , 1971, Optimum consumption and portfolio rules in a continuous-time model, *Journal of Economic Theory* 3, 373–413.
- Meyer, Josefin, Carmen M. Reinhart, and Christoph Trebesch, 2022, Sovereign bonds since Waterloo, *Quarterly Journal of Economics* 137, 1615–1680.
- Michaelides, Alexander, 2003, International portfolio choice, liquidity constraints and the home equity bias puzzle, *Journal of Economic Dynamics and Control* 28, 555–594.
- Michaelides, Alexander, and Yuxin Zhang, 2017, Stock market mean reversion and portfolio choice over the life cycle, *Journal of Financial and Quantitative Analysis* 52, 1183–1209.
- , 2022, Life-cycle portfolio choice with imperfect predictors, *Journal of Banking and Finance* 135, 106357.
- Michaud, Richard O., 1989, The Markowitz optimization enigma: Is ‘optimized’ optimal?, *Financial Analysts Journal* 45, 31–42.
- Mitchell, Olivia S., and Stephen P. Utkus, 2022, Target-date funds and portfolio choice in 401(k) plans, *Journal of Pension Economics & Finance* 21, 519–536.
- Parker, Jonathan A., Antoinette Schoar, and Yang Sun, 2023, Retail financial innovation and stock market dynamics: The case of target date funds, *Journal of Finance* 78, 2673–2723.
- Pástor, Luboš, and Robert F. Stambaugh, 2012, Are stocks really less volatile in the long run?, *Journal of Finance* 67, 431–477.
- Politis, Dimitris N., and Joseph P. Romano, 1994, The stationary bootstrap, *Journal of the American Statistical Association* 89, 1303–1313.
- Poterba, James, Steven Venti, and David Wise, 2011, The composition and drawdown of wealth in retirement, *Journal of Economic Perspectives* 25, 95–118.

- Poterba, James M., 2003, Employer stock and 401(k) plans, *American Economic Review: Papers & Proceedings* 93, 398–404.
- Poterba, James M., and Lawrence H. Summers, 1988, Mean reversion in stock prices: Evidence and implications, *Journal of Financial Economics* 22, 27–59.
- Reinhart, Carmen M., and Kenneth S. Rogoff, 2011, The forgotten history of domestic debt, *Economic Journal* 121, 319–350.
- Roy, Mahi, 2026, 2026 Target-Date Fund Landscape, <https://www.morningstar.com/business/insights/research/tdf-landscape>, Morningstar Research.
- Samuelson, Paul A., 1969, Lifetime portfolio selection by dynamic stochastic programming, *Review of Economics and Statistics* 51, 239–246.
- Sangvinatsos, Antonios, and Jessica A. Wachter, 2005, Does the failure of the expectations hypothesis matter for long-term investors?, *Journal of Finance* 60, 179–230.
- Schnabl, Gunther, 2019, The 1948 German currency and economic reform: Lessons for European monetary policy, *Cato Journal* 39, 607–634.
- Shearer, Ronald A., and Carolyn Clark, 1984, Canada and the interwar gold standard, 1920–35: Monetary policy without a Central Bank, in Michael D. Bordo, and Anna J. Schwartz, eds.: *A Retrospective on the Classical Gold Standard, 1821–1931* (University of Chicago Press), Chicago, IL, pp. 277–310.
- Shen, Jialu, 2024, Countercyclical risks, consumption, and portfolio choice: Theory and evidence, *Management Science* 70, 2862–2881.
- Siegel, Jeremy, 2014, *Stocks for the Long Run*. McGraw-Hill, New York, NY.
- Solnik, Bruno H., 1974, Why not diversify internationally rather than domestically?, *Financial Analysts Journal* 30, 48–54.
- van Binsbergen, Jules H., William F. Diamond, and Marco Grotteria, 2022, Risk-free interest rates, *Journal of Financial Economics* 143, 1–29.
- van Binsbergen, Jules H., Sophia Hua, Jonas Peeters, and Jessica A. Wachter, 2025, Is the United States a lucky survivor? A hierarchical Bayesian approach, *Journal of Finance* 80, 2355–2388.
- van Rooij, Maarten, Annamaria Lusardi, and Rob Alessie, 2011, Financial literacy and stock market participation, *Journal of Financial Economics* 101, 449–472.
- Vanguard, 2026, How America Saves 25th Edition, <https://workplace.vanguard.com/content/ig-transformation/pdf/how-america-saves-2026.html>.
- Venti, Steven F., and David A. Wise, 1991, Aging and the income value of housing wealth, *Journal of Public Economics* 44, 371–397.
- Viceira, Luis M., 2001, Optimal portfolio choice for long-horizon investors with nontradable labor income, *Journal of Finance* 56, 433–470.
- Wachter, Jessica A., 2002, Portfolio and consumption decisions under mean-reverting returns: An exact solution for complete markets, *Journal of Financial and Quantitative Analysis* 37, 63–91.

———, 2010, Asset allocation, *Annual Review of Financial Economics* 2, 175–206.

Zettelmeyer, Jeromin, Christoph Trebesch, and Mitu Gulati, 2013, The Greek debt restructuring: An autopsy, *Economic Policy* 28, 513–563.

Table I: Empirical properties of real returns for bonds and international stocks

The table summarizes the empirical properties of real returns for bonds and international stocks. The underlying data are a monthly panel of real asset class returns for 39 developed countries covering the period from 1890 to 2023. Details of the dataset construction are provided in Section 3. Panel A reports the annualized mean and standard deviation of real returns for each asset class. Panel B reports variance ratios at horizons of one, ten, 20, and 30 years. Panel C reports the correlation of the log return for each asset class with the log return for domestic stocks (based on monthly and 30-year returns) and with log inflation (based on 30-year returns). The statistics reported in Panels B and C are based on a single block bootstrap simulation of 100,000,000 months. All statistics are estimated from a sample that excludes data for Germany in 1923.

Measure	Asset class	
	Bonds	International stocks
Panel A: Moments of annualized real returns		
Mean (%)	0.95	7.03
Standard deviation (%)	9.51	23.26
Panel B: Variance ratios		
VR(1)	1.00	1.00
VR(10)	2.16	0.89
VR(20)	2.34	0.80
VR(30)	2.39	0.75
Panel C: Log real return correlations		
Correlation with domestic stocks (monthly returns)	0.21	0.33
Correlation with domestic stocks (30-year returns)	0.45	0.34
Correlation with inflation (30-year returns)	−0.78	−0.02

Table II: Sample coverage and summary statistics

For each developed country, the table reports the sample period start date, the sample period end date, and summary statistics (i.e., geometric mean return and standard deviation of return) of monthly real returns for domestic stocks, international stocks, bonds, and bills. The development classification and sample formation criteria are described in the internet appendix.

Country	Sample start	Sample end	Asset class returns							
			Domestic stocks		International stocks		Bonds		Bills	
			Mean	StDev	Mean	StDev	Mean	StDev	Mean	StDev
Denmark	1890:01	2023:12	0.46	3.58	0.39	3.93	0.20	1.89	0.17	0.73
France	1890:01	2023:12	0.26	5.40	0.45	6.58	-0.08	2.28	-0.16	1.74
Germany	1890:01	2023:12	0.25	8.29	0.56	10.20	-0.14	45.61	0.15	0.86
United Kingdom	1890:01	2023:12	0.41	4.25	0.43	4.09	0.13	1.98	0.06	0.87
United States	1890:01	2023:12	0.52	5.01	0.34	3.85	0.12	1.76	0.06	0.61
Canada	1891:01	2023:12	0.48	4.26	0.44	3.48	0.18	1.66	0.11	0.57
New Zealand	1896:01	2023:12	0.47	3.66	0.44	4.06	0.13	1.84	0.14	0.58
Belgium	1897:01	2023:12	0.21	5.03	0.39	4.57	0.02	1.79	-0.04	1.13
Australia	1901:01	2023:12	0.57	3.96	0.43	3.75	0.14	1.73	0.06	0.53
Sweden	1910:01	2023:12	0.47	4.85	0.45	4.15	0.14	1.86	0.08	0.96
Netherlands	1914:01	2023:12	0.41	5.19	0.41	4.39	0.13	1.73	0.01	0.80
Norway	1914:01	2023:12	0.37	4.62	0.44	4.19	0.13	1.73	0.02	0.85
Switzerland	1914:01	2023:12	0.38	4.30	0.37	4.47	0.14	1.41	0.02	0.61
Austria	1920:01	2023:12	0.22	7.27	0.57	12.79	-0.38	4.51	-0.46	3.68
Czechoslovakia	1922:05	1945:05	-0.14	6.56	0.41	6.24	0.69	3.28	0.34	3.06
Chile period I	1927:01	1970:12	0.13	6.15	0.60	8.55	-0.92	3.38	-0.86	2.34
Japan	1930:01	2023:12	0.31	6.59	0.51	15.92	-0.18	3.40	-0.32	2.60
Italy	1931:01	2023:12	0.19	7.38	0.43	12.98	-0.14	2.56	-0.25	1.68
Portugal	1934:01	2023:12	0.14	7.82	0.44	4.10	0.02	2.81	-0.06	1.34
Ireland	1936:01	2023:12	0.45	4.77	0.46	4.08	0.16	2.40	0.01	0.59
Argentina	1947:02	1966:12	-0.18	8.53	0.64	15.33	-1.66	2.84	-1.56	2.73
Spain	1959:01	2023:12	0.28	5.58	0.39	4.25	0.14	2.22	0.00	0.69
Finland	1969:01	2023:12	0.73	6.24	0.41	4.35	0.24	2.26	0.03	0.47
Greece	1981:02	2023:12	0.48	10.23	0.55	4.77	0.30	5.41	0.14	1.29
Luxembourg	1982:01	2023:12	0.55	5.75	0.57	4.51	0.28	1.88	0.10	0.57
Singapore	1998:07	2023:12	0.44	5.74	0.29	4.09	0.14	2.04	-0.04	0.48

(Continued on next page)

Table II (Continued)

Country	Sample start	Sample end	Asset class returns							
			Domestic stocks		International stocks		Bonds		Bills	
			Mean	StDev	Mean	StDev	Mean	StDev	Mean	StDev
Hungary	1999:02	2023:12	0.35	6.52	0.26	4.21	0.17	3.49	0.09	0.53
Poland	1999:06	2023:12	0.26	6.20	0.21	3.78	0.22	2.92	0.12	0.53
Slovakia	2000:01	2023:12	0.17	5.10	−0.02	4.26	0.24	2.96	−0.12	0.62
Czech Republic	2000:05	2023:12	0.77	6.92	0.01	4.23	0.08	2.45	−0.10	0.63
South Korea	2000:11	2023:12	0.63	6.18	0.34	3.93	0.25	2.02	0.06	0.36
Mexico	2001:08	2023:12	0.62	4.94	0.44	3.72	0.29	2.67	0.14	0.40
Iceland	2002:01	2023:12	0.02	7.27	0.33	4.80	0.16	3.16	0.14	0.54
Chile period II	2010:01	2023:12	−0.01	4.89	0.70	3.99	−0.05	1.93	−0.02	0.43
Israel	2010:01	2023:12	0.01	5.07	0.68	3.63	0.28	2.14	0.09	0.87
Slovenia	2010:01	2023:12	0.41	4.75	0.72	3.79	0.09	2.69	−0.10	0.80
Türkiye	2010:02	2023:12	0.36	7.42	1.11	6.49	−0.79	6.92	−0.47	1.77
Latvia	2016:01	2023:12	0.47	4.42	0.40	4.09	−0.51	2.23	−0.36	0.82
Lithuania	2018:01	2023:12	0.02	3.85	0.26	4.57	−0.82	2.76	−0.46	0.80
Colombia	2020:01	2023:12	−0.79	9.02	0.46	4.75	−0.44	4.27	−0.08	0.51

Table III: Summary of optimal and benchmark investment strategies

The table summarizes the portfolio weights, policy functions, and welfare implications of the optimal age-based strategy, the optimal fixed-weight strategy, and four benchmark investment strategies. Panel A reports the portfolio weights in domestic stocks, international stocks, bonds, and bills. For the TDF and optimal age-based strategies, the reported weights denote the range of portfolio weights over the lifecycle corresponding to the glide paths shown in Figures 1 and 3, respectively. Panels B and C report the estimated coefficients of the saving rate and withdrawal rate policy functions, respectively. Panel D reports two measures of the welfare cost of each strategy relative to the optimal age-based strategy: the equivalent annual fee and the consumption certainty-equivalent loss (CEL). The welfare comparisons are based on expected lifetime utility from consumption and bequest across 1,000,000 bootstrap simulations.

	Strategy					
	Optimal age-based	Optimal fixed-weight	Bills	Domestic stocks	Balanced	TDF
Panel A: Portfolio weights						
Domestic stocks	[30%, 39%]	34%	0%	100%	60%	[10%, 54%]
International stocks	[61%, 70%]	66%	0%	0%	0%	[7%, 36%]
Bonds	[0%, 0%]	0%	0%	0%	40%	[10%, 73%]
Bills	[0%, 0%]	0%	100%	0%	0%	[0%, 10%]
Panel B: Saving rate policy coefficients						
Intercept	-2.7211	-2.7210	-3.2463	-2.7103	-2.7362	-2.5362
Age	6.1781	6.1781	5.7711	5.4162	5.4770	5.4970
Wealth	-1.1886	-1.1888	-0.7242	-0.9801	-0.8656	-0.8670
Panel C: Withdrawal rate policy coefficients						
Intercept	-2.8654	-2.8659	-2.8240	-2.9056	-2.9494	-2.9845
Age	2.6249	2.6258	3.5289	2.8677	3.0265	3.1128
Wealth	-0.0011	-0.0011	-0.0174	-0.0014	-0.0014	-0.0020
Panel D: Welfare costs						
Equivalent annual fee	—	0.0008%	5.7895%	1.2178%	2.0867%	1.8960%
Consumption CEL	—	0.0007%	5.0209%	0.9729%	1.7278%	1.5589%

Table IV: Retirement saving outcomes

The table summarizes the distributions of real wealth at retirement (Panel A), the real income replacement rate (Panel B), and real wealth at death (Panel C) across 1,000,000 bootstrap simulations for households adopting various asset allocation strategies. For each asset allocation strategy, the table reports the mean, standard deviation, and selected percentiles of each outcome. Real wealth at retirement and real wealth at death are reported in millions of 2022 USD.

Strategy	Moments		Percentiles								
	Mean	StDev	1%	5%	10%	25%	50%	75%	90%	95%	99%
Panel A: Wealth at retirement (\$MM)											
Bills	0.38	0.30	0.00	0.04	0.09	0.18	0.31	0.50	0.74	0.93	1.43
Domestic stocks	0.99	1.57	0.00	0.08	0.16	0.32	0.61	1.14	2.04	2.96	6.45
Balanced	0.80	2.48	0.00	0.08	0.15	0.30	0.53	0.92	1.51	2.08	4.40
TDF	0.86	2.30	0.00	0.10	0.19	0.35	0.60	1.01	1.62	2.19	4.40
Optimal age-based	0.99	1.33	0.00	0.13	0.21	0.38	0.68	1.17	1.95	2.70	5.50
Panel B: Income replacement rate											
Bills	0.68	0.28	0.00	0.40	0.46	0.55	0.66	0.78	0.92	1.03	1.36
Domestic stocks	1.51	1.29	0.00	0.51	0.62	0.83	1.16	1.73	2.73	3.78	7.04
Balanced	1.17	0.85	0.00	0.50	0.60	0.76	0.99	1.33	1.85	2.37	4.51
TDF	1.14	0.71	0.00	0.52	0.62	0.78	0.99	1.30	1.76	2.18	3.87
Optimal age-based	1.59	1.27	0.00	0.63	0.75	0.96	1.28	1.79	2.62	3.48	7.02
Panel C: Wealth at death (\$MM)											
Bills	0.07	0.16	0.00	0.00	0.00	0.00	0.02	0.08	0.20	0.32	0.69
Domestic stocks	0.85	11.46	0.00	0.01	0.02	0.06	0.18	0.52	1.27	2.24	8.08
Balanced	0.51	17.00	0.00	0.01	0.01	0.04	0.13	0.33	0.73	1.18	3.27
TDF	0.42	10.92	0.00	0.00	0.01	0.04	0.11	0.31	0.68	1.07	2.80
Optimal age-based	0.76	7.05	0.01	0.02	0.04	0.10	0.26	0.61	1.27	2.04	6.11

Table V: Portfolio drawdowns

The table summarizes the distributions of the maximum portfolio drawdown during the working period (Panel A) and the maximum portfolio drawdown during the retirement period (Panel B) across 1,000,000 bootstrap simulations for households adopting various asset allocation strategies. For each asset allocation strategy and drawdown period, the table reports the mean, standard deviation, and selected percentiles of the maximum portfolio drawdown.

Strategy	Moments		Percentiles								
	Mean	StDev	1%	5%	10%	25%	50%	75%	90%	95%	99%
Panel A: Working-period drawdown											
Bills	0.42	0.26	0.04	0.09	0.13	0.22	0.36	0.57	0.84	0.96	1.00
Domestic stocks	0.67	0.16	0.32	0.41	0.47	0.55	0.66	0.78	0.91	0.96	0.99
Balanced	0.54	0.19	0.23	0.29	0.32	0.41	0.50	0.66	0.85	0.92	0.99
TDF	0.52	0.16	0.22	0.28	0.32	0.41	0.51	0.61	0.74	0.81	0.94
Optimal age-based	0.55	0.13	0.24	0.31	0.38	0.48	0.56	0.61	0.69	0.77	0.91
Panel B: Retirement-period drawdown											
Bills	0.46	0.30	0.00	0.03	0.09	0.22	0.42	0.69	0.94	0.98	1.00
Domestic stocks	0.62	0.21	0.00	0.27	0.35	0.49	0.63	0.77	0.91	0.96	0.99
Balanced	0.50	0.23	0.00	0.18	0.24	0.34	0.47	0.65	0.85	0.92	0.98
TDF	0.40	0.23	0.00	0.11	0.15	0.23	0.35	0.54	0.76	0.89	0.99
Optimal age-based	0.48	0.17	0.00	0.19	0.25	0.38	0.52	0.59	0.66	0.74	0.88

Table VI: Optimal asset allocation policies with additional background risks

Panel A reports the optimal fixed-weight and optimal age-based asset allocation policies for the baseline specification. For the fixed-weight strategy, the table reports the total equity allocation together with the portfolio weights in domestic stocks, international stocks, bonds, and bills. For the age-based strategy, the table reports the minimum total equity allocation together with the portfolio weight in domestic stocks at selected ages over the lifecycle. Panel B reports the optimal asset allocation policies when labor income is positively correlated with domestic stock returns. Panel C reports the optimal asset allocation policies when the lifecycle model is extended to incorporate health status, mortality risk, and out-of-pocket medical expenditures following the framework of De Nardi, French, and Jones (2010).

Description	Optimal fixed-weight strategy					Optimal age-based strategy				
	Total equity	Fixed asset class weights				Min. equity	Domestic stocks weights			
		Dom. stocks	Intl. stocks	Bonds	Bills		Age 25–34	Age 50–54	Age 70–74	Age 90+
Panel A: Baseline specification										
Base case	100%	34%	66%	0%	0%	100%	35%	36%	33%	30%
Panel B: Dependencies between labor income and stock returns										
Income–domestic stock correlation of 0.1	100%	31%	69%	0%	0%	100%	0%	31%	33%	32%
Income–domestic stock correlation of 0.2	100%	27%	73%	0%	0%	100%	0%	27%	33%	32%
Income–domestic stock correlation of 0.3	100%	24%	76%	0%	0%	100%	0%	23%	33%	32%
Income–domestic stock correlation of 0.4	100%	20%	80%	0%	0%	100%	0%	19%	33%	32%
Income–domestic stock correlation of 0.5	100%	17%	83%	0%	0%	100%	0%	16%	33%	32%
Panel C: Health, mortality, and medical expenditure risk										
Health, mortality, and medical expenditure risk	100%	34%	66%	0%	0%	100%	35%	35%	32%	32%

Table VII: Optimal age-based asset allocation policy conditional on the aggregate price-dividend ratio
The table summarizes the optimal age-based asset allocation policy conditional on the aggregate price-dividend ratio for domestic stocks. For each aggregate price-dividend quintile, the table reports the optimal portfolio weights in total equity, bonds, and bills. Under the optimal policy, these portfolio weights are constant across the lifecycle within each quintile. The table also reports the optimal portfolio weight in domestic stocks at selected ages within each aggregate price-dividend quintile.

Market state	Aggregate P_t/D_t	Asset class weights (all ages)			Domestic stocks weights (selected ages)			
		Total equity	Bonds	Bills	Age 25–34	Age 50–54	Age 70–74	Age 90+
Q1 (low P_t/D_t)	[0, 18.76]	100%	0%	0%	100%	72%	63%	61%
Q2	(18.76, 23.47]	100%	0%	0%	50%	34%	30%	29%
Q3	(23.47, 29.94]	100%	0%	0%	50%	34%	30%	29%
Q4	(29.94, 43.67]	100%	0%	0%	50%	34%	30%	29%
Q5 (high P_t/D_t)	(43.67, ∞)	92%	8%	0%	22%	15%	13%	13%

Table VIII: Optimal asset allocation policies under alternative design assumptions

Panel A reports the optimal fixed-weight and optimal age-based asset allocation policies for the baseline specification. For the fixed-weight strategy, the table reports the total equity allocation together with the portfolio weights in domestic stocks, international stocks, bonds, and bills. For the age-based strategy, the table reports the minimum total equity allocation together with the portfolio weight in domestic stocks at selected ages over the lifecycle. Panels B–I report the optimal fixed-weight and age-based asset allocation policies under the alternative design assumptions described in the table.

Description	Optimal fixed-weight strategy					Optimal age-based strategy				
	Total equity	Fixed asset class weights				Min. equity	Domestic stocks weights			
		Dom. stocks	Intl. stocks	Bonds	Bills		Age 25–34	Age 50–54	Age 70–74	Age 90+
Panel A: Baseline specification										
Base case	100%	34%	66%	0%	0%	100%	35%	36%	33%	30%
Panel B: Alternative samples										
Post-World War II sample	100%	32%	68%	0%	0%	100%	46%	34%	30%	29%
Countries with large populations	100%	28%	72%	0%	0%	100%	21%	29%	27%	26%
Countries with large market cap-to-GDP ratios	100%	27%	73%	0%	0%	100%	7%	28%	27%	24%
Ex-United States sample	100%	49%	51%	0%	0%	100%	98%	50%	44%	46%
Panel C: Calculation of asset class returns										
Equal-weight international stock index	100%	10%	90%	0%	0%	100%	0%	11%	13%	9%
USD-denominated returns	100%	29%	71%	0%	0%	100%	98%	29%	24%	25%
Panel D: Average block length										
Average block length of 60 months	100%	34%	66%	0%	0%	100%	35%	35%	32%	31%
Average block length of 240 months	100%	34%	66%	0%	0%	100%	32%	35%	32%	31%
Panel E: Risk aversion										
Risk aversion parameter $\gamma = 0.5$	100%	32%	68%	0%	0%	100%	0%	32%	32%	42%
Risk aversion parameter $\gamma = 1$ (log utility)	100%	34%	66%	0%	0%	100%	25%	35%	34%	34%
Risk aversion parameter $\gamma = 2$	100%	34%	66%	0%	0%	100%	32%	36%	33%	30%
Risk aversion parameter $\gamma = 5$	100%	34%	66%	0%	0%	100%	36%	35%	33%	30%
Risk aversion parameter $\gamma = 7.5$	100%	34%	66%	0%	0%	100%	39%	34%	32%	28%
Risk aversion parameter $\gamma = 10$	100%	34%	66%	0%	0%	100%	39%	34%	33%	27%

(Continued on next page)

Table VIII (Continued)

Description	Optimal fixed-weight strategy					Optimal age-based strategy				
	Total equity	Fixed asset class weights				Min. equity	Domestic stocks weights			
		Dom. stocks	Intl. stocks	Bonds	Bills		Age 25–34	Age 50–54	Age 70–74	Age 90+
Panel F: Household utility specification										
No bequest motive ($\theta = 0$)	100%	34%	66%	0%	0%	100%	35%	36%	33%	30%
Consumption scaled by household size	100%	34%	66%	0%	0%	100%	32%	36%	33%	32%
Panel G: Retirement timing										
Claim Social Security and retire at age 62	100%	34%	66%	0%	0%	100%	37%	35%	32%	31%
Claim Social Security and retire at age 67	100%	34%	66%	0%	0%	100%	33%	35%	33%	32%
Claim Social Security and retire at age 70	100%	34%	66%	0%	0%	100%	31%	35%	33%	32%
Endogenous retirement	100%	34%	66%	0%	0%	100%	34%	36%	32%	29%
Panel H: Household type										
Single female	100%	34%	66%	0%	0%	100%	36%	35%	33%	33%
Single male	100%	35%	65%	0%	0%	100%	34%	36%	33%	34%
Same-sex couple, both female	100%	34%	66%	0%	0%	100%	35%	35%	33%	31%
Same-sex couple, both male	100%	34%	66%	0%	0%	100%	34%	35%	33%	31%
Panel I: Investor type										
Low initial income and low human capital	100%	35%	65%	0%	0%	100%	30%	36%	34%	35%
Low initial income and high human capital	100%	34%	66%	0%	0%	100%	0%	37%	33%	32%
High initial income and low human capital	100%	34%	66%	0%	0%	100%	40%	34%	32%	30%
High initial income and high human capital	100%	33%	67%	0%	0%	100%	43%	36%	31%	29%

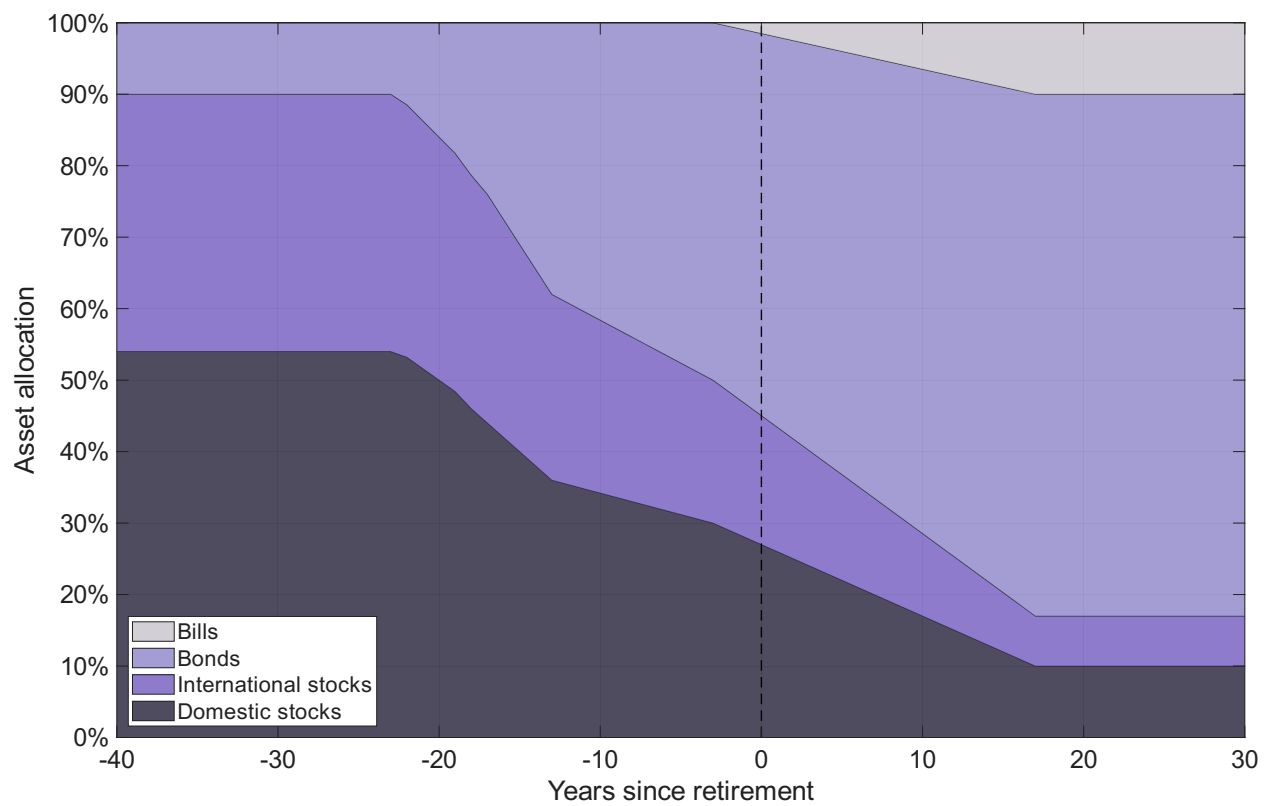


Figure 1. Glide path weights for the target-date fund. The figure shows the asset allocation of the target-date strategy as a function of time since retirement. The strategy invests in domestic stocks, international stocks, bonds, and bills.

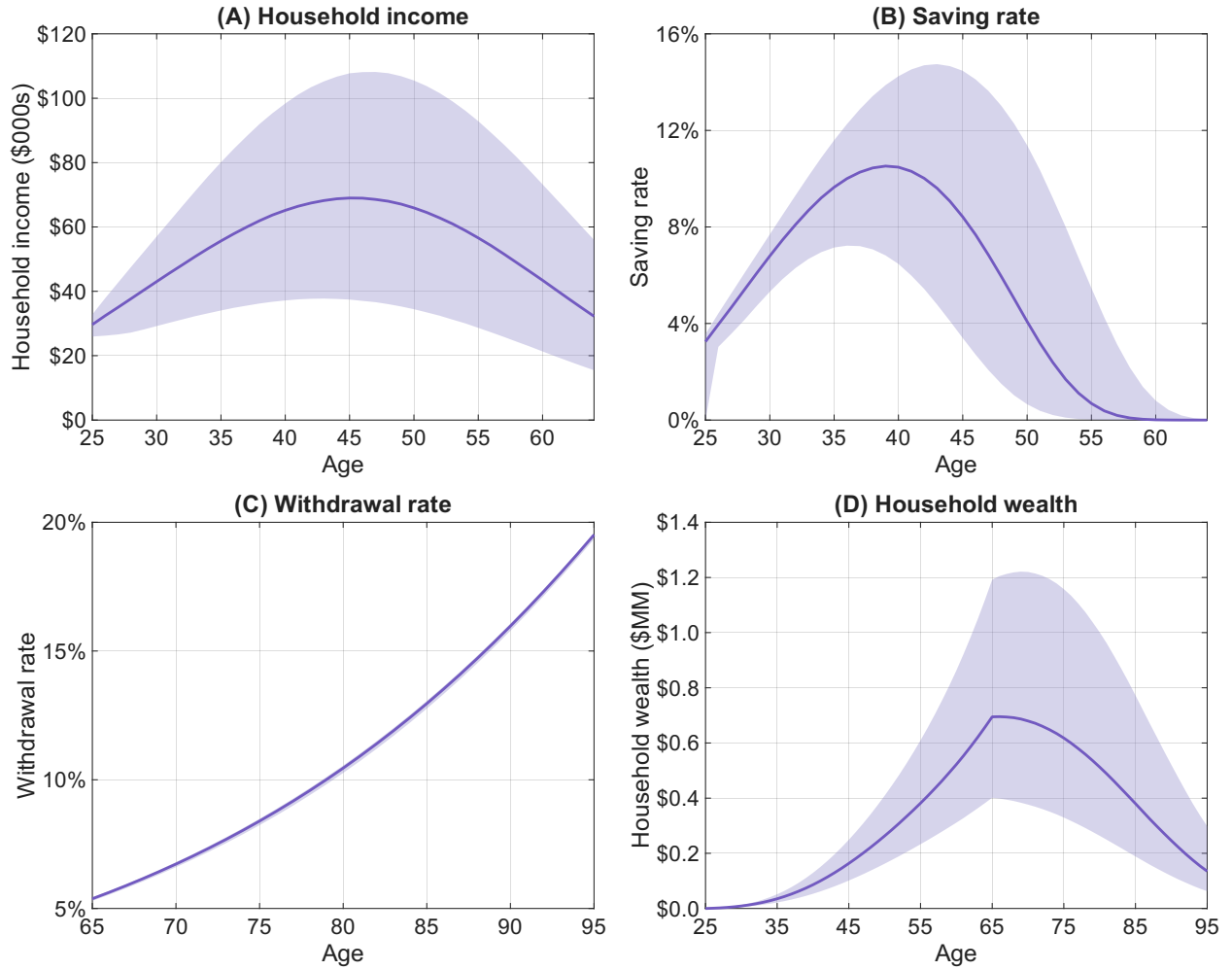


Figure 2. Lifecycle profiles of household income, saving, withdrawals, and wealth. The figure illustrates the distributions of real household income (Panel A), the optimal saving rate (Panel B), the optimal withdrawal rate (Panel C), and real household wealth (Panel D) across 1,000,000 bootstrap simulations as a function of age. Household income follows the stochastic process estimated by Guvenen, Karahan, Ozkan, and Song (2021) with an initial income parameter of $z_0^i = 0$ and human capital parameters of $(\alpha^i, \beta^i) = (0, 0)$. The saving rate, withdrawal rate, and household wealth are based on the optimal age-based strategy. Real household income and real household wealth are reported in 2022 USD. In each panel, the solid line corresponds to the median at each age, and the shaded region covers the 25th through 75th percentiles of the distribution.

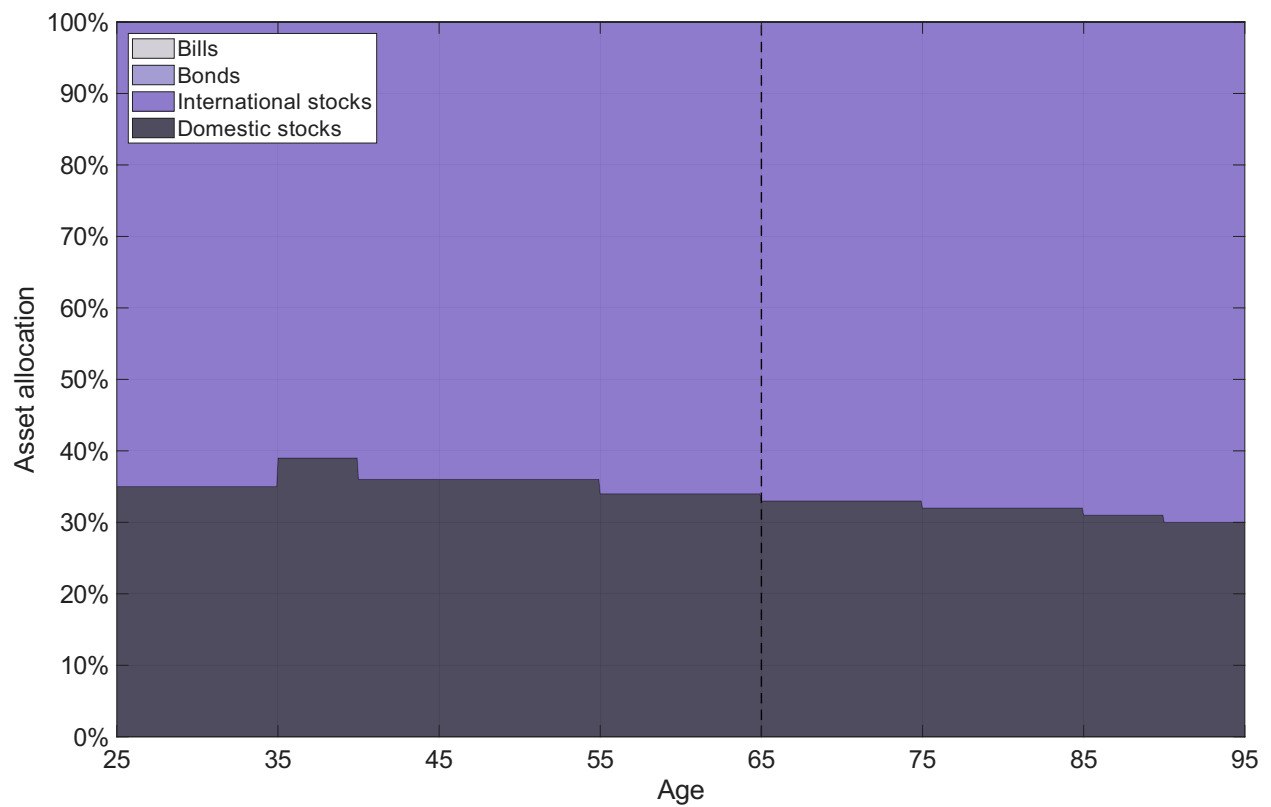


Figure 3. Optimal age-based asset allocation policy. The figure shows the portfolio weights in domestic stocks, international stocks, bonds, and bills for the optimal asset allocation policy as a function of household age. The vertical dashed line indicates the household's exogenous retirement age. The optimal policy assigns zero weight to bonds and bills at all ages.

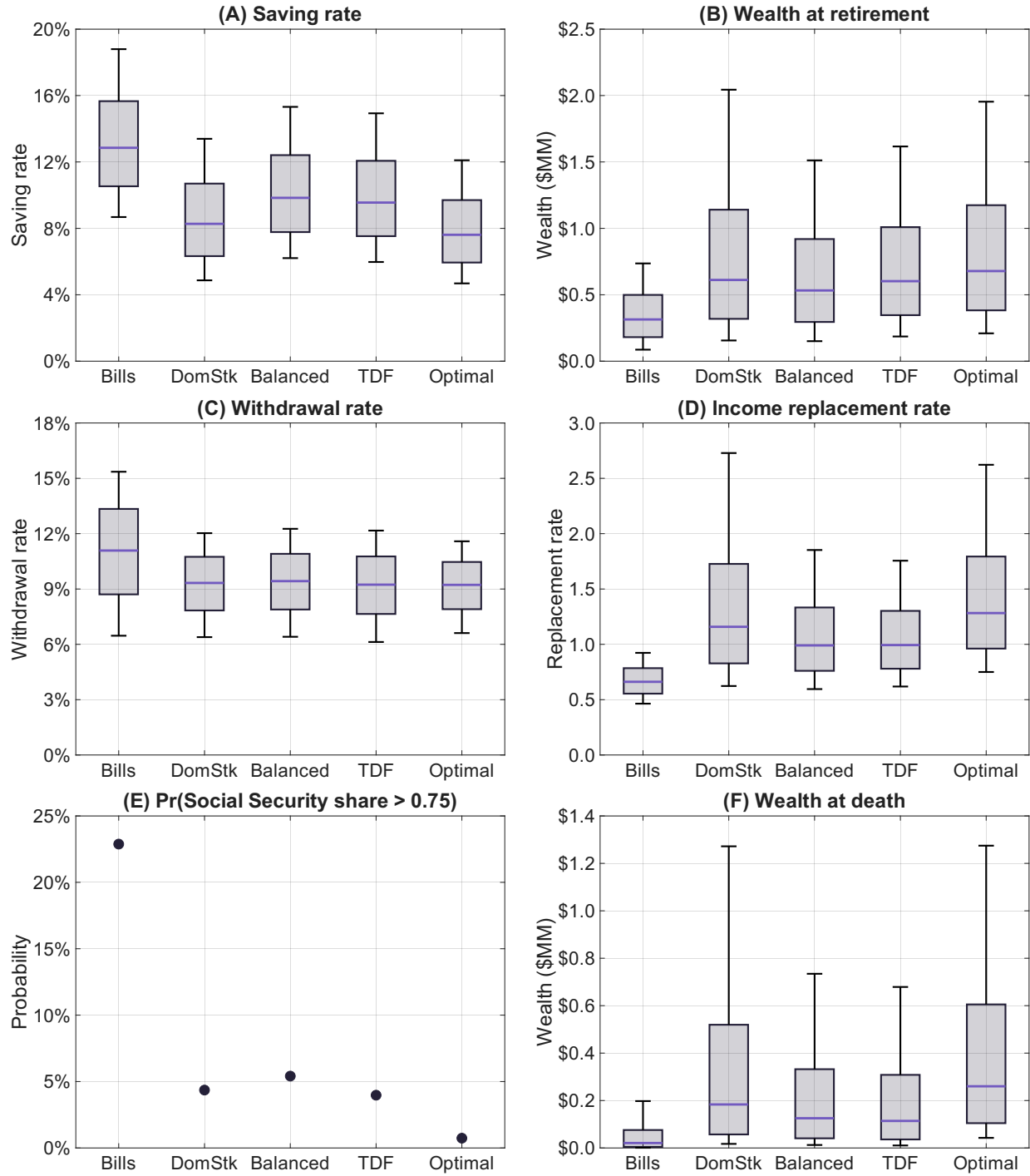


Figure 4. Measures of investment performance. The figure summarizes the distribution of the total dollar-weighted saving rate (Panel A), the distribution of real wealth at retirement (Panel B), the distribution of the withdrawal rate during retirement (Panel C), the distribution of the real income replacement rate (Panel D), the probability that Social Security income accounts for more than 75% of retirement income (Panel E), and the distribution of real wealth at death (Panel F) across 1,000,000 bootstrap simulations for households adopting the optimal age-based strategy and four benchmark asset allocation strategies. In each box-and-whiskers plot, the middle line corresponds to the median, the box covers the interquartile range, and the whiskers cover the 10th through 90th percentiles.

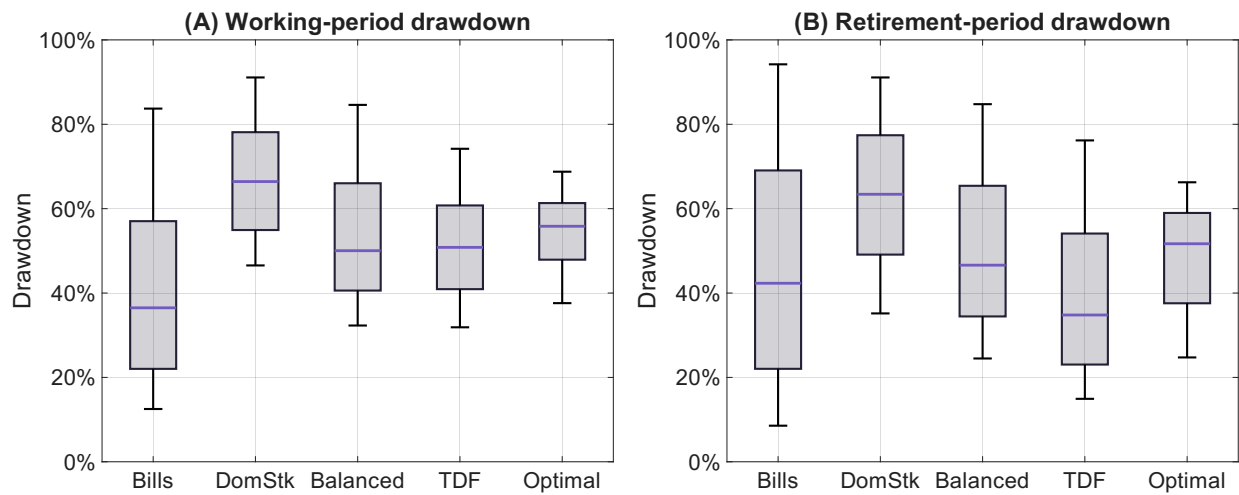


Figure 5. Portfolio drawdowns. The figure summarizes the distributions of the maximum working-period drawdowns (Panel A) and retirement-period drawdowns (Panel B) across 1,000,000 bootstrap simulations for households adopting the optimal age-based strategy and four benchmark asset allocation strategies. In each box-and-whiskers plot, the middle line corresponds to the median, the box covers the interquartile range, and the whiskers cover the 10th through 90th percentiles.

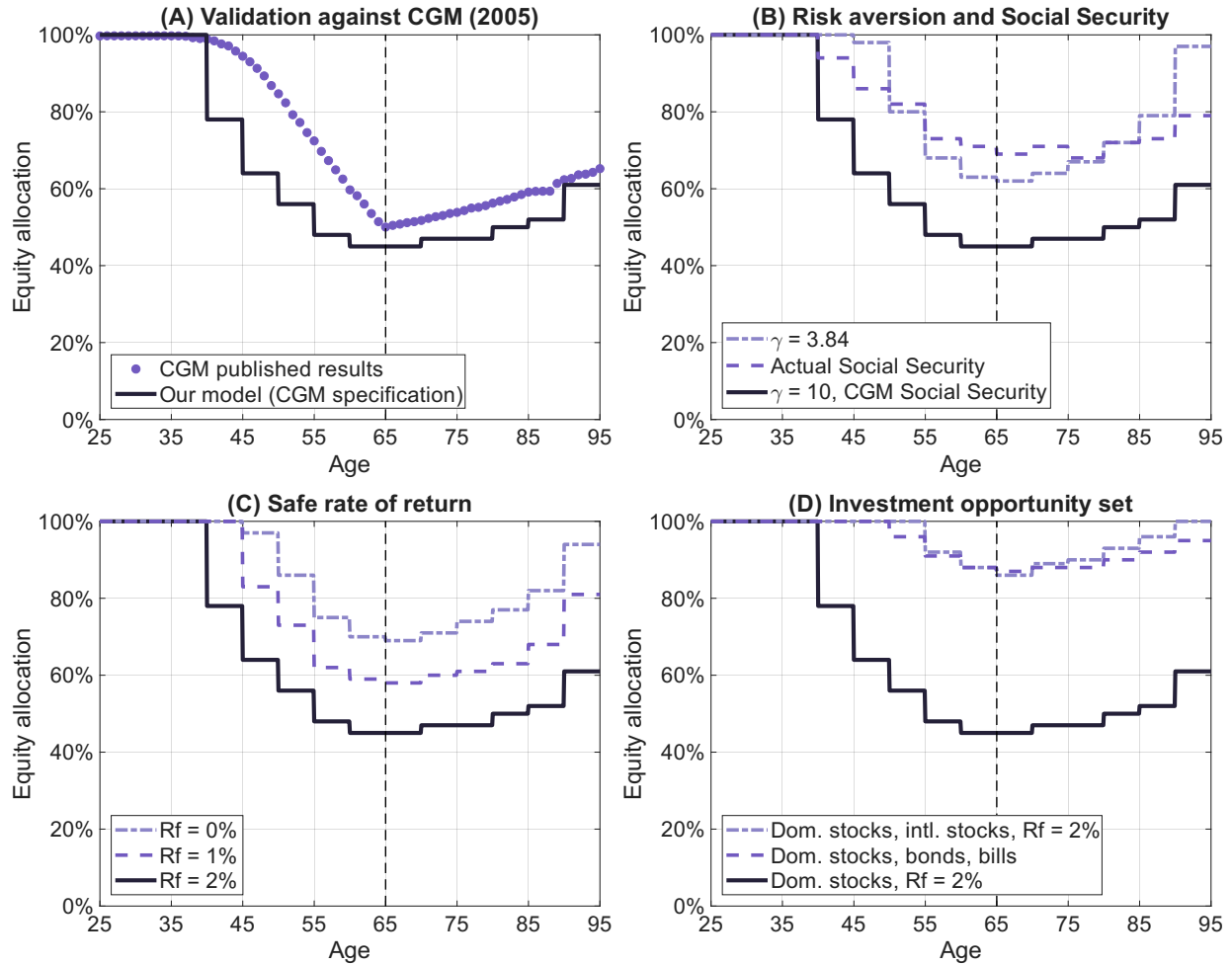


Figure 6. Reconciliation with Cocco, Gomes, and Maenhout (2005). The figure shows the total equity share in the optimal asset allocation policy as a function of household age under alternative model specifications. Panel A compares the average equity share by age reported in Figure 3(C) of Cocco, Gomes, and Maenhout (2005) with the glide path generated by our model under a specification that matches the assumptions in Cocco, Gomes, and Maenhout (2005). The specification adopts a risk aversion coefficient of $\gamma = 10$, a stylized approximation to Social Security benefits, and an investment opportunity set consisting of domestic stocks and a risk-free asset paying a guaranteed real annual return of 2%. The remaining panels illustrate the sensitivity of the optimal glide path to alternative assumptions regarding risk aversion and Social Security (Panel B), the risk-free rate of return (Panel C), and the investment opportunity set (Panel D). In each panel, the vertical dashed line indicates the household's exogenous retirement age.

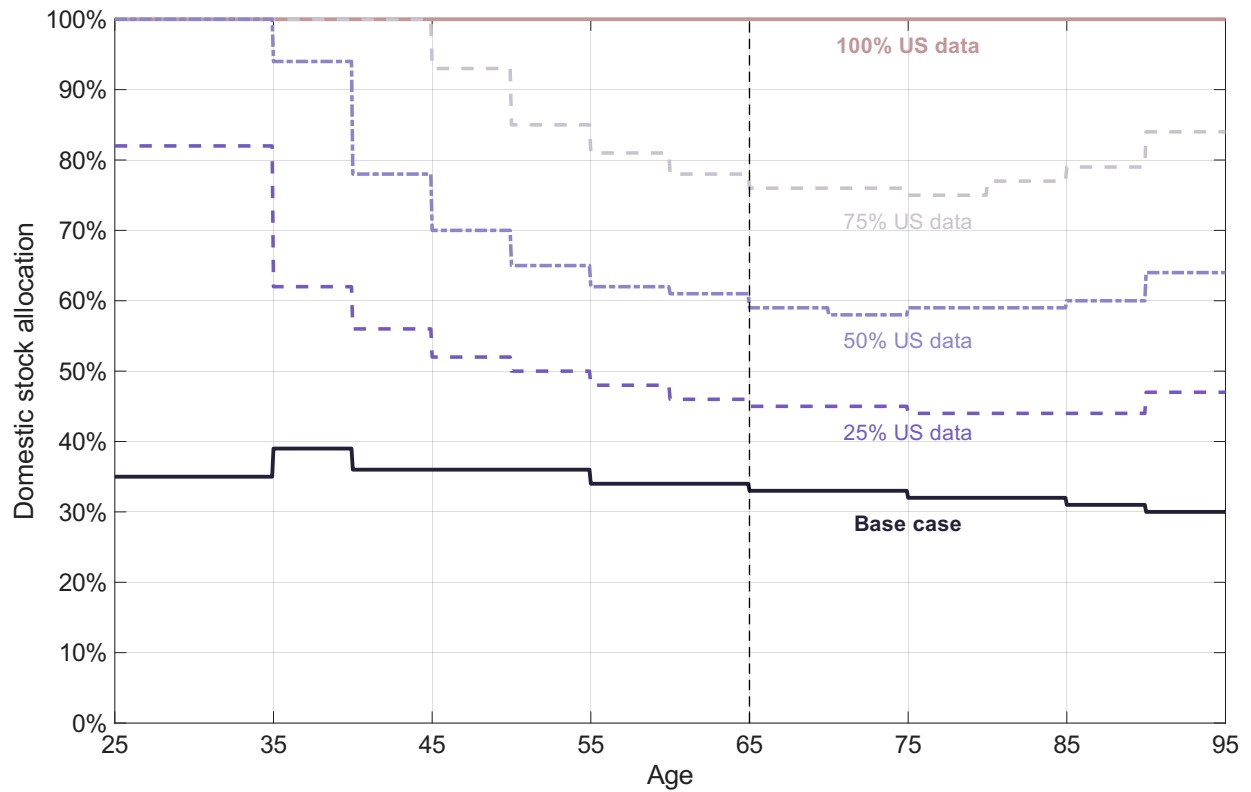


Figure 7. Optimal age-based asset allocation policies under alternative subjective beliefs. The figure shows the portfolio weight in domestic stocks under the optimal asset allocation policy as a function of household age under alternative subjective beliefs regarding the extent to which historical US returns and historical developed-country returns reflect the forward-looking return distribution. The base case corresponds to subjective beliefs estimated from the developed-country sample. The remaining cases assume that 25%, 50%, 75%, or 100% of the 1,000,000 simulated lifetimes draw returns from the historical US sample, with the remainder from the developed-country sample. The vertical dashed line indicates the household's exogenous retirement age. In all cases, the optimal policy invests entirely in domestic and international stocks, assigning zero weight to bonds and bills at all ages.

Internet Appendix

“Beyond the Status Quo: A Critical Assessment of Lifecycle Investment Advice”

A Data

This appendix describes our development classification approach, data sources, return calculations, special data issues, and dataset validation. Section A.1 outlines our development classification and the data sources used to compute asset class returns. Sections A.2–A.5 provide details on the calculations of returns for domestic stocks, international stocks, government bonds, and government bills, respectively, and special data issues related to each asset class. Section A.6 presents data construction details for other variables. Section A.7 compares our data on stock and bond returns with data from alternative sources.

A.1 Development classification and data sources

We follow Anarkulova, Cederburg, and O’Doherty (2022) to classify countries as developed. We classify a given country as developed early in the sample period if its agricultural labor share is less than 50% based on evidence about labor patterns from the economics literature [e.g., Kuznets (1973)]. Beginning with the formation of the Organisation for European Economic Co-operation (OEEC) in 1948, we use membership in the OEEC and the Organisation for Economic Co-operation and Development (OECD) to identify development dates.

To form a balanced panel, a given developed country is not included in our sample until its government issues ten-year bonds. Sample eligibility postdates development for several countries on this basis. The sample eligibility date is the latest of 1890 (i.e., the sample period start date for our study), the country development year, and the year in which the country first issued long-term bonds.

Table A.I displays the development date, reason for classification, sample eligibility date, and data coverage for each country. Three countries are reclassified from developed to developing during the sample period. These instances occur in Argentina, Chile, and Czechoslovakia, and each reclassification results from substantial changes in governments and markets in these countries. Chile, the Czech Republic, and Slovakia reenter the sample with membership in the OECD. We include the early periods in these countries to avoid survivor bias.

For some countries, we have missing data at the beginning of the eligible period. Returns on a diversified domestic stock index are the binding data constraint in each of these instances. No country has data gaps in the middle or at the end of its series.

The primary data source for our study is the GFDatabase from Global Financial Data (GFD). The database contains long time series of total return indexes, price indexes, dividend-price ratios, and total market capitalization for stocks; yields for ten-year government bonds and short-term bills; consumer price indexes; and exchange rates for a broad set of countries. Table A.II reports the data series that we use to compute monthly stock, bond, and bill returns for each country. As described in the footnotes to Table A.II, we supplement the data from GFD with data from other sources.

A.2 Domestic stocks

The GFDDatabase contains data for total return indexes, price indexes, and dividend-price ratios. It includes stock market indexes that are created and calculated by stock exchanges (e.g., the Tokyo Stock Price Index from the Tokyo Stock Exchange), by well-known index providers (e.g., the S&P 500 Index), or by GFD directly from original source documents. Multiple stock indexes are available in the database for some countries and periods. We select a single index in these cases by considering the breadth of market coverage and the length of historical coverage. We use a total return index whenever one is available, and we otherwise use a price index and a dividend-price ratio to calculate returns.

For sample months in which a total return index is available, we calculate the monthly nominal return,

$$R_{i,t}^{Nominal\ stocks} = \frac{I_{i,t}^{Total}}{I_{i,t-1}^{Total}}, \quad (A1)$$

where $I_{i,t}^{Total}$ is the total return index for country i at the end of month t and $R_{i,t}^{Nominal\ stocks}$ is the gross nominal return for country i in month t . If no total return index is available, we use price index and dividend-price ratio data to calculate returns. We assume that the annual dividend reflected by the reported dividend-price ratio is paid equally across months in the year. If $I_{i,t}^{Price}$ is the price index and $\hat{D}_{i,t}$ is the estimated dividend (appropriately scaled to the level of the price index) for country i in month t , then we calculate the monthly nominal return,

$$R_{i,t}^{Nominal\ stocks} = \frac{I_{i,t}^{Price} + \hat{D}_{i,t}}{I_{i,t-1}^{Price}}. \quad (A2)$$

Nominal returns reflect diversified investments in a broad country-level index. To calculate real returns, we first calculate gross inflation,

$$\Pi_{i,t} = \frac{I_{i,t}^{CPI}}{I_{i,t-1}^{CPI}}, \quad (A3)$$

where $I_{i,t}^{CPI}$ is the consumer price index (CPI) for country i at the end of month t . We then calculate the gross real return on domestic stocks,

$$R_{i,t}^{Domestic\ stocks} = \frac{R_{i,t}^{Nominal\ stocks}}{\Pi_{i,t}}. \quad (A4)$$

This return calculation produces real returns that are denominated in the local currency of country i .

A.2.1 Data issues related to domestic stocks

Our treatments of special data issues mirror those in Anarkulova, Cederburg, and O'Doherty (2022) with minor exceptions. Details on the data adjustments required to compute nominal and real stock returns for our developed country sample are available in Anarkulova, Cederburg, and O'Doherty (2022) and the corresponding internet appendix. Additional adjustments are described below.

We measure returns that are denominated in the primary home currency with one exception. Our real returns for Germany are denominated in gold marks (rather than paper marks) for the 1917 to 1923 period. Extraordinary hyperinflation during this period complicates the calculation of

real returns based on nominal returns in paper marks, and the GFDdatabase reports a series of stock market returns denominated in gold marks.

The internet appendix for Anarkulova, Cederburg, and O'Doherty (2022) outlines the smoothing procedures used to fill gaps in return series for short periods in a few sample countries. In addition to those cases, we apply a smoothing procedure to convert five- and seven-month return data for Austria over the period from January 1920 to February 1922 and quarterly return data for Belgium over the period from May 1919 to January 1926 into time series of monthly returns. In particular, we assume constant monthly returns within each period. We apply the same procedure to quarterly return data for Switzerland from February 1914 to July 1914 and from August 1916 to January 1921. Czechoslovakia is missing return data for July 1921. We estimate returns for July 1921 and August 1921 using price index data for June 1921 and August 1921 under the assumption of equal monthly returns in July and August of 1921.

Our treatment of multi-month return observations associated with stock market disruptions and closures differs from that in Anarkulova, Cederburg, and O'Doherty (2022). There are 35 instances in which stock exchanges closed for extended periods, typically as the result of a major war, political revolution, or banking crisis. Investors tend to earn negative real returns in these periods, such that omitting countries or periods because of these stock return data gaps induces an easy data bias. Table A.III reports cases of exchange closures or heavily restricted trading during our sample period along with the corresponding nominal and real returns. The bootstrap procedure in Anarkulova, Cederburg, and O'Doherty (2022) treats each of these events as a single return observation covering a multi-month period. This treatment reflects that most investors would have been unable to trade during these periods, such that they could only wait for the eventual realizations of the longer-period returns. Such treatment is not ideal for our multi-asset analysis, which requires a balanced panel of monthly asset returns for each country. At the same time, we need the data to reflect the economic outcomes of stock market investors.

In our current approach to handling multi-month returns, we take the perspective of an investor in a hypothetical fund attempting to track the market index for a given country. Although this investor could not directly liquidate her stock holdings via exchange trades during times of market closure, she could sell her shares in the hypothetical fund. The fund's managers, in turn, could either rely on black market data for valuation purposes or estimate the event's impact on asset prices at the beginning of the closure period. Based on this perspective, we apply one of two approaches to handling multi-month returns:

1. For events during which GFD provides black market prices, we use these values to estimate stock market index returns.
2. For events without corresponding data in GFD, we assign the total multi-month real return to the first monthly observation and zero real return to the remaining monthly observations.

The three exceptions to this general approach correspond to Austria's 113-month return from July 1939 to November 1948, Switzerland's 24-month return from August 1914 to July 1916, and Czechoslovakia's 26-month return from April 1943 to May 1945. For Austria, GFD reports limited black market data in January 1943, April 1946, and from November 1946 to November 1948. We use these intermittent values and assign the remaining part of the total real return to July 1939. Similarly, for Switzerland, we use GFD's black market data in January 1916 and July 1916 and assign the remaining part of the total real return to August 1914. For Czechoslovakia, the April 1943 to May 1945 period corresponds to an episode that starts with severe trading restrictions and price controls and ends with the permanent stock exchange closure in Prague on May 5, 1945. For this period, we assign a terminal nominal return of -90.00% to May 1945 and zero nominal returns

to the other months. This treatment of Czechoslovakia's multi-month return is consistent with the economic experience of investors over this period, as detailed in Anarkulova, Cederburg, and O'Doherty (2022).

A.3 International stocks

We calculate real returns on an international stock portfolio from the perspective of an investor in a developed country. For each country, the international stock portfolio is a weighted investment across all developed stock markets excluding the local stock market. The international stock portfolio is value weighted by total market capitalization, and the returns are expressed in the domestic currency such that they reflect the exchange rate risk incurred by investing in assets denominated in foreign currencies.

The return calculation for international stocks uses the gross nominal stock market returns described in Section A.2. We convert the nominal return for each country $j \neq i$ into a real return denominated in the domestic currency of country i and calculate the weighted average across countries $j \neq i$,

$$R_{i,t}^{International\ stocks} = \sum_{j \neq i} w_{j,t-1} \frac{R_{j,t}^{Nominal\ stocks}}{\Pi_{i,t}} \left(\frac{E_t^{i,j}}{E_{t-1}^{i,j}} \right), \quad (A5)$$

where $E_t^{i,j}$ is the exchange rate at the end of month t expressed in country i 's currency per country j 's currency, $w_{j,t-1}$ is country j 's weight in the international stock portfolio in month t ,

$$w_{j,t-1} = \frac{M_{j,t-1}}{\sum_{j \neq i} M_{j,t-1}}, \quad (A6)$$

and $M_{j,t-1}$ is the total market capitalization for the stock market in country j at the end of month $t-1$ expressed in USD.

A.4 Bonds

We calculate bond returns using monthly data on bond yields. For comparability across countries and periods, we focus on ten-year government bonds. The GFDDatabase has variables for ten-year bond yields for most countries and periods in our sample, and we supplement these data to achieve full data coverage.

We first estimate ten-year bond prices given bond yields. We assume that each bond has exactly ten years to maturity, semiannual coupons, and a coupon rate equal to the greater of the bond yield and zero at the end of month $t-1$. We then reprice the bond at the end of month t given the month- t yield and the one month shorter maturity. We calculate the gross nominal return,

$$R_{i,t}^{Nominal\ bonds} = \frac{P_{i,t}}{P_{i,t-1}}, \quad (A7)$$

where $P_{i,t}$ is the calculated dirty bond price (i.e., inclusive of accrued interest) for country i at the end of month t . Finally, we calculate the gross real bond return,

$$R_{i,t}^{Bonds} = \frac{R_{i,t}^{Nominal\ bonds}}{\Pi_{i,t}}. \quad (A8)$$

The return calculation requires assumptions about the maturity and the coupon rate of the underlying bond. We validate this calculation in Section A.7 by comparing our calculated returns with returns from Datastream over the period of overlap between the two data samples. Our return calculations are highly correlated with and have similar moments to those from Datastream.

Sections A.4.1–A.4.8 describe several issues related to the underlying bond yield data.

A.4.1 Bond data availability

For several countries in our sample, there are no ten-year government bonds in circulation at the time the country is initially classified as developed. For example, ten-year government bonds are first issued in Iceland in 1992, Singapore in 1998, Hungary in 1999, Poland in 1999, the Czech Republic in 2000, South Korea in 2000 [Kang, Kim, and Rhee (2005)], Mexico in 2001 [Jeanneau and Pérez Verdia (2005)], and Türkiye in 2010.²⁷ These circumstances create gaps between the development dates and the sample eligibility dates for these countries.

Estonia issued its only domestic bond in 1993, and all tranches were redeemed by 2004.²⁸ As a result, Estonia is excluded from our sample because the country has no domestic bond data for the developed period.

A.4.2 Data gaps and errors

Table A.IV shows periods over which we are missing monthly bond yields. In these cases, we use a smoothing procedure to fill gaps in the monthly bond return series. The procedure uses the country-level yields immediately before and after the missing observations to produce a series of constant monthly returns across a given period.

Because there are no data from GFD or alternative sources, we use the last non-missing yield of 4.33% in June 1944 to fill the gap in bond yields for Czechoslovakia from July 1944 to May 1945.

We adjust an apparent error in the GFD bond yield data for Switzerland. The source for the GFD data is the Swiss National Bank. In comparing the GFD data with the Swiss National Bank data, however, the yields match only through December 1941. The Swiss National Bank reports yields of 3.11% in January 1942, 3.14% in February 1942, 3.12% in March 1942, and 3.08% in April 1942. GFD reports yields of 3.14%, 3.12%, and 3.07% for January through March 1942. From April 1942 to December 1990, the GFD data lead the Swiss National Bank data by one month. We adjust the GFD data by entering a 3.11% yield for January 1942 and shifting the original GFD data from January 1942 to November 1990 so that it covers February 1942 to December 1990.

A.4.3 Merging multiple sources

As shown in Table A.II, constructing a series of bond returns for a given country often requires us to combine yield data from multiple sources. We make additional adjustments when linking the data series for several sample countries. The GFD data for Chile end in March 2015, and we use data from Federal Reserve Economic Data (FRED) from April 2015 to December 2023. GFD reports

²⁷See http://www.lanamal.is/asset/12732/special-report-markadsvidskipti_agust-2019.pdf for Iceland, <https://eservices.mas.gov.sg/statistics/fdanet/BenchmarkPricesAndYields.aspx> for Singapore, <https://www.elibrary.imf.org/view/journals/001/2008/174/article-A001-en.xml> for Hungary, <https://www.gov.pl/web/finance/transaction-database> for Poland, https://www.cnb.cz/export/sites/cnb/en/about_cnb/.galleries/performance/annual_reports/download/vz_2000_en.pdf for the Czech Republic, and <https://www.tcmb.gov.tr/wps/wcm/connect/EN/TCMB+EN/Main+Menu/Statistics/Markets+Data/Treasury+Auction/> for Türkiye.

²⁸See <https://www.riigiteataja.ee/et/akt/768065>.

a yield of 2.23% for March 2015, whereas the yields from FRED are 4.34% for March 2015 and 4.49% for April 2015. Merging these data series without adjustment would result in a return calculation of -17.76% for April 2015. This return likely provides a poor characterization of investment outcomes, given the relative stability in yields in the FRED data. To address this issue, we use March 2015 and April 2015 yields from FRED to compute the April 2015 bond return. We make an analogous adjustment for Iceland in March 2004, as well as for Lithuania, Luxembourg, Poland, Portugal, Slovakia, and Türkiye in the corresponding months of 2023 when we merge data from GFD and FRED.

A.4.4 Alternative bond return calculations

Our primary bond return calculations use yield data and assume that the coupon rate equals the bond yield for a hypothetical new ten-year bond. In the cases described below, we use an alternative approach of separately measuring the capital gain and the coupon income due to data availability. We use data on current yields and coupon rates from the Central Bank of Argentina to infer bond prices for each month end from January 1947 to December 1966. We compute the capital gain based on the change in bond price and add one month of coupon income based on the 3% coupon rate from February 1947 to July 1960 and the 8% coupon rate from August 1960 to December 1966. We use London quotes from the International Center for Finance at Yale for Chile (December 1926 to September 1929) and Czechoslovakia (April 1922 to January 1927). We compute monthly bond returns based on price changes and monthly coupon income at the coupon rate of 4.5% for Chile and 8.0% for Czechoslovakia. Similarly, we compute monthly bond returns based on price changes and monthly coupon income at the coupon rate of 4% for Chile from April 1965 to December 1970. We use bond price data from the Central Bank of Chile over this period.

A.4.5 Bond conversion in Argentina

Argentina issued a 3% bond in 1955. In August 1960, the government allowed for a voluntary conversion of these old bonds to new 8% bonds. The conversion was favorable for bondholders, as they could receive bonds with higher interest payments. According to Duggan (1963), the 3% bonds were exchanged at 79 pesos for the nominal value of 100 pesos. Because the terms of the conversion were favorable, the majority of existing bondholders took the offer. In constructing our bond series for Argentina, we assume conversion at the 79:100 rate. We compute the price change and multiply by 0.79 to reflect the conversion when computing the capital gain for August 1960, and we add one month of coupon income at the 8% coupon rate to calculate the return.

A.4.6 Bond default in Greece

The bond return calculation must be adjusted in the event of a default or bond exchange that produces a change in par value. Defaults on domestic sovereign bonds are rare relative to external defaults, particularly for developed countries [Reinhart and Rogoff (2011)]. Rather, inflation is a more commonly used tool for eroding the real value of debt.

A notable event that produced a change in par value is the Greek default in 2012. Greece undertook a debt exchange in March 2012 in which creditors exchanged their existing bonds for a package of new government securities with a lower face value. Zettelmeyer, Trebesch, and Gulati (2013) provide an issue-by-issue estimate of the haircut for existing bondholders. We use the 53.8% haircut estimate for the bond with maturity closest to ten years. The ten-year bond yield declined substantially from 36.6% to 21.0% in March 2012, such that our calculation based on bond

yields produces a nominal net return of 67.1%. Our calculation of the nominal gross return that incorporates the haircut is $1.671 \times (1 - 0.538) = 0.772$ to produce a nominal net return of -22.8% for ten-year bonds in March 2012.

A.4.7 Germany in 1919 to 1924 and 1948

To maintain consistency with our treatment of stock returns in Germany in the inflationary period from 1917 to 1923, we also compute bond returns in gold marks. We use bond prices in paper marks from Fischer (1923, 1924, 1925) and convert paper mark prices to gold marks by using the USD exchange rate because the United States was on the gold standard during that period. The change in gold mark bond prices provides an estimate of the capital appreciation of the bonds. We compute the total bond return by including interest payments based on the 3% coupon rate of the bonds. We use this approach from February 1919 to January 1924.

Germany exchanged Reichsmarks for Deutschemarks in June 1948. For government bonds, the exchange was 10:1 [Schnabl (2019)]. To reflect the economic value of the currency exchange, we adjust the bond price at the end of June 1948 by dividing the price of the bond by ten. The resulting nominal bond return in June 1948 is -90.0% .

A.4.8 International bonds

For one robustness exercise described in Section 5.7, we add international bonds to the base case investment opportunity set of domestic stocks, international stocks, domestic bonds, and domestic bills. We compute international bond returns analogously to the approach for international stocks in equation (A5), with the exception that the international bond portfolio is equally weighted across countries. This approach accounts for exchange rate risk incurred by investing in foreign assets. For most country-periods, we assume that domestic and foreign bondholders earn the same return in the bond's original currency. This assumption breaks down during sovereign default episodes, when foreign private creditors typically face suspended or reduced coupon payments and ultimately receive less than full repayment at restructuring.

To identify the relevant default episodes, we compare the sovereign external default spell database of Graf von Luckner, Meyer, Reinhart, and Trebesch (2025) to our sample of countries and years. Graf von Luckner, Meyer, Reinhart, and Trebesch (2025) define a default spell as beginning with a missed debt payment or distressed debt exchange with foreign private creditors and ending when a restructuring resolves the default for at least 24 consecutive months without a relapse. The comparison identifies eight episodes requiring additional attention: Austria (1914 to 1923 and 1938 to 1952), Chile (1931 to 1948), Czechoslovakia (1939 to 1946), Germany (1938 to 1953), Greece (2011 to 2012), Italy (1940 to 1947), and Japan (1941 to 1952). For Greece, both domestic and foreign private creditors participated in the March 2012 debt restructuring, so we assume that the domestic return series already captures the foreign bondholder experience and no further adjustment is needed.

For the remaining seven episodes, we construct monthly bond returns using the externally traded sovereign bond price and coupon data of Meyer, Reinhart, and Trebesch (2022). For each episode, we select a single representative bond, favoring issues with the best price coverage over the spell and a remaining maturity close to ten years at the spell start. If a bond's month-end price is missing, we carry forward the most recently observed price. The monthly gross nominal return for country j during a default episode is

$$R_{j,t}^{\text{Nominal bonds}} = \frac{P_{j,t} + \phi_{j,t} C_{j,t}}{P_{j,t-1}}, \quad (\text{A9})$$

where $P_{j,t}$ is the end-of-month price per 100 face value of the representative bond for country j in month t , $C_{j,t}$ is the scheduled monthly coupon payment per 100 face value, and $\phi_{j,t} \in [0, 1]$ is the fraction of the scheduled coupon actually paid in month t as reported by Meyer, Reinhart, and Trebesch (2022). Outside of default episodes, returns are computed from the domestic yield series as in equation (A7). The international bond portfolio return for country i is the equal-weighted average of $R_{j,t}^{Nominal\ bonds}$ across all $j \neq i$, with the appropriate currency and exchange rate adjustments. Because the representative bonds are denominated in GBP or USD, the currency conversions for these bonds use the exchange rate between country i 's currency and GBP or USD as appropriate for country j .

The representative bonds for the default episodes are as follows:

- Austria, 1914–1923: 4.5% Treasury Notes of 1914 (GBP)
- Austria, 1938–1952: 4.5% Bonds of 1934 (GBP)
- Chile, 1931–1948: External 6% Bonds of 1926, due 1960 (USD)
- Czechoslovakia, 1939–1946: 8% Bonds of 1922/1924, due 1960 (GBP)
- Germany, 1938–1953: 5% (5.5) Sterling International Loan of 1930, due 1965 (GBP)
- Italy, 1940–1947: External 7% Gold Loan of 1925, due 1951 (USD)
- Japan, 1941–1952: 6% Bonds of 1924, due 1959 (GBP)

A.5 Bills

We estimate bill returns using short-term yields and rates. For most countries and periods, the GFDDatabase provides yield data on short-term (typically three-month) government bills. When these data are missing, we next use central bank rates when available and then interbank rates from the GFDDatabase. We supplement these data with hand-collected, short-term rates from original source documents to achieve full coverage. We convert annual nominal rates on bills into monthly nominal returns denoted by $R_{i,t}^{Nominal\ bills}$ and then calculate real returns,

$$R_{i,t}^{Bills} = \frac{R_{i,t}^{Nominal\ bills}}{\Pi_{i,t}}. \quad (A10)$$

We compute monthly nominal bill returns from annual yields or rates as

$$R_{i,t}^{Nominal\ bills} = (R_{i,t-1}^{Annual\ rate})^{1/12}, \quad (A11)$$

where $R_{i,t-1}^{Annual\ rate}$ is the annualized short-term government bill yield, central bank rate, or interbank rate reported at the end of month $t - 1$.

A.5.1 Data issues related to bills

For a few periods, there are no bill data from GFD or alternative sources, requiring us to make assumptions to fill these gaps. For Canada, we use a yield of 5.75% for the seven-month period from January 1914 to July 1914. This value is an average of the 6.50% interbank rate for December 1913 from GFD and the 5.00% advance rate for August 1914 from Shearer and Clark (1984). The Netherlands is missing data for February 2014, so we average the short-term government bill yields from GFD of 0.09% for January 2014 and 0.13% for March 2014.

For New Zealand from January 1896 to December 1914, we use short-term yields on bills held by the Post Office Savings Bank Fund. The Post Office Savings Bank Fund did not hold Treasury

bills in 1913, so we are missing data for that year. The yields are 3.00% in December 1912 and 4.00% in January 1914, and we use the average of 3.50% to fill the data gap. We are also missing yield data for New Zealand from January 1915 to December 1919. The yields for December 1914 and January 1920 are both 4.00%, so we assume a 4.00% yield throughout the intervening period with missing data.

A.6 Other variables

We follow the data adjustments noted in Anarkulova, Cederburg, and O’Doherty (2022) and the corresponding internet appendix to estimate country-level inflation and exchange rate changes.

The data for market capitalization are from GFD. These series are typically reported at an annual frequency. Because market capitalization is reported in USD, we fill data gaps by interpolating changes in proportion to USD nominal stock index returns. We use market capitalization series for Germany from 1917 to 1923 that are denominated in gold marks rather than paper marks. This approach is consistent with the calculation of the total return index for domestic stocks for Germany over this period. We fill the 2023 data gap in GFD’s market capitalization series for the UK using an alternative source.²⁹

Table A.V shows dividend-price ratio data for each country in the sample. We use annual dividend-price ratio data. We rely on external sources to calculate dividend-price ratios for Slovakia and Latvia. In both cases, GFD does not contain sufficient information to compute these ratios. For Slovakia, we use dividend-price ratio data from the Bratislava Stock Exchange’s official website.³⁰ For Latvia, we calculate dividend-price ratios using data on total dividends paid by companies from Nasdaq Baltic and data on total market capitalization from GFD.³¹ We use dividend-price ratios from Jordà, Knoll, Kuvshinov, Schularick, and Taylor (2019) for Portugal from 1934 to 1987.

We follow Anarkulova, Cederburg, and O’Doherty (2022) to fill the data gaps in GFD dividend-price ratio data for Chile from January 1967 to December 1970 and Czechoslovakia from April 1938 to March 1943. For Chile, we fill the data gap with a 7.0% yield based on the dividend yield observation in December 1966. The dividend yield in Czechoslovakia fluctuates between 1.4% and 2.6% in the three years before the break in the data, so we assume a 2.0% dividend yield for the missing observations. Table A.VI shows additional periods over which annual dividend-price ratio data are missing in the GFD database. We estimate dividend-price ratios for these missing periods using the methods described in the table.

A.7 External validation tests

This section presents external validation tests for our stock and bond return data.

A.7.1 Comparison of stock data from GFD and Jordà et al. (2019)

Anarkulova, Cederburg, and O’Doherty (2022) compare their data on stock returns from GFD with the stock returns from the overlapping periods in Jordà, Knoll, Kuvshinov, Schularick, and Taylor (2019). They find that the country-level samples from these two sources have very similar average returns, standard deviations, and extreme returns. They also show that the return correlation across the two sources exceeds 0.90 for nearly all countries. Given that our approach to constructing

²⁹The data are available at <https://www.ceicdata.com/en/indicator/united-kingdom/market-capitalization>.

³⁰See <https://www.bsse.sk/bcpb/en/market-data/statistics/monthly/>.

³¹The dividend data for Latvia are available at <https://nasdaqbaltic.com/statistics/en/statistics>.

country-level stock returns closely follows the approach in Anarkulova, Cederburg, and O'Doherty (2022), these tests also provide external validation of our stock data.

A.7.2 Comparison of bond data from GFD and Datastream

As described in Section A.4, we calculate bond returns using bond yield data from GFD and other sources. In this section, we perform an external validation exercise by comparing our bond returns with those from Datastream over the periods and countries for which they are available. The analysis serves both to ensure that our approach to converting bond yields to returns is empirically accurate and to assess whether our bond return data and those from a popular alternative source exhibit common characteristics.

Table A.VII shows results from the external validation analysis. The table reports statistics for real returns. Our sample overlaps with Datastream for 27 countries. Datastream data begin in 1989 for several countries and more recently for others. The table reports the sample size, the arithmetic and geometric means, standard deviation, and minimum and maximum returns for our data, the corresponding statistics for the Datastream data, and the correlation between our returns and those from Datastream.

Table A.VII indicates a close correspondence between our bond return data and those from Datastream. For nearly all countries, the means, standard deviations, and extreme returns are highly similar across the two data sources. The return correlations are above 0.90 for 24 of the 27 countries. Only Hungary, Mexico, and Singapore have correlations below 0.90. Of the 24 countries with high correlations, Greece is unique in Table A.VII as the only country with economically significant differences in the remaining summary statistics. We discuss these four exceptions below.

Hungary and Singapore appear to be the simplest cases. We examine bond yields and returns across the two datasets. The GFDDatabase and Datastream bond yields differ, sometimes substantially, for these two countries. To reconcile the differences, we collect ten-year historical bond yield data from the Magyar Nemzeti Bank (the central bank of Hungary) and the Monetary Authority of Singapore.³² For Hungary, the correlation between yield changes in the central bank data and our data is near one, whereas the correlation between yield changes in the central bank data and the Datastream data is much lower. For Singapore, the GFDDatabase and Singaporean government data exactly match. The large deviations between Datastream and these other sources primarily occur in the first seven months of the sample, and the reported returns in Datastream imply changes in yields that are not reflected in the data from the Monetary Authority. Excluding the first seven months, the correlation between returns in our data and Datastream is 0.98. Our data appear reliable for these countries.

The bond yields for Mexico in our data and in Datastream are relatively similar. For several months in the sample, the reported Datastream return seems inconsistent with the reported yield change. For example, the reported yield increases by 0.08% in June 2015, but the reported return is 8.12%. We compare our calculated returns and the reported Datastream returns with the returns on the S&P/BMV Mexico Sovereign Bond Index in these months.³³ The S&P/BMV index tracks bonds with several maturities, and its duration is low compared with the other two series. Nonetheless, the returns from this index are much more consistent with our data than with Datastream. In June 2015, for example, the S&P/BMV index reports a return of -0.15% , which is close to our return calculation of -0.10% but far from the 8.12% reported return in Datastream. Given the

³²See https://statisztika.mnb.hu/statistical-topics/interest-rates_-money-and-capital-market-data and <https://eservices.mas.gov.sg/statistics/fdanet/BenchmarkPricesAndYields.aspx>.

³³See <https://www.spglobal.com/spdji/en/indices/fixed-income/sp-bmv-mexico-sovereign-bond-index/>.

consistency between the GFDatabase and the S&P/BMV index, the deviations between our data and Datastream appear to be reporting errors for returns in Datastream.

The largest deviations in bond returns for Greece are related to the Greek bond default in 2012. As discussed in Section A.4.6, we calculate a bond return in March 2012 that accounts for the bond exchange and the associated haircut. Our return calculation, which reflects information from ten-year bond yields and the default, is -22.80% in this month. This return differs substantially from the -4.16% return reported by Datastream. Our study focuses on domestic debt, so we take the perspective of a hypothetical domestic investor. Participation rates in the exchange were higher among domestic investors compared with international investors [Zettelmeyer, Trebesch, and Gulati (2013)]. We do not know the methodology underlying Datastream's return calculation for this month, but the difference could arise from a different assumption about participation in the exchange. Late in 2012, Greece announced a voluntary bond buyback to be executed in December 2012, and the buyback led to an increase in market prices [Zettelmeyer, Trebesch, and Gulati (2013)]. We observe a 4.32% decrease in bond yield in December 2012 and calculate a return of 26.12% . Datastream reports a 3.21% decrease in bond yield and reports a return of 41.47% , such that the return is much larger than that implied by the yield change. Given that the buyback occurred at prevailing market prices, our view is that any effect of the buyback should be reflected in the change in yields. The return differences for these two months account for much of the difference in average returns for Greece in Table A.VII.

Table A.I: Developed country sample periods

The table shows developed countries, initial development dates, classification reasons for development, sample eligibility details, and sample coverage. The development year classifications are based on agricultural labor share or organizational membership in the Organisation for European Economic Co-operation (OEEC) or the Organisation for Economic Co-operation and Development (OECD). Sample eligibility for a given developed country requires that the country has issued long-term government bonds. The sample start date is the later of the sample eligibility date and the first month with return data for stocks, bonds, and bills.

Country	Development details		Sample eligibility details		Sample coverage		
	Year	Reason for classification	Year	Reason for delayed sample eligibility	Start date	End date	Coverage (%)
United Kingdom	1841	Agricultural labor share	1890	Sample for study starts in 1890	1890:01	2023:12	100.0
Netherlands	1849	Agricultural labor share	1890	Sample for study starts in 1890	1914:01	2023:12	82.1
Belgium	1856	Agricultural labor share	1890	Sample for study starts in 1890	1897:01	2023:12	94.8
France	1866	Agricultural labor share	1890	Sample for study starts in 1890	1890:01	2023:12	100.0
Norway	1875	Agricultural labor share	1890	Sample for study starts in 1890	1914:01	2023:12	82.1
Germany	1882	Agricultural labor share	1890	Sample for study starts in 1890	1890:01	2023:12	100.0
Denmark	1890	Agricultural labor share	1890	n/a	1890:01	2023:12	100.0
Switzerland	1890	Agricultural labor share	1890	n/a	1914:01	2023:12	82.1
United States	1890	Agricultural labor share	1890	n/a	1890:01	2023:12	100.0
Canada	1891	Agricultural labor share	1891	n/a	1891:01	2023:12	100.0
Argentina	1895	Agricultural labor share	1895	n/a	1947:02	1966:12	27.7
New Zealand	1896	Agricultural labor share	1896	n/a	1896:01	2023:12	100.0
Australia	1901	Agricultural labor share	1901	n/a	1901:01	2023:12	100.0
Sweden	1910	Agricultural labor share	1910	n/a	1910:01	2023:12	100.0
Austria	1920	Agricultural labor share	1920	n/a	1920:01	2023:12	100.0
Chile period I	1920	Agricultural labor share	1920	n/a	1927:01	1970:12	86.3
Greece	1920	Agricultural labor share	1920	n/a	1981:02	2023:12	41.3
Czechoslovakia	1921	Agricultural labor share	1921	n/a	1922:05	1945:05	94.5
Japan	1930	Agricultural labor share	1930	n/a	1930:01	2023:12	100.0
Portugal	1930	Agricultural labor share	1930	n/a	1934:01	2023:12	95.7
Italy	1931	Agricultural labor share	1931	n/a	1931:01	2023:12	100.0
Ireland	1936	Agricultural labor share	1936	n/a	1936:01	2023:12	100.0
Singapore	1947	Agricultural labor share	1998	Long-term bonds first available in 1998	1998:07	2023:12	100.0
Iceland	1948	OEEC membership	1992	Long-term bonds first available in 1992	2002:01	2023:12	68.8
Luxembourg	1948	OEEC membership	1948	n/a	1982:01	2023:12	55.3
Türkiye	1948	OEEC membership	2010	Long-term bonds first available in 2010	2010:02	2023:12	100.0

(Continued on next page)

Table A.I (Continued)

Country	Development details		Sample eligibility details		Sample coverage		
	Year	Reason for classification	Year	Reason for delayed sample eligibility	Start date	End date	Coverage (%)
Spain	1959	OEEC membership	1959	n/a	1959:01	2023:12	100.0
Finland	1969	OECD membership	1969	n/a	1969:01	2023:12	100.0
Mexico	1994	OECD membership	2001	Long-term bonds first available in 2001	2001:08	2023:12	100.0
Czech Republic	1995	OECD membership	2000	Long-term bonds first available in 2000	2000:05	2023:12	100.0
Hungary	1996	OECD membership	1999	Long-term bonds first available in 1999	1999:02	2023:12	100.0
Poland	1996	OECD membership	1999	Long-term bonds first available in 1999	1999:06	2023:12	100.0
South Korea	1996	OECD membership	2000	Long-term bonds first available in 2000	2000:11	2023:12	100.0
Slovakia	2000	OECD membership	2000	n/a	2000:01	2023:12	100.0
Chile period II	2010	OECD membership	2010	n/a	2010:01	2023:12	100.0
Estonia	2010	OECD membership	—	No qualifying long-term bonds	—	—	—
Israel	2010	OECD membership	2010	n/a	2010:01	2023:12	100.0
Slovenia	2010	OECD membership	2010	n/a	2010:01	2023:12	100.0
Latvia	2016	OECD membership	2016	n/a	2016:01	2023:12	100.0
Lithuania	2018	OECD membership	2018	n/a	2018:01	2023:12	100.0
Colombia	2020	OECD membership	2020	n/a	2020:01	2023:12	100.0

Table A.II: Data sources

The table summarizes the data sources used to compute returns for each developed country in the sample. For each country and asset class, the table reports the underlying data series used to construct returns and the corresponding start and end dates. The data series symbols correspond to those in the GFDdatabase from Global Financial Data. For stocks and bills, the table also indicates the data series type. Stock returns are based on either total return indexes (TRI) or combinations of price indexes and dividend-price ratios (PI/DP). Bill returns are based on short-term Treasury yields (TBY), central bank interest rates (CBR), interbank interest rates (IIR), one-year government bond yields (GBY-1), deposit interest rates (DIR), interest rates on advances (IRA), overnight interest rates (OIR), or time money rates (TMR). All bond returns correspond to returns on ten-year government bonds. Additional details on alternative data sources are provided in the footnotes.

Country	Stocks				Bonds			Bills			
	Series	Type	Start	End	Series	Start	End	Series	Type	Start	End
Argentina	_IBGD, SYARGYM	PI/DP	1947:02	1966:12	3	1947:02	1966:12	IDARGD	CBR	1947:02	1966:12
Australia	_AORDAD	TRI	1901:01	2023:12	IGAUS10D	1901:01	2023:12	2	TBY	1901:01	1920:06
								IDAUSD	CBR	1920:07	1928:06
								ITAUS3D	TBY	1928:07	2023:12
Austria	_ATXTRD	TRI	1920:01	2023:12	IGAUT10D	1920:01	2023:12	IDAUTD	CBR	1920:01	1959:12
								ITAUT3M	TBY	1960:01	1990:12
								IGAUT1D	GBY-1	1991:01	2023:12
Belgium	_BCSHD	TRI	1897:01	2023:12	IGBEL10D	1897:01	2023:12	IDBELD	CBR	1897:01	1947:12
								ITBEL3D	TBY	1948:01	2023:12
Canada	_TRGSPTSE	TRI	1891:01	2023:12	IGCAN10D	1891:01	2023:12	11	DIR	1891:01	1901:12
								IMCANMOM	OIR	1902:01	1913:12
								12	IRA	1914:08	1934:02
								ITCAN3D	TBY	1934:03	2023:12
Chile period I	_IGPAD, SYCHLYM	PI/DP	1927:01	1970:12	4	1927:01	1929:09	IDCHLD	CBR	1927:01	1970:12
					5	1929:10	1930:12				
					IGCHLCM	1931:01	1956:02				
					6	1956:03	1970:12				
Chile period II	_IPSAD	TRI	2010:01	2023:12	IGCHLIM	2010:01	2015:03	ITCHL3D	TBY	2010:01	2012:09
					7	2015:04	2023:12	IDCHLD	CBR	2012:10	2023:12
Colombia	TRCOLSTM	TRI	2020:01	2023:12	IGCOL10D	2020:01	2023:12	ITCOL3W	TBY	2020:01	2023:12
Czech Republic	_PXTRD	TRI	2000:05	2023:12	IGCZE10D	2000:05	2023:12	ITCZE3D	TBY	2000:05	2017:02
								IDCZED	CBR	2017:03	2023:12
Czechoslovakia	CZINDXM, SYCZEYM	PI/DP	1922:05	1937:11	4	1922:05	1927:01	IDCZED	CBR	1922:05	1945:05
	CZINDEXM, SYCZEYM	PI/DP	1937:12	1943:03	5	1927:02	1944:06				

(Continued on next page)

Table A.II (Continued)

Country	Stocks				Bonds			Bills			
	Series	Type	Start	End	Series	Start	End	Series	Type	Start	End
Denmark	_OMXCGID	TRI	1890:01	2023:12	IGDNK10D	1890:01	2023:12	IDDNKD	CBR	1890:01	1975:12
Finland	_OMXHGID	TRI	1969:01	2023:12	IGFIN10D	1969:01	2023:12	ITDNK3D	TBY	1976:01	2023:12
								IDFIND	CBR	1969:01	1996:08
								IGFIN1D	GBY-1	1996:09	2012:01
								ITFIN1D	TBY	2012:02	2013:05
								IGFIN1D	GBY-1	2013:06	2014:07
France	TRSBF250D	TRI	1890:01	2023:12	IGFRA10D	1890:01	2023:12	IDFIND	CBR	2014:08	2023:12
								IDFRAD	CBR	1890:01	1930:12
Germany	_CDAXD	TRI	1890:01	2023:12	IGDEU10D	1890:01	2023:12	ITFRA3D	TBY	1931:01	2023:12
								IDDEUD	CBR	1890:01	1952:12
Greece	_RETMD	TRI	1981:02	2023:12	IGGRC10D	1981:02	2014:01	ITDEU3D	TBY	1953:01	2023:12
								ITGRC3D	TBY	1981:02	2023:12
Hungary	_BUXD	TRI	1999:02	2023:12	IGGRC10D	2014:02	2014:02	ITHUN3D	TBY	1999:02	2023:12
Iceland	_OMXIPID, SYISLYM	PI/DP	2002:01	2002:06	IGHUN10D	2002:01	2004:02	ITISL3D	TBY	2002:01	2013:01
Ireland	_OMXIGID	TRI	2002:07	2023:12	IGISL10D	2004:03	2023:12	IDISLD	CBR	2013:02	2023:12
Israel	_IVRTD	TRI	1936:01	2023:12	IGIRL10D	1936:01	2023:12	IDIRLD	CBR	1936:01	1969:11
Italy	_TRISRSTM	TRI	2010:01	2023:12	IGISL10IM	2010:01	2014:12	ITIRL3M	TBY	1969:12	2008:12
Japan	_BCIPRD	TRI	1931:01	2019:12	IGITA10D	1931:01	2023:12	IIR	IIR	2009:01	2023:12
Latvia	_FTITLMS, SYITAYM	PI/DP	2020:01	2023:12	IGJPN10D	1930:01	2023:12	IDITAD	CBR	1931:01	1939:12
Lithuania	_TOPXDVD	TRI	1930:01	2023:12	IGISL10IM	2015:01	2023:12	ITITA3D	TBY	1940:01	2023:12
Luxembourg	_OMXRGID	TRI	2016:01	2023:12	IGJPN10D	1930:01	2023:12	IDJPND	CBR	1930:01	1959:12
Mexico	_OMXVGID	TRI	2018:01	2023:12	IGLUX10D	1982:01	2023:03	ITJPN3D	TBY	1960:01	2023:12
Mexico	_LUXXD, SYLUXYM	PI/DP	1982:01	1984:12	IGLTU10D	2016:01	2023:12	IIR	IIR	2016:01	2023:12
Mexico	_LUXXR	TRI	1985:01	2016:11	IGLUX10D	2018:01	2023:01	IIR	IIR	2018:01	2023:12
Mexico	_1	TRI	2016:12	2023:12	IGLUX10D	2023:02	2023:12	IMLUXM	OIR	1982:01	1998:12
Mexico	_IRT	TRI	2001:08	2023:12	IGLUX10D	2023:04	2023:12	IIR	IIR	1999:01	2023:12
Mexico	_IRT	TRI	2001:08	2023:12	IGMEX10D	2001:08	2023:12	ITMEX3D	TBY	2001:08	2023:12

(Continued on next page)

Table A.II (Continued)

Country	Stocks				Bonds			Bills			
	Series	Type	Start	End	Series	Start	End	Series	Type	Start	End
Netherlands	_AAXGRD	TRI	1914:01	2023:12	IGNLD10D	1914:01	1944:08	IDNLDBD	CBR	1914:01	1940:12
					2	1944:09	1945:12	ITNLD3D	TBY	1941:01	2023:12
					IGNLD10D	1946:01	2023:12				
New Zealand	_NZGID	TRI	1896:01	2023:12	IGNZL10D	1896:01	2023:12	13	TBY	1896:01	1919:12
								13	DIR	1920:01	1922:12
								IDNZLD	CBR	1923:01	1978:02
								ITNZL3D	TBY	1978:03	2023:12
Norway	_OSEAXD	TRI	1914:01	2023:12	IGNOR10D	1914:01	2023:12	IDNORD	CBR	1914:01	1941:11
								ITNOR3D	TBY	1941:12	2023:12
Poland	_WIGD	TRI	1999:06	2023:12	IGPOL10D	1999:06	2023:03	ITPOL3D	TBY	1999:06	2023:12
					7	2023:04	2023:12				
Portugal	_IBTAD, 2	PI/DP	1934:01	1988:01	IGPRT10D	1934:01	1974:04	IDPRTD	CBR	1934:01	1988:12
	_BVLGD	TRI	1988:02	2023:12	2	1974:05	1975:12	ITPRT6D	TBY	1989:01	1999:01
					IGPRT10D	1976:01	2023:06	IDPRTD	CBR	1999:02	2001:12
					7	2023:07	2023:12	2	TBY	2002:01	2010:09
								ITPRT6D	TBY	2010:10	2022:09
								7	IIR	2022:10	2023:12
Singapore	_TFTFSTD	TRI	1998:07	2023:12	IGSGP10D	1998:07	2023:12	ITSGP3D	TBY	1998:07	2023:12
Slovakia	_SAXD	TRI	2000:01	2023:12	IGSVK10D	2000:01	2023:03	IDSVKD	CBR	2000:01	2008:12
					7	2023:04	2023:12	7	IIR	2009:01	2023:12
Slovenia	_SBITOPD, SYSVNYM	PI/DP	2010:01	2023:12	7	2010:01	2023:12	ITSVN3M	TBY	2010:01	2023:12
South Korea	TRKORSTM	TRI	2000:11	2023:12	IGKOR10D	2000:11	2023:12	IGKOR1D	GBY-1	2000:11	2023:12
Spain	_BCNPR30	TRI	1959:01	2023:12	IGESP10D	1959:01	2023:12	IDESPD	CBR	1959:01	1978:12
								ITESP12D	TBY	1979:01	2023:12
Sweden	_OMXSBI	TRI	1910:01	2023:12	IGSWE10D	1910:01	2018:12	IDSWED	CBR	1910:01	1954:12
					7	2019:01	2019:01	ITSWE3D	TBY	1955:01	2023:12
					IGSWE10D	2019:02	2023:12				
Switzerland	_SSHID	TRI	1914:01	2023:12	IGCHE10D	1914:01	1941:12	IDCHED	CBR	1914:01	1979:12
					8	1942:01	1942:01	ITCHE3D	TBY	1980:01	2023:12
					IGCHE10D	1942:02	2023:12				
Türkiye	TRRBILED	TRI	2010:02	2023:12	IGTUR10D	2010:02	2023:06	ITTUR3D	TBY	2010:02	2014:09
					9	2023:07	2023:12	IGTUR1D	GBY-1	2014:10	2023:03
								10	TBY	2023:04	2023:12

(Continued on next page)

Table A.II (Continued)

Country	Stocks				Bonds			Bills			
	Series	Type	Start	End	Series	Start	End	Series	Type	Start	End
United Kingdom	_TFTASD	TRI	1890:01	2023:12	IGGBR10D	1890:01	2023:12	IDGBRD	CBR	1890:01	1899:12
United States	_SPXTRD	TRI	1890:01	2023:12	IGUSA10D	1890:01	2023:12	ITGBR3D	TBY	1900:01	2023:12
								14	TMR	1890:01	1914:10
								IDUSAD	CBR	1914:11	1919:12
								ITUSA3CMD	TBY	1920:01	2023:12

Table footnotes:

- 1 Stock returns for Luxembourg for the period from 2016:12 to 2023:12 are from the Luxembourg Stock Exchange. See <https://www.bourse.lu/home>.
- 2 Dividend-price ratios, bond returns, and bill returns for several countries are from Jordà, Knoll, Kuvshinov, Schularick, and Taylor (2019).
- 3 Bond returns for Argentina for the period from 1947:02 to 1966:12 are based on data from the Central Bank of Argentina. See https://www.bcra.gob.ar/catalogo_de_datos/boletin-estadistico-con-series-monetarias-y-financieras-historicas/.
- 4 Bond returns for select periods in Chile and Czechoslovakia are based on London quotes from the International Center for Finance at Yale. See <https://som.yale.edu/faculty-research/our-centers-initiatives/international-center-finance/data/historical-financial-research-data/london-stock-exchange>.
- 5 Bond returns for select periods in Chile and Czechoslovakia are based on London quotes from the League of Nations reports.
- 6 Bond returns for Chile for the period from 1956:03 to 1970:12 are based on data from the Central Bank of Chile. See <https://repositoriodigital.bcentral.cl/xmlui/handle/20.500.12580/26/browse?type=dateissued>.
- 7 Bond and bill returns for several countries are based on data from Federal Reserve Economic Data (FRED) at the Federal Reserve Bank of St. Louis. See <https://fred.stlouisfed.org/>.
- 8 The bond return in Switzerland for 1942:01 is based on data from the Swiss National Bank (https://www.snb.ch/de/iabout/stat/statrep/statpubdis/id/statpub_hiszt_arch#t2). We also shift a portion of the series IGCH10D from GFD, which originally covers the period from 1942:01 to 1990:11, to cover the period from 1942:02 to 1990:12.
- 9 Bond returns in Türkiye for the period from 2023:07 to 2023:12 are based on data from <http://www.worldgovernmentbonds.com/bond-historical-data/turkey/10-years/>.
- 10 Bill returns for Türkiye for the period from 2023:04 to 2023:12 are based on data from <https://www.worldgovernmentbonds.com/bond-historical-data/turkey/3-months/>.
- 11 Bill returns for Canada for the period from 1891:01 to 1901:12 are based on interest rates on deposits in government savings banks. See p. 363 of <https://www66.statcan.gc.ca/eng/1901-eng.htm>.
- 12 Bill returns for Canada for the period from 1914:08 to 1934:02 are based on data from Shearer and Clark (1984).
- 13 Bill returns for New Zealand for the period from 1896:01 to 1922:12 are based on data from the annual New Zealand Official Year-book. See, e.g., https://www3.stats.govt.nz/New_Zealand_Official_Yearbooks/1896/NZOYB_1896.html.
- 14 Bill returns for the United States for the period from 1890:01 to 1914:10 are based on data from Macaulay (1938).

Table A.III: Multi-month stock returns

The table reports periods of multi-month stock returns associated with exchange closures and describes our approach to converting each observation into a series of monthly returns. For each multi-month return observation, the table reports the number of months, the start and end dates of the period, the nominal and real net stock returns over the period, and the adjustment method. Under adjustment method 1, we use alternative data sources from GFD (e.g., black market trading data) to fill in a complete series of monthly returns. Under adjustment method 2, we assign the full multi-month real return to the first month of the period and assign zero real returns to the remaining months. Cases requiring custom treatment are discussed in Section A.2.1. Panels A and B show events associated with World War I and World War II, respectively. Panel C shows periods with revolutions, Panel D shows financial and banking crises, and Panel E shows labor strikes.

Country	Months	Start date	End date	Nominal return (%)	Real return (%)	Adjustment
Panel A: World War I						
Australia	6	1914:08	1915:01	-0.45	-0.39	Method 1
Belgium	52	1914:08	1918:11	25.12	-55.91	Method 2
Canada	7	1914:08	1915:02	1.38	-3.59	Method 1
Denmark	4	1914:08	1914:11	0.72	-0.27	Method 1
France	6	1914:08	1915:01	-10.89	-27.54	Method 1
Germany	42	1914:08	1918:01	20.03	-38.87	Method 1
Netherlands	7	1914:08	1915:02	-1.23	-3.50	Method 1
Norway	3	1914:08	1914:10	-3.80	-4.36	Method 2
Sweden	4	1914:08	1914:11	-5.91	-8.96	Method 2
Switzerland	24	1914:08	1916:07	0.17	-18.71	Custom
United Kingdom	6	1914:08	1915:01	0.11	-2.94	Method 1
United States	5	1914:08	1914:12	-2.14	-3.11	Method 1
Panel B: World War II						
Austria	2	1938:04	1938:05	6.01	5.64	Method 2
Austria	113	1939:07	1948:11	315.61	-16.90	Custom
Belgium	5	1940:06	1940:10	22.38	12.54	Method 2
Belgium	11	1944:08	1945:06	-0.29	-17.08	Method 2
Czechoslovakia	16	1938:10	1940:01	32.23	16.91	Method 2
Czechoslovakia	4	1942:01	1942:04	20.66	12.32	Method 2
Denmark	2	1940:05	1940:06	-7.64	-10.67	Method 2
France	2	1939:09	1939:10	-2.96	0.53	Method 1
France	10	1940:06	1941:03	94.57	75.61	Method 2
Germany	67	1943:01	1948:07	-87.62	-91.10	Method 2
Japan	45	1945:09	1949:05	449.38	-87.15	Method 1
Netherlands	5	1940:05	1940:09	20.63	15.21	Method 2
Netherlands	21	1944:09	1946:05	-14.33	-33.15	Method 2
Norway	2	1940:04	1940:05	-2.07	-3.52	Method 1
Switzerland	2	1940:06	1940:07	-3.57	-5.11	Method 1
Panel C: Revolution						
Czechoslovakia	26	1943:04	1945:05	-90.00	-88.59	Custom
Portugal	35	1974:05	1977:03	-80.40	-89.24	Method 2

(Continued on next page)

Table A.III (Continued)

Country	Months	Start date	End date	Nominal return (%)	Real return (%)	Adjustment
Panel D: Financial or banking crisis						
Austria	2	1931:10	1931:11	6.25	5.60	Method 2
Germany	2	1931:08	1931:09	−24.58	−23.01	Method 2
Germany	7	1931:10	1932:04	−8.22	1.78	Method 2
Greece	2	2015:07	2015:08	−21.53	−20.13	Method 2
Panel E: Labor strike						
France	2	1974:04	1974:05	−6.17	−8.76	Method 1
France	2	1979:03	1979:04	12.79	10.69	Method 1

Table A.IV: Bond return smoothing

The table summarizes periods with missing government bond yield data. In each case, we use the country-level yield data from before and after the missing observations to produce a series of constant monthly returns across the period reported in the table. For each period, the table reports the country, the number of missing observations, and the corresponding start and end dates.

Country	Months	Start date	End date
Argentina	4	1948:08	1948:11
	11	1949:01	1949:11
	11	1950:01	1950:11
	11	1951:01	1951:11
	11	1952:01	1952:11
	11	1953:01	1953:11
	11	1954:01	1954:11
	24	1955:01	1956:12
	1	1958:02	1958:02
	1	1958:08	1958:08
	1	1959:05	1959:05
	1	1959:08	1959:08
Belgium	3	1940:05	1940:07
Czechoslovakia	15	1938:10	1939:12
Finland	1	1991:06	1991:06
Germany	8	1931:08	1932:03
	25	1943:12	1945:12
Greece	44	1989:01	1992:08
Netherlands	2	1940:05	1940:06
	3	1944:09	1944:11
	11	1945:01	1945:11
Portugal	7	1974:05	1974:11
	11	1975:01	1975:11
	1	2014:02	2014:02
Switzerland	5	1914:08	1914:12

Table A.V: Dividend-price ratio data

The table reports the annual dividend-price ratio data used in the analysis for each country in the sample. The data series symbols correspond to those in the GFDDatabase from Global Financial Data. For countries requiring custom data series, the table refers the reader to Section A.6, which describes the underlying data sources and construction procedures.

Country	Series	Start year	End year
Argentina	SYARGYM	1947	1966
Australia	SYAUSYM	1901	2023
Austria	SYAUTYM	1920	2023
Belgium	SYBELYM	1897	2023
Canada	SYCANYTM	1891	2023
Chile period I	SYCHLYM	1927	1970
Chile period II	SYCHLYM	2010	2023
Colombia	SYCOLYM	2020	2023
Czech Republic	SYCZEYM	2000	2023
Czechoslovakia	SYCZEYM	1922	1945
Denmark	SYDNKYM	1890	2023
Finland	SYFINYM	1969	2023
France	SYFRAYM	1890	2023
Germany	SYDEUYM	1890	2023
Greece	SYGRCYM	1981	2023
Hungary	SYHUNYM	1999	2023
Iceland	SYISLYM	2002	2023
Ireland	SYIRLYM	1936	2023
Israel	SYISRYM	2010	2023
Italy	SYITAYM	1931	2023
Japan	SYJPNYM	1930	2023
Latvia	See Section A.6	2016	2023
Lithuania	SYLTUYM	2018	2023
Luxembourg	SYLUXYM	1982	2023
Mexico	SYMEXYM	2001	2023
Netherlands	SYNLDYAM	1914	2023
New Zealand	SYNZLYM	1896	2023
Norway	SYNORYM	1914	2023
Poland	SYPOLYM	1999	2023
Portugal	See Section A.6	1934	1987
	SYPRTYM	1988	2023
Singapore	SYSGPYM	1998	2023
Slovakia	See Section A.6	2000	2023
Slovenia	SYSVNYM	2010	2023
South Korea	SYKORYM	2000	2023
Spain	SYESPYM	1959	2023
Sweden	SYSWEYM	1910	2023
Switzerland	SYCHEYM	1914	2023
Türkiye	SYTURYM	2010	2023
United Kingdom	_DFTASD	1890	2023
United States	SYUSAYM	1890	2023

Table A.VI: Data gaps for dividend-price ratios

The table summarizes periods with missing annual dividend-price ratio data. For each period, the table reports the country, the number of missing annual observations, the start and end dates for the period, and the method used to estimate the missing values. Under method 1, we infer annual dividend-price ratios using total return index and price index data. Under method 2, we fill in missing dividend-price ratios using the last non-missing dividend-price ratio. Under method 3, we replace missing values with the following year's dividend-price ratio.

Country	Years	Start year	End year	Method
Denmark	31	1938	1968	Method 1
France	1	1940	1940	Method 2
Germany	7	1944	1950	Method 2
Iceland	1	2001	2001	Method 3
	17	2007	2023	Method 1
Italy	1	1945	1945	Method 1
Japan	4	1945	1948	Method 2
Luxembourg	1	1981	1981	Method 3
	29	1995	2023	Method 1
Mexico	1	2014	2014	Method 1
Singapore	1	2021	2021	Method 1
Switzerland	4	1914	1917	Method 2

Table A.VII: External validation test results

The table reports summary statistics for monthly real net bond returns for each developed country with return data that overlap the Datastream sample. For each country, the table shows the number of sample months. The table also reports the following summary statistics for our sample and for the Datastream sample: the arithmetic average return ($\bar{R}_a$), the geometric average return ($\bar{R}_g$), the standard deviation of return (SD), the minimum (Min) and the maximum (Max) return, and the correlation between the return samples (Corr).

Country	Months	Our data					Datastream					Corr
		$\bar{R}_a$ (%)	$\bar{R}_g$ (%)	SD (%)	Min (%)	Max (%)	$\bar{R}_a$ (%)	$\bar{R}_g$ (%)	SD (%)	Min (%)	Max (%)	
Australia	420	0.39	0.36	2.19	-5.63	6.80	0.40	0.38	2.22	-6.75	7.84	0.97
Austria	420	0.21	0.19	1.85	-7.31	5.99	0.26	0.25	1.67	-7.10	4.99	0.96
Belgium	414	0.27	0.25	1.94	-6.49	7.47	0.33	0.31	1.92	-7.48	8.33	0.98
Canada	420	0.33	0.31	2.02	-6.33	6.47	0.34	0.32	1.99	-6.11	6.36	0.97
Czech Republic	284	0.11	0.08	2.45	-8.47	9.62	0.17	0.15	2.35	-8.49	8.76	0.94
Denmark	419	0.31	0.29	2.11	-7.05	6.40	0.34	0.32	2.00	-7.02	5.83	0.98
Finland	388	0.34	0.32	2.16	-7.16	8.59	0.38	0.36	1.96	-7.55	8.01	0.97
France	420	0.31	0.29	1.92	-7.15	5.86	0.35	0.34	1.81	-5.85	5.19	0.97
Germany	420	0.23	0.21	1.86	-6.84	5.31	0.26	0.24	1.78	-6.93	5.08	0.97
Greece	297	0.33	0.14	5.95	-30.84	26.45	0.54	0.28	7.08	-41.35	41.84	0.93
Hungary	299	0.23	0.17	3.49	-9.83	12.42	0.24	0.18	3.41	-16.31	15.34	0.86
Ireland	420	0.34	0.30	2.62	-14.79	15.45	0.35	0.32	2.42	-14.16	14.54	0.97
Italy	393	0.42	0.38	2.51	-8.03	10.26	0.44	0.41	2.38	-11.27	8.62	0.94
Japan	420	0.19	0.18	1.45	-7.76	5.30	0.24	0.23	1.33	-6.15	5.19	0.95
Mexico	162	0.13	0.09	2.60	-7.25	5.76	0.12	0.09	2.38	-7.18	7.95	0.87
Netherlands	420	0.22	0.20	1.93	-8.54	6.71	0.27	0.25	1.83	-8.22	5.21	0.97
New Zealand	393	0.38	0.36	2.10	-8.40	7.28	0.40	0.38	2.04	-7.11	6.93	0.95
Norway	373	0.29	0.27	1.96	-6.24	6.01	0.28	0.27	1.85	-6.48	5.50	0.97
Poland	276	0.32	0.28	2.85	-11.06	12.51	0.31	0.27	2.92	-11.71	13.92	0.94
Portugal	365	0.40	0.35	3.18	-13.23	14.98	0.45	0.40	3.16	-14.84	20.14	0.95
Singapore	180	-0.01	-0.03	1.89	-6.13	4.94	0.15	0.13	1.90	-4.66	11.07	0.88
South Korea	141	0.12	0.10	1.84	-4.75	6.05	0.14	0.12	1.72	-3.90	5.88	0.98
Spain	397	0.37	0.34	2.48	-9.92	9.47	0.42	0.40	2.38	-9.76	9.65	0.96
Sweden	420	0.35	0.32	2.30	-7.82	6.77	0.38	0.36	2.12	-8.48	5.62	0.98
Switzerland	420	0.19	0.18	1.65	-4.86	5.88	0.23	0.21	1.53	-4.43	5.06	0.94
United Kingdom	420	0.23	0.20	2.27	-10.62	8.23	0.24	0.22	2.16	-10.79	7.03	0.98
United States	420	0.24	0.22	2.23	-7.34	11.71	0.23	0.20	2.20	-6.24	11.69	0.98

B Methods

This appendix provides additional methodological details for two components of the lifecycle model. Section B.1 presents the Social Security benefit and Supplemental Security Income calculations for the baseline lifecycle model. Section B.2 describes the implementation of the health, mortality, and medical expenditure processes for the extended background risk specification.

B.1 Social Security benefits calculations

We compute Social Security benefit payments using the formulas effective in 2022. Given that Social Security formulas are inflation-indexed, our calculations use real 2022 USD. As such, the amount subject to Social Security taxes for each year in the investor's working life is the minimum of annual income and \$147,000, which is the maximum amount of taxable earnings. The average indexed monthly earnings (AIME) for each investor is the average of their highest 35 years of taxed earnings divided by 12. An investor who retires at the normal retirement age of 67 has a personal Social Security benefit according to a formula that sums 90% of AIME up to \$1,024, 32% of AIME between \$1,024 and \$6,172, and 15% of AIME in excess of \$6,172. The retirement benefit reduces by (5/9)% per month of early retirement up to 36 months and further reduces by (5/12)% per month of early retirement between ages 62 (the earliest allowed retirement age) and 64 (normal retirement age minus 36 months). The retirement benefit increases by (2/3)% per month of late retirement between ages 67 and 70 (the latest allowed retirement age).

Spouses may be eligible for additional benefits under the Social Security system. Some investors may optimally choose to take spousal and/or survivor benefits. When both members of the couple are living, each spouse may elect to receive half of their spouse's personal retirement benefit as a spousal benefit instead of their own retirement benefit. This option is advantageous if one of the two household members earns substantially more than the other during their working years. The spousal benefit is reduced by (25/36)% per month of early retirement up to 36 months and further reduced by (5/12)% per month between ages 62 and 64. Upon the death of one spouse, the surviving spouse qualifies for a survivor benefit. The potential survivor benefit is the personal benefit of the deceased spouse if the surviving spouse has reached full retirement age. If not, the survivor benefit is reduced by up to 28.5%. The surviving spouse may take the larger of their personal retirement benefit and their survivor benefit.

Finally, the SSA administers the Supplemental Security Income (SSI) program, which provides payments to retirees (and certain other individuals) who have little income from other sources. SSI payments are reduced by earnings, and the maximum monthly benefit amounts in 2022 are \$1,261 for couples and \$841 for singles. We impose minimum monthly consumption levels of \$1,261 for couples and \$841 for singles to reflect SSI payments. This modeling choice reflects the social safety net and avoids numerical issues in computing utility when consumption approaches zero.

B.2 Health and medical expense processes

This appendix provides full details of the household health risk specification described in Section 5.4.

The De Nardi, French, and Jones (2010) processes, which apply beginning at age 70, depend on sex, age, health status, and permanent income. We measure permanent income for individual i as the rank of realized lifetime earnings,

$$q^i = F\left(\sum_{\tau=1}^{40} Y_{\tau}^i\right), \quad (\text{B1})$$

where F is the cumulative distribution function of lifetime earnings in a large reference simulation of the income model in Section 4.2. The estimated profiles are continuous functions of the rank, and we evaluate them at each individual's q^i .

Health status h_τ^i equals one if individual i is sick at age $\tau + 24$ and zero if the individual is healthy. De Nardi, French, and Jones (2010) estimate health transitions at a two-year horizon:

$$\Pr(h_\tau^i = 1 \mid h_{\tau-2}^i) = \Lambda(\kappa_0(\tau) + \kappa_1(\tau) h_{\tau-2}^i + \kappa_2(\tau) \mathbb{1}\{i = m\} + \kappa_3(\tau) q^i + \bar{\kappa} (q^i)^2), \quad (\text{B2})$$

where $\bar{\kappa}$ is constant and the remaining coefficients vary with age. The survival and medical expense profiles below share this structure. We follow De Nardi, French, and Jones (2010) to convert the two-year transition probabilities to annual probabilities. Specifically, let P_{11} denote the two-year probability of remaining sick and P_{00} the two-year probability of remaining healthy. The annual probabilities of remaining sick and remaining healthy are

$$p_{11} = \frac{-B + \sqrt{B^2 - 4AC}}{2A} \quad \text{and} \quad p_{00} = \frac{p_{11}^2 + 1 - p_{11} - P_{11}}{1 - p_{11}}, \quad (\text{B3})$$

where $A = 2 - P_{11} - P_{00}$, $B = 2(P_{00} - 1)$, and $C = 1 + P_{11}^2 - P_{11} - P_{00}$. De Nardi, French, and Jones (2010) estimate the health-transition, survival, and medical-expense profiles as continuous functions of an individual's permanent-income percentile rank, which enters linearly and squared. We follow them directly, evaluating the profiles at each household's exact rank. The initial health state at age 70 is drawn from the stationary distribution of the annual chain evaluated at the age 70 coefficients, such that the initial probability of being sick is $\pi_1 = (1 - p_{00}) / [(1 - p_{00}) + (1 - p_{11})]$.

The annual survival probability for individual i is

$$\Pr(\text{alive at } \tau + 1 \mid \text{alive at } \tau) = [\Lambda(\psi_0(\tau) + \psi_1(\tau) h_\tau^i + \psi_2(\tau) \mathbb{1}\{i = m\} + \psi_3(\tau) q^i + \bar{\psi} (q^i)^2)]^{1/2}, \quad (\text{B4})$$

where the term in brackets is the estimated two-year survival probability. Health-dependent mortality replaces the SSA life tables beginning at age 70. Survival probabilities reach zero by age 102, so T_{max} does not exceed the final month of age 102 in this specification.

Out-of-pocket medical expenses for individual i are lognormally distributed with persistent and transitory shocks:

$$\log M_\tau^i = \mu(\tau, h_\tau^i, \mathbb{1}\{i = m\}, q^i) + \sigma(\tau, h_\tau^i, \mathbb{1}\{i = m\}, q^i) (\zeta_\tau^i + \omega_\tau^i), \quad (\text{B5})$$

$$\zeta_\tau^i = \rho_\zeta \zeta_{\tau-1}^i + u_\tau^i, \quad (\text{B6})$$

where the mean μ and variance σ^2 of log expenses are linear in health status, a male indicator, q^i , and $(q^i)^2$, $\omega_\tau^i \sim N(0, \sigma_\omega^2)$ is a transitory shock, and $u_\tau^i \sim N(0, \sigma_u^2)$ is the innovation to the persistent component, which begins at age 70 from its stationary distribution, $N(0, \sigma_u^2 / (1 - \rho_\zeta^2))$. The parameters are $\rho_\zeta = 0.922$, $\sigma_u = 0.224$, and $\sigma_\omega = 0.815$ [Table I in Duarte, Fonseca, Goodman, and Parker (2024), from De Nardi, French, and Jones (2010)]. The profiles are estimated in 1998 USD, and we convert expenses to 2022 USD by adjusting for the change in the CPI, a factor of 1.7949. Household medical expenses are the sum of M_τ^i across living members, subject to a cap:

$$M_\tau = \min \left\{ M_\tau^f + M_\tau^m, M_{max} \right\}, \quad (\text{B7})$$

where $M_\tau^i = 0$ for a deceased member and M_{max} is the \$1 million annual household cap of Duarte, Fonseca, Goodman, and Parker (2024).

For $t + 1 > T_{ret}$, the wealth dynamics in equation (10) become

$$\widetilde{W}_t = W_t - \min\{W_t, \frac{1}{12}M_\tau\}, \quad (\text{B8})$$

$$D_{t+1} = \min\left\{\frac{1}{12}\Lambda(c_0 + c_1a_{t+1}^r + c_2\widehat{W}_{t+1}^{SS})W_t, \widetilde{W}_t\right\}, \quad (\text{B9})$$

$$W_{t+1} = \left(\widetilde{W}_t - D_{t+1}\right)R_{t+1}^p, \quad (\text{B10})$$

where τ denotes the year containing month $t + 1$ and consumption follows equation (11). The withdrawal rate applies to wealth before medical expenses, and the withdrawal amount is capped by wealth net of expenses. Unpaid excess expenses capture means-tested coverage of catastrophic costs in reduced form. Setting $M_\tau = 0$ recovers the base case.

C Supplemental results

This appendix contains supplemental empirical results.

C.1 Variance ratios

In Section 1, we show that the long-horizon properties of real returns differ markedly across asset classes. International stock returns exhibit substantial mean reversion, whereas bond returns exhibit persistence, as reflected in long-horizon variance ratios below one for international stocks and above one for bonds. These return properties, in combination with bonds' greater long-horizon correlation with domestic stocks and stronger exposure to inflation, play a central role in the paper's finding that households optimally diversify using international stocks rather than bonds.

Table C.I examines the robustness of the variance ratio results. For domestic stocks, international stocks, bonds, and bills, the table reports variance ratios calculated from the actual data and from block bootstrap simulations with average block lengths ranging from five to 30 years. The variance ratios based on the actual data are computed using the pooled mean return across all country-month observations, consistent with the pooled bootstrap design used throughout the paper. The variance ratios exhibit similar qualitative patterns across the actual data and all bootstrap specifications, with long-horizon variance ratios remaining below one for domestic and international stocks and above one for bonds and bills. The baseline bootstrap simulation with a 10-year average block length produces variance ratios that are generally similar to those observed in the actual data. The key variance ratio patterns are also largely insensitive to the choice of bootstrap specification.

C.2 Household longevity

Table C.II summarizes the distribution of age at death conditional on survival to age 25 based on the SSA data and our simulation procedure described in Section 4. The table reports the mean, standard deviation, and selected distributional percentiles for age at death for heterosexual couples, single females, single males, female couples, and male couples. The statistics for the couples correspond to the age at death of the last surviving member of the couple. In the base case of a heterosexual couple, the mean age of the last survivor at death is 87.6 years, and the median age is 88.9 years. There is considerable uncertainty about longevity outcomes. The 5th percentile of age at death for the couple is 70.8 years, and the 95th percentile is 100.0 years. This uncertainty is an important consideration when assessing the ability of investment strategies to fund consumption through retirement. The last column of Table C.II reports the probability that each investor type dies before reaching retirement age. There is a 19.5% (11.8%) chance that the male (female) dies before age 65, and there is a 2.3% chance that neither member of the heterosexual couple survives into the retirement period.

C.3 Optimization of the age- and wealth-dependent asset allocation policy

In Section 5.1, we present the optimal age-based strategy. In this section, we describe the construction and optimization of an asset allocation policy that conditions on household wealth in addition to age. We expand the strategy space so that portfolio weights may differ across wealth groups within each window, re-optimize, and measure the welfare gain of considering wealth.

The age-based policy assigns one portfolio weight vector to each of the 13 age windows. The age- and wealth-dependent policy additionally partitions the households in each window into

thirds of the within-window wealth distribution, assigning a separate weight vector to each age-window/wealth-tercile cell. Wealth is measured by the same normalized state variables the saving and withdrawal rules condition on. A household's tercile is determined once per age window, at the window's first month, and held throughout the window. To fix the tercile breakpoints, we use the boundaries generated under the optimal age-based strategy from the base case. The first age window (ages 25–34) has a single cell, as all households enter this window with zero wealth. The resulting policy has $12 \times 3 + 1 = 37$ cells with four asset weights each, which produces 111 free weight parameters, while the saving and withdrawal rules remain common to all cells.

Given the dimensionality of the problem, our base optimization procedure is impractical. We therefore optimize in stages, using initial evidence that a corner solution is likely, restricted parameterizations to locate the optimum's neighborhood, and certification of the final policy in the full, unrestricted weight space. The intermediate parameter restrictions are only used to narrow the search window, and none is imposed on the final policy.

The optimization proceeds in the following steps:

1. We first evaluate whether any cell benefits from holding bonds or bills. For each of the 37 cells and each fixed income asset, we perturb the certified age-based optimum by moving one percentage point of the cell's equity (reduced pro rata across domestic and international stocks) into the fixed income asset, holding the saving and withdrawal rules at their optimal base values. By the envelope theorem, the value of re-optimizing the rules against a one-percentage-point weight perturbation is second order, so these evaluations isolate the first-order welfare effect of the allocation change. Only two of the 74 perturbations increase welfare (adding 1% in bonds for rich households at ages 70–74 and 75–79), and the utility gains are very small. To rule out allocations beyond one percentage point, we repeat the exercise on the 18 most favorable cells with fixed income allocations of 1% to 15% in 1% increments. Welfare declines monotonically with the allocation in every cell. We conclude that the optimum is unlikely to contain a substantial fixed income allocation in any cell.
2. We next parameterize wealth dependence parsimoniously: each wealth tercile tilts the domestic-international split of the optimal age-based strategy by a constant number of percentage points at every age. We evaluate the full joint grid of tilts from -5% to 5% in 1% increments ($11^3 = 1,331$ combinations). The jointly welfare-maximizing tilts are -4% for poor households, 2% for middle, and 3% for rich. At this winner, we re-evaluate potential fixed income allocations for older, wealthier households.
3. Finally, we free every cell. Starting from the Step-2 winner, we run the third stage from our base optimization procedure. On a one-percentage-point portfolio-weight grid that covers all age- and wealth-based cells, we evaluate all neighboring weight vectors on common lifecycle draws, moving to the best neighbor that improves expected utility and re-optimizing the saving and withdrawal parameters after each move, until no neighboring weight vector offers an improvement. This procedure allows the optimization to flexibly find any pattern in age- and wealth-based weights and is free from the restrictions imposed in the earlier steps.

At the optimum, every cell remains fully invested in equities; at no age and no wealth level does the optimal policy hold bonds or bills. Figure C.1 shows the optimal allocation to domestic stocks at each age and wealth tercile. A household may shift across terciles over its lifetime, so its optimal strategy would involve switching across the lines in the figure over time. The figure shows that the domestic share is ordered in wealth at every age. Poor households hold the most in international stocks and rich households the most in domestic.

The welfare gain from conditioning on wealth is economically small. Using the equivalent annual fee measure of Section 4.3, the optimal age- and wealth-based policy improves on the optimal age-based policy by less than 0.0040% per year. The gain is larger than that of the optimal age-based policy over the optimal fixed-weight policy, but negligible relative to the equivalent annual fee of any of the benchmark strategies. Overall, conditioning on age or on age and wealth has a negligible benefit, and the case for internationally diversified all-equity portfolios is consistent across designs.

C.4 Utility costs of deviating from the optimal fixed-weight asset allocation

In Section 5.1, we show that an optimal fixed-weight allocation of 34% to domestic stocks, 66% to international stocks, and 0% to bonds and bills delivers nearly the same expected lifetime utility as the optimal age-based strategy. We quantify below the welfare losses associated with deviations from this optimal fixed-weight allocation using equivalent annual fees.

Figure C.2 illustrates the equivalent annual fees associated with alternative fixed-weight portfolios. Panel A varies the allocation to domestic stocks from 0% to 100% within an all-equity portfolio, with the remaining portfolio weight allocated to international stocks. Expected lifetime utility as a function of the allocation to domestic stocks is relatively flat around the optimal allocation, with allocations ranging from 15% to 53% in domestic stocks generating equivalent annual fees of less than 10 basis points. Larger deviations are considerably more costly, particularly when the portfolio is concentrated in domestic stocks. An allocation of 100% to domestic stocks produces an equivalent annual fee of 122 basis points.

Panels B and C introduce fixed allocations to bonds and bills, respectively, while preserving the optimal 34/66 split between domestic and international stocks within the equity allocation. The equivalent annual fees increase rapidly with the allocation to either fixed income asset. Bond allocations of 10%, 20%, 30%, and 40% produce equivalent annual fees of 31, 68, 110, and 156 basis points, respectively. The corresponding fees for bill allocations are even larger, at 43, 91, 144, and 200 basis points. These results indicate that the welfare costs of modest deviations from the optimal domestic-international stock mix are relatively small, whereas replacing equities with bonds or bills imposes substantial welfare losses.

C.5 Performance of optimal age-based and fixed-weight strategies

In Section 5.1, we demonstrate that the optimal fixed-weight policy with 34% allocated to domestic stocks, 66% to international stocks, and 0% to bonds and bills delivers nearly the same expected lifetime utility as the optimal age-based allocation policy. Table C.III compares the optimal age-based and fixed-weight strategies based on three retirement saving outcomes: the distribution of wealth at retirement (Panel A), the distribution of the income replacement rate (Panel B), and the distribution of wealth at death (Panel C). Table C.IV compares the two strategies based on the distribution of the maximum drawdown during the working years (Panel A) and the distribution of the maximum drawdown during the retirement years (Panel B). The results confirm that the optimal fixed-weight strategy generates retirement saving outcomes and drawdown distributions that are nearly identical to those of the optimal age-based strategy.

C.6 Conditional strategy performance

In Section 5.2, we show that the optimal age-based strategy provides stronger retirement income security than the benchmark strategies, as measured by the probability that Social Security income accounts for more than 75% of retirement income. This measure reflects both wealth accumulation

before retirement and the ability of the investment strategy to sustain wealth throughout retirement. In this section, we further characterize these results by examining the probability that Social Security income exceeds 75% of retirement income conditional on household and market outcomes.

Figure C.3 plots the conditional probability that Social Security income accounts for more than 75% of retirement income for three benchmark strategies and the optimal age-based strategy (the bills strategy is omitted for readability because its probabilities are substantially higher than those of the other strategies). In each panel, the 1,000,000 bootstrap simulations are divided into quintiles based on the indicated household or market outcome, and the conditional probability is computed within each quintile.

Panel A conditions on household longevity. Greater longevity is generally associated with a higher probability that Social Security income accounts for more than 75% of retirement income because longer retirements place greater demands on accumulated retirement wealth. The optimal age-based strategy produces the lowest probability in every quintile. The differences across strategies are largest for households in the highest longevity quintile. The optimal age-based strategy has a probability of only 1.1%, compared with 5.9% for the domestic stock strategy, 7.6% for the balanced strategy, and 6.1% for the TDF. These results indicate that the optimal strategy is particularly effective at sustaining retirement income for the longest-lived households.

Panel B conditions on the cumulative real return for domestic stocks during the retirement period. Although the optimal strategy produces the lowest probability in every return quintile, the differences across strategies are most pronounced when domestic stock returns are weakest. In the lowest return quintile, the optimal age-based strategy has a probability of 1.7%, compared with 11.1% for the domestic stock strategy, 11.9% for the balanced strategy, and 7.9% for the TDF. These results suggest that the larger fixed income allocations of the balanced strategy and the TDF do not improve retirement income security when stock market performance is poor.

Panel C examines the role of inflation. When realized inflation during retirement is low, all strategies perform relatively well, with probabilities ranging from 0.4% (optimal) to 3.4% (domestic stocks). When inflation is high, the optimal age-based strategy has a probability of only 1.1%, compared with 8.0% for the domestic stock strategy, 13.6% for the balanced strategy, and 11.7% for the TDF. The low absolute correlation between inflation and international stock returns over long horizons (Table I) implies that international stocks provide important diversification benefits during periods of high inflation.

The benefits of international diversification depend on the correlation between returns on domestic stocks and international stocks. This correlation varies over time [e.g., Longin and Solnik (2001)], so the value of international diversification could also vary. Because the optimal age-based strategy relies heavily on international stocks, its performance could be particularly sensitive to changes in the realized domestic-international correlation. Panel D examines strategy performance conditional on the realized correlation between domestic and international stock returns during the retirement period. The optimal age-based strategy produces the lowest probability in every correlation quintile, with probabilities ranging from 0.6% to 0.8%. The probabilities are meaningfully higher in the lowest correlation quintile for the benchmark strategies. The low correlations seem to proxy for adverse economic environments and wars, in which domestic markets perform relatively poorly, whereas high correlations tend to coincide with periods of stronger asset class returns. Whatever the underlying causes may be, the results in Panel D alleviate concerns that periods of unusually high domestic-international correlations substantially diminish the benefits of international diversification within the optimal age-based strategy.

C.7 Dependencies between labor income and domestic stock returns

In Section 5.4, we link innovations to the persistent component of labor income to domestic stock returns through a Gaussian copula. A natural question is whether this specification generates meaningful dependence in the states that matter for portfolio choice, since the Gaussian copula is asymptotically tail independent. Figure C.4 addresses this question by characterizing the joint behavior of income growth and returns that the model produces.

Panel A reports conditional percentiles of log earnings growth by stock return quintile for the countercyclical income risk process estimated by Catherine (2022), which we simulate from his published parameter estimates. Catherine (2022) estimates this process to match the cyclical skewness moments of Guvenen, Ozkan, and Song (2014), so Panel A serves as a benchmark for the state dependence of earnings risk in US administrative data. Panels B and C report the same percentiles for our model among employed individuals at copula correlations of 0.1 and 0.2. In both panels, the dashed lines show the zero-correlation case, in which the percentiles are flat across return states. The percentile gradients in Panel B, with correlation 0.1, closely track those in Panel A. Earnings growth deteriorates in low-return states, and the deterioration is concentrated in the left tail while the median moves modestly. In terms of Kelley skewness, $\rho = 0.1$ generates a Q1-to-Q5 swing of 0.14, which is comparable to the swing of 0.16 in the simulated Catherine (2022) process and to the decline of 0.11 to 0.16 that Guvenen, Ozkan, and Song (2014) estimate between expansions and recessions. The Panel C results, with correlation 0.2, show steeper gradients, with a Kelley skewness swing of 0.28 that exceeds these benchmarks, so a correlation of 0.1 aligns the model with US evidence. The state dependence in our model therefore arises from the interaction of the copula with the heavy-tailed distributions of income shocks and returns rather than from either the copula or the distributions alone.

The Guvenen, Karahan, Ozkan, and Song (2021) nonemployment channel provides a second source of dependence with no counterpart in Catherine (2022), whose income process is continuous. Because the nonemployment hazard in our model depends on the permanent income component, workers become more likely to lose a full year of earnings when returns are low. Panel D shows that the probability of new nonemployment for a previously employed worker rises from 6.5% in the best return quintile to 8.9% in the worst at $\rho = 0.1$, with a sharper rise at $\rho = 0.2$. This pattern is consistent with the countercyclical employment-to-nonemployment transitions that Guvenen, Ozkan, and Song (2014) document. Taken together, the panels show that a modest copula correlation reproduces the state-dependent earnings dynamics of estimated countercyclical risk processes while adding an extensive margin those processes do not incorporate.

C.8 Alternative samples

In Section 5.7, we consider the optimal age-based and fixed-weight asset allocation policies estimated from samples restricted to countries with large populations or large market capitalization-to-GDP ratios. Table C.V reports the sample start dates for developed countries in each of these alternative samples.

C.9 Leverage

Our baseline specification assumes that households cannot use leverage in their retirement savings accounts. This constraint is realistic for most investors and can affect the optimal portfolio allocation [e.g., Frazzini and Pedersen (2014)]. Asness (1996) argues, however, that investors should use leverage to invest in a balanced portfolio with 60% domestic stocks and 40% bonds rather than

invest 100% in equities without leverage. We relax the borrowing constraint to study hypothetical households that can use leverage in their retirement savings accounts.

We model leverage as follows. We assume that households borrow at the prevailing yield on local government bills plus a spread. We consider (i) a high spread of 6.50%, estimated by Davis, Kubler, and Willen (2006) from household borrowing costs and close to the median broker margin spread in April 2024, (ii) a medium spread of 1.40%, equal to the lowest broker margin spread in April 2024, and (iii) a low spread of 0.37%, estimated by van Binsbergen, Diamond, and Grotteria (2022) from risk-free rates implied by derivative prices. Households jointly optimize over asset allocation, saving and withdrawal policies, and leverage. Borrowing is capped at 100% of household wealth, implying a maximum gross portfolio exposure of 200%. This limit is consistent with US margin regulations for stocks.

Figure C.5 illustrates the optimal age-based asset allocation policies under the three borrowing spreads. Under the high spread of 6.50% (Panel A), the optimal asset allocation is identical to that in our baseline specification, which precludes leverage. That is, the household optimally chooses to forgo leverage, and the portfolio remains fully invested in equities at all ages. Thus, access to borrowing at a cost representative of household credit or typical margin rates does not induce an allocation to bonds or bills.

At the medium spread of 1.40% (Panel B), the household uses substantial leverage throughout the lifecycle. Borrowing varies modestly over the lifecycle, producing gross portfolio exposure ranging from 132% to 200% of household wealth. Contrary to the argument that investors should lever a balanced stock-bond portfolio, the optimal portfolio remains almost entirely invested in domestic and international stocks. Bonds receive only a negligible allocation at some ages after age 50, and bills receive zero weight at all ages.

At the low spread of 0.37% (Panel C), borrowing reaches the cap at every age, producing gross portfolio exposure of 200% of household wealth. Bonds enter the optimal portfolio beginning around age 50, but the allocation remains small relative to the combined allocation to domestic and international stocks. Thus, even when borrowing is available at rates close to the risk-free rate, the household remains heavily invested in equities. More broadly, the optimal age-based strategy bears little resemblance to the levered balanced portfolio advocated by Asness (1996). We also note that the borrowing spread estimated by van Binsbergen, Diamond, and Grotteria (2022) reflects the risk-free rates implicit in futures and other derivatives. Few households use derivatives in their retirement accounts. Leveraged strategies must also be actively managed to maintain their target leverage, and leveraged products are often viewed as unsuitable for long-term investors because frequent re-leveraging can reduce performance [e.g., Cheng and Madhavan (2009)]. To the extent that households access these borrowing rates through financial intermediaries, management fees would further increase the cost of leverage.³⁴

³⁴Expense ratios for leveraged ETFs vary. As of August 2026, the largest leveraged ETF (ProShares UltraPro QQQ) charges an annual expense ratio of 0.82%, and the average leveraged ETF expense ratio is 1.12% (<https://www.etf.com/topics/leveraged>).

Table C.I: Variance ratios in the actual data and block bootstrap simulations

The table reports variance ratios of real returns for domestic stocks, international stocks, bonds, and bills. The underlying data are a monthly panel of real asset class returns for 39 developed countries covering the period from 1890 to 2023. Each panel reports variance ratios at horizons of one, ten, 20, and 30 years for the actual data and for block bootstrap simulations with average block lengths ranging from five to 30 years. Under each bootstrap design, the variance ratios are estimated from a simulated sample of 100,000,000 months. All statistics are estimated from a sample that excludes data for Germany in 1923.

Variance ratio	Actual data	Bootstrap data: Average block length				
		5 years	10 years	15 years	20 years	30 years
Panel A: Domestic stocks						
VR(1)	1.00	1.00	1.00	1.00	1.00	1.00
VR(10)	0.79	0.95	0.90	0.87	0.86	0.84
VR(20)	0.59	0.91	0.82	0.78	0.75	0.72
VR(30)	0.47	0.90	0.79	0.74	0.70	0.66
Panel B: International stocks						
VR(1)	1.00	1.00	1.00	1.00	1.00	1.00
VR(10)	0.86	0.92	0.89	0.87	0.86	0.85
VR(20)	0.64	0.88	0.80	0.75	0.72	0.69
VR(30)	0.36	0.87	0.75	0.68	0.63	0.56
Panel C: Bonds						
VR(1)	1.00	1.00	1.00	1.00	1.00	1.00
VR(10)	1.88	1.97	2.16	2.22	2.27	2.30
VR(20)	2.01	2.09	2.34	2.42	2.47	2.51
VR(30)	1.80	2.13	2.39	2.47	2.52	2.56
Panel D: Bills						
VR(1)	1.00	1.00	1.00	1.00	1.00	1.00
VR(10)	2.49	2.46	2.70	2.79	2.85	2.91
VR(20)	2.69	2.65	2.96	3.08	3.15	3.22
VR(30)	2.39	2.71	3.04	3.18	3.25	3.33

Table C.II: Distribution of age at death

The table summarizes the distribution of age at death in years conditional on survival to age 25 based on the actuarial life tables from the SSA. For each investor type (heterosexual couple, single female, single male, female couple, or male couple), the table reports the mean, standard deviation, and selected percentiles of the age at death. The statistics for the couples correspond to the age at death of the last surviving member of the couple. The final column shows the probability that the individual, or the last surviving member of a couple, dies before reaching the retirement age of 65.

Investor	Moments		Percentiles									$\Pr(T_{\max} \leq 480)$
	Mean	StDev	1%	5%	10%	25%	50%	75%	90%	95%	99%	
Female and male	87.6	9.1	59.4	70.8	76.1	83.0	88.9	93.8	97.7	100.0	104.5	0.023
Single female	81.9	13.8	36.2	54.3	62.8	75.5	84.9	91.5	96.3	98.9	103.8	0.118
Single male	77.1	15.3	30.6	46.1	55.8	68.8	80.6	88.3	93.5	96.2	101.1	0.195
Both female	89.3	8.5	62.7	74.0	78.7	84.9	90.3	95.0	98.8	101.1	105.5	0.014
Both male	85.5	9.7	56.0	67.1	72.8	80.5	87.0	92.1	96.1	98.4	102.8	0.038

Table C.III: Retirement saving outcomes: Age-based and fixed-weight asset allocation policies

The table summarizes the distributions of real wealth at retirement (Panel A), the real income replacement rate (Panel B), and real wealth at death (Panel C) across 1,000,000 bootstrap simulations for households adopting the optimal age-based and optimal fixed-weight asset allocation strategies. For each asset allocation strategy, the table reports the mean, standard deviation, and selected percentiles of each outcome. Real wealth at retirement and real wealth at death are reported in millions of 2022 USD.

Strategy	Moments		Percentiles								
	Mean	StDev	1%	5%	10%	25%	50%	75%	90%	95%	99%
Panel A: Wealth at retirement (\$MM)											
Optimal age-based	0.99	1.33	0.00	0.13	0.21	0.38	0.68	1.17	1.95	2.70	5.50
Optimal fixed-weight	0.99	1.34	0.00	0.13	0.21	0.38	0.68	1.17	1.95	2.70	5.52
Panel B: Income replacement rate											
Optimal age-based	1.59	1.27	0.00	0.63	0.75	0.96	1.28	1.79	2.62	3.48	7.02
Optimal fixed-weight	1.59	1.28	0.00	0.63	0.75	0.96	1.28	1.79	2.62	3.49	7.06
Panel C: Wealth at death (\$MM)											
Optimal age-based	0.76	7.05	0.01	0.02	0.04	0.10	0.26	0.61	1.27	2.04	6.11
Optimal fixed-weight	0.76	7.16	0.01	0.02	0.04	0.10	0.26	0.60	1.27	2.03	6.11

Table C.IV: Portfolio drawdowns: Age-based and fixed-weight asset allocation policies

The table summarizes the distributions of the maximum portfolio drawdown during the working period (Panel A) and the maximum portfolio drawdown during the retirement period (Panel B) across 1,000,000 bootstrap simulations for households adopting the optimal age-based and optimal fixed-weight asset allocation strategies. For each asset allocation strategy and drawdown period, the table reports the mean, standard deviation, and selected percentiles of the maximum portfolio drawdown.

Strategy	Moments		Percentiles								
	Mean	StDev	1%	5%	10%	25%	50%	75%	90%	95%	99%
Panel A: Working-period drawdown											
Optimal age-based	0.55	0.13	0.24	0.31	0.38	0.48	0.56	0.61	0.69	0.77	0.91
Optimal fixed-weight	0.55	0.12	0.24	0.31	0.38	0.48	0.56	0.61	0.68	0.76	0.91
Panel B: Retirement-period drawdown											
Optimal age-based	0.48	0.17	0.00	0.19	0.25	0.38	0.52	0.59	0.66	0.74	0.88
Optimal fixed-weight	0.48	0.17	0.00	0.19	0.25	0.38	0.52	0.59	0.67	0.74	0.88

Table C.V: Country-level start dates for alternative samples

The table reports the sample start dates for developed countries included in the alternative samples based on population and equity market size.

Country	Base sample	Country population $\geq$ $0.5\% \times$ world population	Market-cap/GDP ratio ≥ 0.5
United Kingdom	1890:01	1890:01	1890:01
Netherlands	1914:01	—	1919:01
Belgium	1897:01	—	1897:01
France	1890:01	1890:01	1890:01
Norway	1914:01	—	2006:01
Germany	1890:01	1890:01	1912:01
Denmark	1890:01	—	1890:01
Switzerland	1914:01	—	1928:01
United States	1890:01	1890:01	1928:01
Canada	1891:01	1933:01	1902:01
Argentina	1947:02	1947:02	—
New Zealand	1896:01	—	1900:01
Australia	1901:01	—	1932:01
Sweden	1910:01	—	1911:01
Austria	1920:01	—	1923:01
Chile period I	1927:01	—	1927:01
Greece	1981:02	—	1999:01
Czechoslovakia	1922:05	1922:05	—
Japan	1930:01	1930:01	1938:01
Portugal	1934:01	—	1999:01
Italy	1931:01	1931:01	2000:01
Ireland	1936:01	—	1945:01
Singapore	1998:07	—	1998:07
Iceland	2002:01	—	2002:01
Luxembourg	1982:01	—	1985:01
Türkiye	2010:02	2010:02	—
Spain	1959:01	1959:01	1998:01
Finland	1969:01	—	1998:01
Mexico	2001:08	2001:08	—
Czech Republic	2000:05	—	—
Hungary	1999:02	—	—
Poland	1999:06	1999:06	—
South Korea	2000:11	2000:11	2000:11
Slovakia	2000:01	—	—
Chile period II	2010:01	—	2010:01
Estonia	—	—	—
Israel	2010:01	—	2010:01
Slovenia	2010:01	—	—
Latvia	2016:01	—	—
Lithuania	2018:01	—	—
Colombia	2020:01	2020:01	2020:01

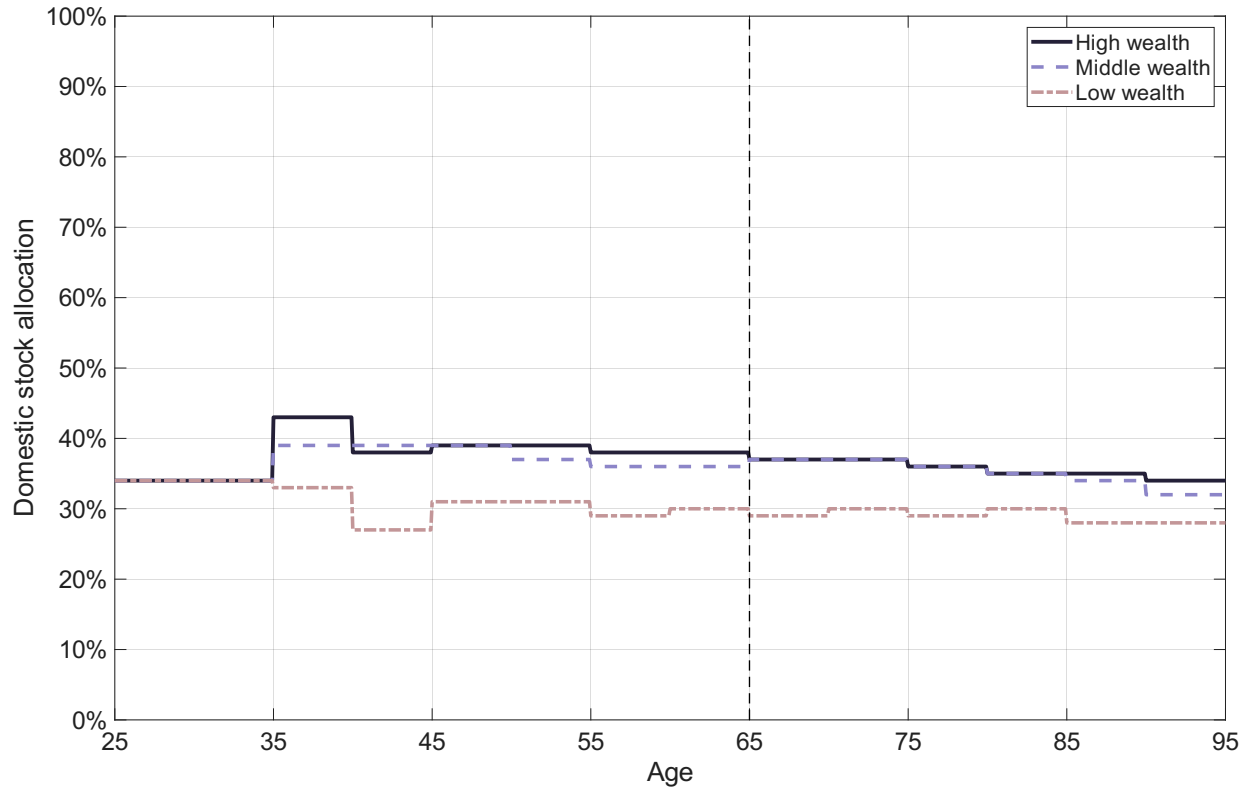


Figure C.1. Optimal age- and wealth-dependent asset allocation policy. The figure shows the portfolio weight in domestic stocks under the optimal asset allocation policy as a function of household age for low-, middle-, and high-wealth households. Wealth groups correspond to terciles of the within-window wealth distribution, with tercile assignment determined at the beginning of each age window. The first age window has a single allocation because all households enter age 25 with zero wealth. The vertical dashed line indicates the household's exogenous retirement age. In all wealth terciles, the optimal policy invests entirely in domestic and international stocks, assigning zero weight to bonds and bills at all ages.

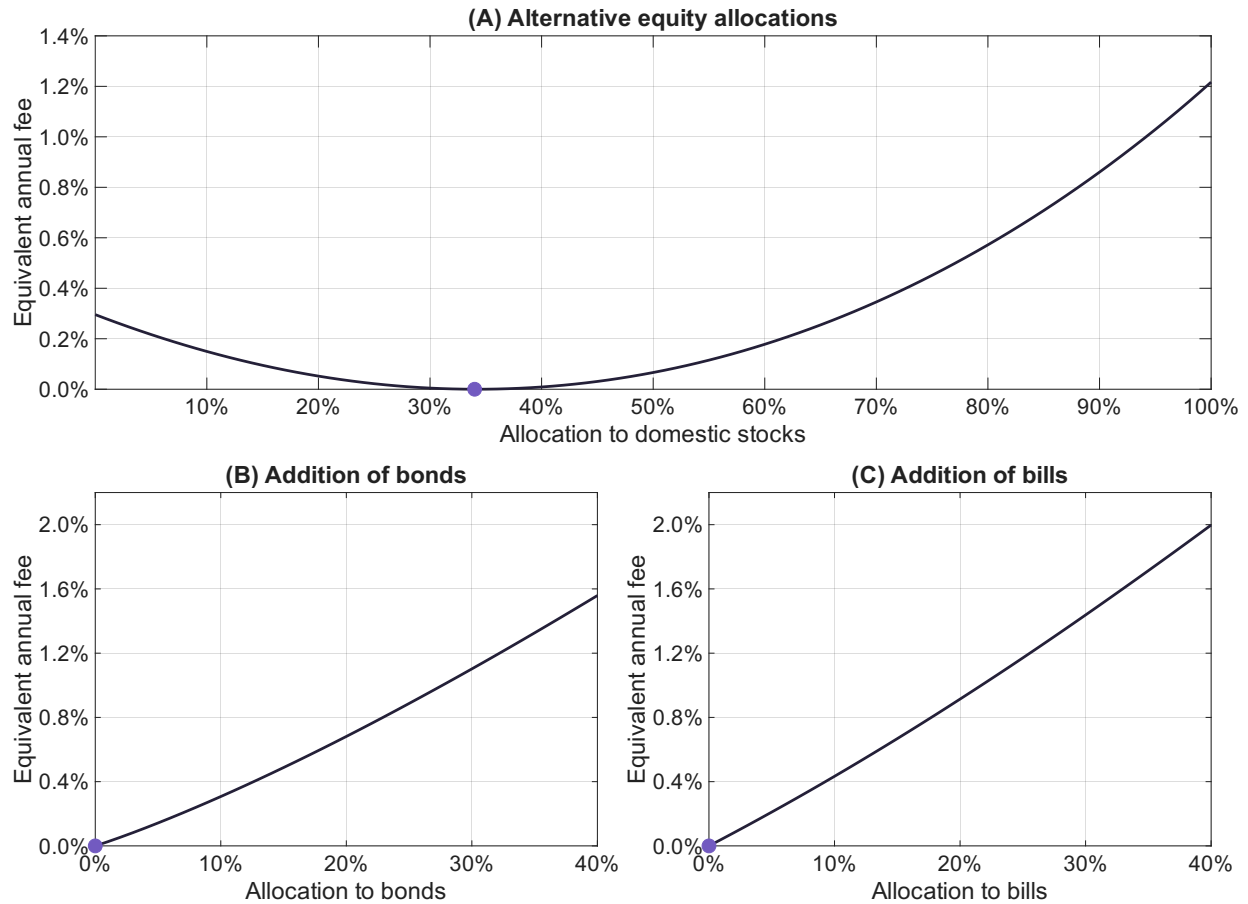


Figure C.2. Equivalent annual fees for deviations from the optimal fixed-weight portfolio. The figure illustrates the equivalent annual fees associated with deviations from the optimal fixed-weight asset allocation strategy. The welfare comparisons are based on expected lifetime utility from consumption and bequest across 1,000,000 bootstrap simulations. In each panel, the reference portfolio corresponds to the optimal fixed-weight allocation of 34% in domestic stocks and 66% in international stocks. Panel A considers alternative all-equity portfolios with different allocations to domestic and international stocks. Panels B and C consider portfolios that preserve the optimal 34/66 split between domestic and international stocks within the equity allocation while introducing fixed allocations to bonds and bills, respectively. The marker in each panel denotes the optimal fixed-weight asset allocation.

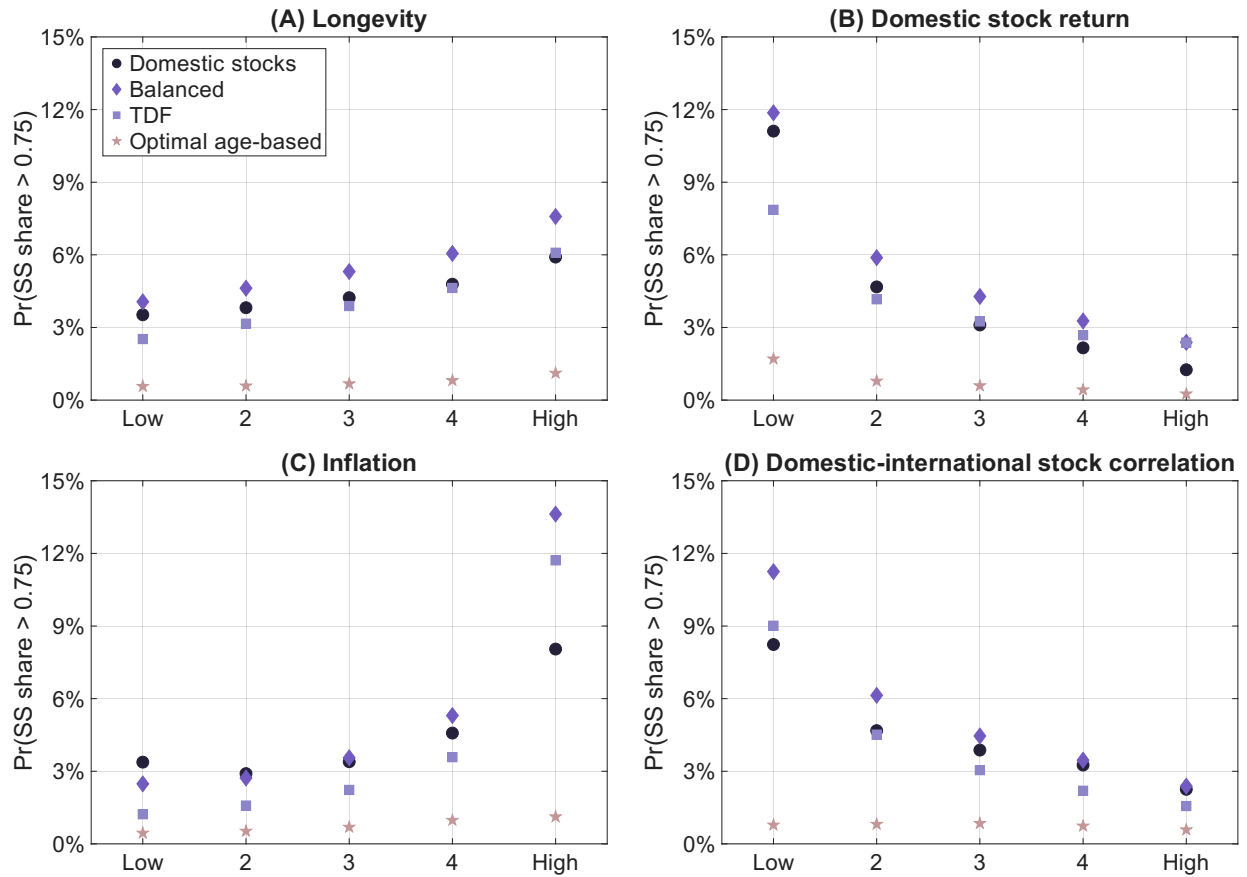


Figure C.3. Conditional probability that Social Security income exceeds 75% of retirement income. The figure shows the probability that Social Security income accounts for more than 75% of retirement income conditional on quintile outcomes of household longevity (Panel A), realized returns for domestic stocks (Panel B), realized inflation (Panel C), and realized correlation between real returns for domestic stocks and international stocks (Panel D) across 1,000,000 bootstrap simulations for households adopting various asset allocation strategies. The probabilities in Panels B, C, and D are conditional on realizations during the retirement period.

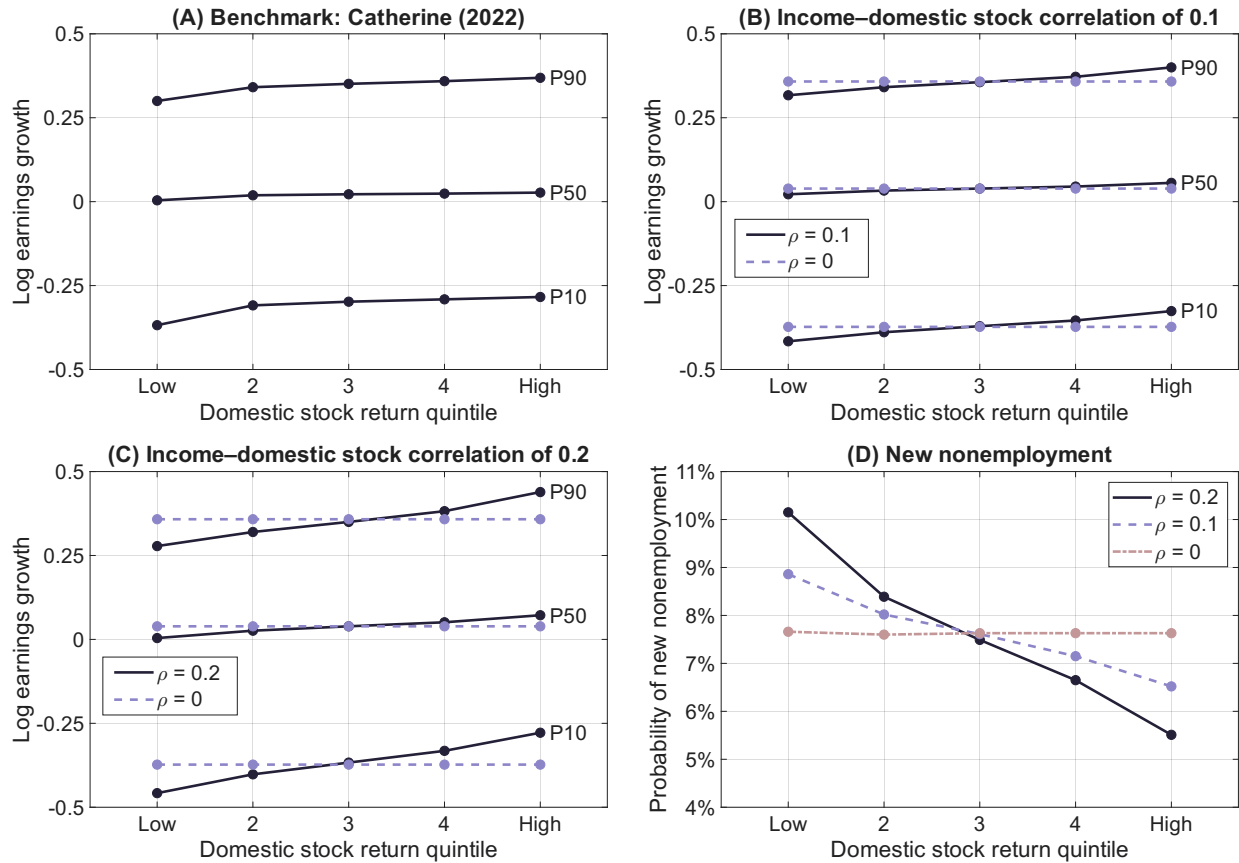


Figure C.4. Labor income growth and new nonemployment across stock return states. The figure shows how the distribution of labor income growth varies with domestic stock returns. Panel A reports the 10th, 50th, and 90th percentiles of log earnings growth by return quintile in a simulation of the estimated countercyclical income risk process of Catherine (2022), constructed from his published parameter estimates and including the aggregate wage index component of earnings growth. Panels B and C report the same percentiles for our model at income-return correlations of 0.1 and 0.2, computed among workers employed in consecutive years, with the zero-correlation case shown as a dashed baseline. Panel D reports the probability that a worker employed in the prior year has nonemployment in the current year.

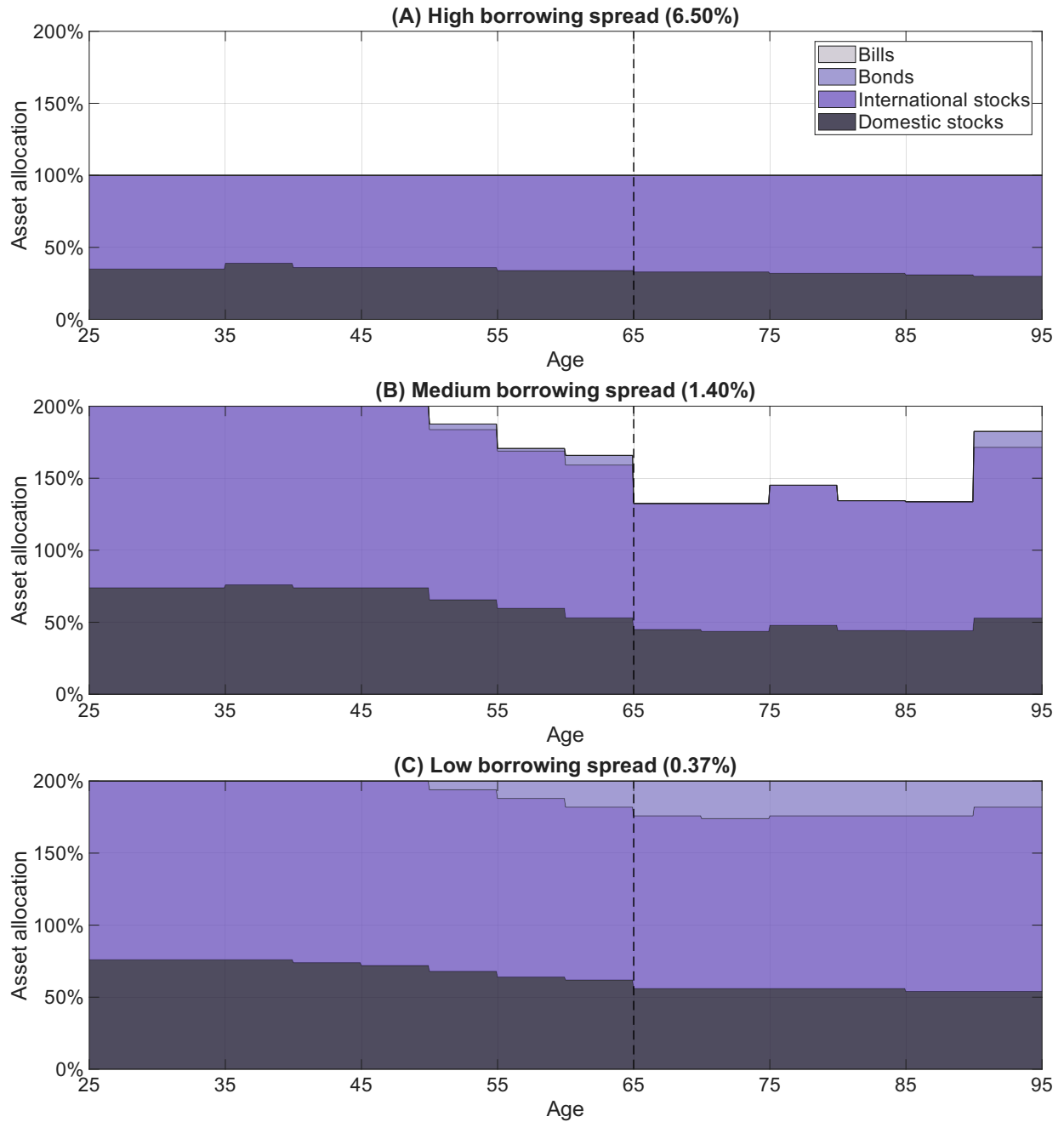


Figure C.5. Optimal age-based asset allocation policies with leverage. The figure shows the portfolio weights in domestic stocks, international stocks, bonds, and bills for the optimal asset allocation policy as a function of household age under three alternative borrowing spreads. Panels A, B, and C report the optimal policies for borrowing spreads of 6.50%, 1.40%, and 0.37%, respectively. The vertical dashed line indicates the household's exogenous retirement age. When leverage is employed, the portfolio weights sum to more than 100% because the household borrows to finance additional investments.